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ICIISTM-2026

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Proceedings of the

1st International Conference on Inventions & Innovations in Science, Technology

& Management (ICIISM-2026)

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International Conference Coordinator

Prof. Prasad R. Mutkule

International Conference Coordinator

CONVENOR

Dr. Y. R. Kharde

Principal, Shri Chhatrapati Shivaji Maharaj College of Engineering, Ahilyanagar

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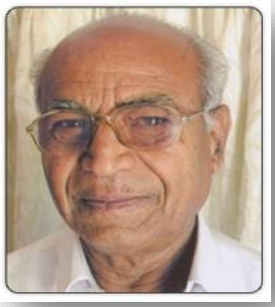

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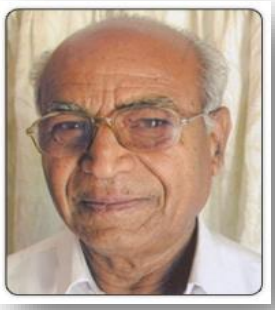
All Honorable members of Management!!!

Message from the President



It gives me immense pleasure to know that Shri Chhatrapati Shivaji Maharaj College of Engineering, Ahilyanagar**,** under the esteemed banner of Ahmednagar Jilha Maratha Vidya Prasarak Samaj (AJMVPS)**,** is organizing the *1st International Conference on Inventions and Innovations in Science, Technology & Management (ICIISTM-2026)*. Such academic initiatives play a vital role in promoting research, innovation, and intellectual exchange among scholars, academicians, industry professionals, and students.

In today's rapidly evolving technological landscape, platforms that encourage interdisciplinary dialogue and knowledge sharing are essential for addressing global challenges. The theme of this conference, *"Transforming Ideas into Impact: Innovation Across Disciplines,"* reflects the spirit of progress and collaboration required to drive meaningful advancements in science, engineering, technology, and management.

I appreciate the efforts of the Principal, Convenor, Coordinator, faculty members, and the entire organizing committee for their dedication in arranging this international forum. Events like ICIISTM-2026 not only strengthen the research culture within the institution but also enhance the reputation of the organization at national and international levels.

I extend my heartfelt best wishes to all participants, authors, and delegates. I am confident that this conference will provide a valuable platform for sharing knowledge, fostering collaborations, and inspiring innovative ideas that contribute to the development of society.

I congratulate the organizing team for their commendable initiative and wish the conference a grand success.

Hon. Shri. Ramchandra Haribhau Dare
President
Ahmednagar Jilha Maratha Vidya Prasarak Samaj (AJMVPS)
Ahilyanagar

Message from the Vice-President

It is a matter of great pleasure to know that Shri Chhatrapati Shivaji Maharaj College of Engineering, Ahilyanagar**,** under the aegis of Ahmednagar Jilha Maratha Vidya Prasarak Samaj (AJMVPS)**,** is organizing the *1st International Conference on Inventions and Innovations in Science, Technology & Management (ICIISTM-2026)*. I extend my warm greetings and best wishes to the organizing committee, participants, and all stakeholders associated with this important academic event.

In today's knowledge-driven world, innovation and interdisciplinary collaboration are key factors for sustainable development and technological advancement. Conferences such as ICIISTM-2026 provide an excellent platform for academicians, researchers, industry experts, and students to exchange ideas, present their research findings, and explore new opportunities for collaboration.

The theme of the conference, *"Transforming Ideas into Impact: Innovation Across Disciplines,"* reflects the spirit of modern research and highlights the importance of translating innovative ideas into practical solutions that benefit society. I believe that this conference will inspire young researchers and scholars to pursue excellence in their respective fields.

I sincerely appreciate the dedicated efforts of the Principal, Convenor, Coordinator, faculty members, and organizing team for their commitment in successfully organizing this international conference. Such initiatives significantly contribute to strengthening the research culture and academic excellence of the institution.

I wish ICIISTM-2026 great success and hope that the conference will foster meaningful discussions, collaborations, and innovative outcomes for the advancement of science, technology, and management.

Hon. Shri. Dr. Vivek Prabhakar Bhapkar
Vice-President
Ahmednagar Jilha Maratha Vidya Prasarak Samaj (AJMVPS)
Ahilyanagar

Message from the Secretary

It is a great pleasure to know that Shri Chhatrapati Shivaji Maharaj College of Engineering, Ahilyanagar**,** under the esteemed guidance of Ahmednagar Jilha Maratha Vidya Prasarak Samaj (AJMVPS)**,** is organizing the *1st International Conference on Inventions and Innovations in Science, Technology & Management (ICIISTM-2026)*. I extend my heartfelt greetings and best wishes to the organizers, participants, and all associated with this significant academic initiative.

In the present era of rapid technological advancement and global connectivity, research and innovation play a crucial role in shaping a sustainable and progressive society. Platforms such as ICIISTM-2026 provide valuable opportunities for researchers, academicians, industry professionals, and students to exchange ideas, present their innovative work, and engage in meaningful discussions that lead to knowledge creation and technological development.

The theme of the conference, *"Transforming Ideas into Impact: Innovation Across Disciplines,"* highlights the importance of collaborative and interdisciplinary approaches to solving real-world challenges. I am confident that this conference will encourage the sharing of creative ideas, strengthen academic partnerships, and inspire young researchers to pursue excellence in science, technology, and management.

I sincerely appreciate the dedicated efforts of the Principal, Convenor, Coordinator, faculty members, and the entire organizing committee for their commitment and hard work in organizing this international conference. Such initiatives greatly contribute to enhancing the research culture and academic reputation of the institution.

I extend my best wishes to all delegates and participants for a successful and enriching conference experience. I am sure that ICIISTM-2026 will achieve great success and contribute meaningfully to the advancement of knowledge and innovation.

Hon. Adv. Vishwasrao Dattatraya Athare
Secretary
Ahmednagar Jilha Maratha Vidya Prasarak Samaj (AJMVPS)
Ahilyanagar

Message from the Joint Secretary

It gives me great pleasure to know that Shri Chhatrapati Shivaji Maharaj College of Engineering, Ahilyanagar**,** under the aegis of Ahmednagar Jilha Maratha Vidya Prasarak Samaj (AJMVPS)**,** is organizing the *1st International Conference on Inventions and Innovations in Science, Technology & Management (ICIISTM-2026)*. I extend my sincere greetings and best wishes to the organizers, participants, and contributors of this prestigious international conference.

In the rapidly evolving world of science and technology, research and innovation are the driving forces behind sustainable development and societal progress. Conferences like ICIISTM-2026 provide a valuable platform for academicians, researchers, industry experts, and students to exchange knowledge, present innovative ideas, and explore collaborative opportunities across diverse disciplines.

The theme of the conference, *"Transforming Ideas into Impact: Innovation Across Disciplines,"* reflects the importance of interdisciplinary research and its role in addressing real-world challenges. Such academic gatherings inspire creativity, encourage intellectual discussions, and contribute significantly to the advancement of science, technology, and management.

I appreciate the dedicated efforts of the Principal, Convenor, Coordinator, faculty members, and the entire organizing committee for their commitment and hard work in successfully organizing this conference. Initiatives like this greatly strengthen the research culture and academic excellence of the institution.

I wish ICIISTM-2026 a grand success and hope that it will lead to meaningful discussions, innovative research outcomes, and lasting academic collaborations.

Hon. Shri. Mukesh Madhavrao Mulay
Joint Secretary
Ahmednagar Jilha Maratha Vidya Prasarak Samaj (AJMVPS)
Ahilyanagar

Message from the Treasurer

It is a matter of great pleasure to know that Shri Chhatrapati Shivaji Maharaj College of Engineering, Ahilyanagar**,** under the esteemed guidance of Ahmednagar Jilha Maratha Vidya Prasarak Samaj (AJMVPS)**,** is organizing the *1st International Conference on Inventions and Innovations in Science, Technology & Management (ICIISTM-2026)*. I extend my warm greetings and best wishes to the organizing committee, participants, and all contributors associated with this significant academic event.

In today's rapidly changing world, innovation and research play a crucial role in addressing complex challenges and driving technological progress. Conferences like ICIISTM-2026 provide a valuable platform for scholars, researchers, industry professionals, and students to share their ideas, present their research findings, and explore new avenues for collaboration and development.

The conference theme, *"Transforming Ideas into Impact: Innovation Across Disciplines,"* highlights the importance of creativity, interdisciplinary thinking, and the practical application of knowledge to solve real-world problems. Such initiatives not only promote academic excellence but also encourage young minds to engage in meaningful research and innovation.

I appreciate the dedicated efforts of the Principal, Convenor, Coordinator, faculty members, and the entire organizing team for their commitment and hard work in organizing this international conference. Their collective efforts will undoubtedly contribute to the success of this event and enhance the academic reputation of the institution.

I convey my best wishes to all delegates and participants for a fruitful and enriching conference. I am confident that ICIISTM-2026 will provide an excellent platform for knowledge sharing, collaboration, and innovative thinking.

Hon. Adv. Deeplakshmi Sambhajirao Mhase
Treasurer
Ahmednagar Jilha Maratha Vidya Prasarak Samaj (AJMVPS)
Ahilyanagar

Principal's Message

It gives me immense pleasure to welcome all distinguished researchers, academicians, industry professionals, and students to the *1st International Conference on Inventions and Innovations in Science, Technology & Management (ICIISTM-2026)*, organized by Shri Chhatrapati Shivaji Maharaj College of Engineering, Ahilyanagar. This conference marks an important step in our institution's mission to foster a vibrant research ecosystem and promote interdisciplinary collaboration among emerging and established scholars.

In an era defined by rapid technological growth and global challenges, innovation has become the cornerstone of progress. The conference theme, *"Transforming Ideas into Impact: Innovation Across Disciplines,"* reflects our belief that meaningful research emerges when diverse minds come together to explore new pathways, challenge existing boundaries, and translate knowledge into real-world solutions.

The wide range of papers presented in ICIISTM-2026 showcases the creativity, dedication, and analytical depth of the contributors. These research works span across Science, Engineering, Technology, Management, and various emerging multidisciplinary domains, demonstrating both academic rigor and practical relevance.

I extend my heartfelt appreciation to our keynote speaker, advisory committee members, session chairs, reviewers, coordinators, and the entire organizing team, whose tireless efforts have ensured the success of this conference. I also congratulate all authors and participants for their valuable contributions and enthusiasm.

I am confident that ICIISTM-2026 will inspire new ideas, strengthen academic partnerships, and pave the way for impactful innovations in the years ahead. I wish all the delegates a fruitful and enriching conference experience.

Dr. Y. R. Kharde
Principal
Shri Chhatrapati Shivaji Maharaj College of Engineering, Ahilyanagar

ABOUT THE CONFERENCE

The *1st International Conference on Inventions and Innovations in Science, Technology & Management (ICIISTM-2026)*, organized by Shri Chhatrapati Shivaji Maharaj College of Engineering (SCSMCOE), Ahilyanagar**,** provides a global platform for the exchange of cutting-edge research, innovative ideas, and interdisciplinary collaboration. Held on 27th–28th March 2026 in hybrid mode, the conference brings together academicians, researchers, industry professionals, and students from diverse domains.

With the central theme *"Transforming Ideas into Impact: Innovation Across Disciplines"* ICIISTM-2026 aims to address emerging challenges, inspire creativity, and contribute solutions that drive societal and technological advancements. The conference invites original and unpublished research contributions across a wide range of tracks, including Science, Engineering, Computer Science, IT, Management, and various interdisciplinary fields.

The objectives of ICIISTM-2026 include:

1. Promoting interdisciplinary collaboration and fostering a culture of innovation.
2. Bridging academia and industry by facilitating meaningful knowledge exchange.
3. Encouraging research-driven and sustainable solutions to real-world challenges.
4. Strengthening research culture among academicians, students, and professionals.

The conference features keynote sessions, technical presentations, expert interactions, and discussions on emerging technologies aligned with global research trends. All accepted papers undergo a stringent review process and are published in the conference proceedings with an ISBN. Selected high-quality papers are considered for publication in SCOPUS-indexed journals**.**

ICIISTM-2026 stands as a testament to SCSMCOE's commitment to academic excellence, technological growth, and the advancement of innovative research across disciplines.

ICIISTM-2026 Organizing Committee

PREFACE

It gives us immense pleasure to present the Proceedings of the *1st International Conference on Inventions and Innovations in Science, Technology & Management (ICIISTM-2026)*, organized by Shri Chhatrapati Shivaji Maharaj College of Engineering, Ahilyanagar**,** under the aegis of Ahmednagar Jilha Maratha Vidya Prasarak Samaj**.** The conference, held on 27th & 28th March 2026 in hybrid mode, serves as a multidisciplinary platform bringing together researchers, academicians, industry professionals, innovators, and students from India and abroad.

The central theme of ICIISTM-2026, *"Transforming Ideas into Impact: Innovation Across Disciplines"* reflects our commitment to fostering creativity, scientific inquiry, and technological advancement across emerging domains. The papers included in this volume highlight original contributions spanning Science, Engineering, Technology, and Management, offering new insights, methodologies, and solutions to contemporary global challenges.

We extend our sincere gratitude to the keynote speaker, advisory committee, reviewers, session chairs, and the dedicated organizing team for their invaluable contributions. We also thank all authors for their active participation and for choosing ICIISTM-2026 as a forum to share their research.

We hope that these proceedings serve as a valuable scholarly resource and inspire continued innovation and collaboration in the years ahead.

ICIISTM-2026 Organizing Committee

FOREWORD

I am delighted to present the *Proceedings of the 1st International Conference on Inventions and Innovations in Science, Technology & Management (ICIISTM-2026)*, organized by Shri Chhatrapati Shivaji Maharaj College of Engineering (SCSMCOE), Ahilyanagar**,** under the esteemed banner of the Ahmednagar Jilha Maratha Vidya Prasarak Samaj (AJMVPS)**.** This conference marks a significant milestone in our institution's ongoing commitment to promoting academic excellence, research culture, and technological progress.

The theme of this year's conference, *"Transforming Ideas into Impact: Innovation Across Disciplines,"* aptly captures the spirit of modern scientific endeavor. In a rapidly evolving global landscape, interdisciplinary collaboration and innovation are no longer optional—they are fundamental drivers of societal advancement. ICIISTM-2026 provides a dynamic platform for scholars, industry experts, and young researchers to exchange ideas, present findings, and explore emerging technologies shaping the future.

The papers compiled in this volume represent rigorous scholarly work across diverse fields, including Science, Engineering, Technology, Computer Science, Management, and Multidisciplinary Innovations. Each contribution reflects the dedication, creativity, and analytical rigor of the authors, and collectively, they illuminate current trends and future directions across multiple domains.

I extend my heartfelt appreciation to our distinguished keynote speaker, the advisory and review committees, session chairs, coordinators, and the entire organizing team for their unwavering efforts. I also congratulate all participating authors and delegates for enriching this conference with their valuable research.

It is my sincere hope that these proceedings inspire further research, foster meaningful collaborations, and serve as a lasting academic resource for scholars worldwide.

ICIISTM-2026,
Shri Chhatrapati Shivaji Maharaj College of Engineering, Ahilyanagar

ACKNOWLEDGEMENTS

The Organizing Committee of the 1st International Conference on Inventions and Innovations in Science, Technology & Management (ICIISTM-2026) *expresses its deep sense of gratitude to all individuals and institutions whose support, contributions, and encouragement made this conference a success.*

We extend our sincere thanks to the management of Ahmednagar Jilha Maratha Vidya Prasarak Samaj (AJMVPS) for their continued guidance, motivation, and institutional support. We are especially grateful to all Hon'ble Patrons for their encouragement in organizing this international event.

Our heartfelt appreciation goes to the Principal and Convenor, Dr. Y. R. Kharde, for his visionary leadership, unwavering support, and constant encouragement throughout the planning and execution of ICIISTM-2026.

We express our sincere gratitude to the Keynote Speaker, Dr. Mukesh K. Kumawat, for delivering insightful perspectives and enriching the academic discourse of the conference.

We are thankful to the distinguished Advisory Committee Members, Reviewers, and Session Chairs for their scholarly inputs, rigorous evaluation of papers, and invaluable guidance that ensured the academic quality of this conference.

We acknowledge the dedication and collaborative efforts of the Coordinator, Dr. V. V. Patil, the Co-Coordinator, Prof. P. R. Mutkule, and all Heads of Departments for their tireless work in streamlining various conference activities and ensuring smooth coordination across teams.

A special note of appreciation is extended to the Organizing Committee Members, faculty, administrative staff, and student volunteers of Shri Chhatrapati Shivaji Maharaj College of Engineering, Ahilyanagar, whose commitment, teamwork, and enthusiasm played a crucial role in the successful conduct of ICIISTM-2026.

Finally, we sincerely thank all authors, participants, and delegates—from India and abroad—for their valuable contributions, active engagement, and overwhelming response. The success of ICIISTM-2026 is a reflection of their interest in advancing innovation and interdisciplinary research.

We hope that the conference and these proceedings serve as a catalyst for continued collaboration, research excellence, and impactful innovations.

ICIISTM-2026 Organizing Committee

EDITORIAL NOTE

It gives us immense pleasure to present the *Proceedings of the 1st International Conference on Inventions and Innovations in Science, Technology & Management (ICIISTM-2026)*. This edited volume is the outcome of a rigorous review process, collective academic effort, and the enthusiastic participation of researchers from India and abroad.

The conference theme, *"Transforming Ideas into Impact: Innovation Across Disciplines,"* guided the selection and curation of research papers included herein. Each accepted manuscript has undergone scholarly evaluation to ensure originality, relevance, and alignment with the conference tracks spanning Science, Engineering, Technology, Management, and multidisciplinary domains. The diversity of papers reflects the dynamic nature of contemporary research and the growing need for cross-disciplinary collaboration.

We extend our sincere appreciation to all reviewers whose timely and insightful feedback contributed significantly to enhancing the quality of the research presented. We also thank the authors for their academic rigor, responsiveness, and contribution to this scholarly compilation.

Our gratitude goes to the Organizing Committee, Coordinator, Co-Coordinator, and the leadership of Shri Chhatrapati Shivaji Maharaj College of Engineering, Ahilyanagar**,** for providing the support and framework required to bring this publication to fruition. The seamless coordination across teams has ensured the timely completion of this proceedings volume.

It is our hope that these proceedings serve as a valuable reference for researchers, practitioners, academicians, and students. May this body of work inspire further innovation, foster new collaborations, and contribute meaningfully to the advancement of global knowledge.

Editorial Board

DEDICATION

This volume of the ICIISTM-2026 Conference Proceedings *is lovingly dedicated to*

All Innovators, Educators, and Knowledge Seekers

whose curiosity fuels discovery,
whose perseverance shapes progress,
and whose ideas inspire a better future.

We also dedicate this work to the students and young researchers—
the torchbearers of tomorrow—
whose passion for learning and innovation remains the driving force behind academic advancement.

May this conference and its outcomes continue to light the path toward excellence, creativity, and meaningful impact.

VISION & MISSION

OUR VISION

- To create a global platform that inspires innovation, promotes interdisciplinary research, and empowers scholars, scientists, technologists, and professionals to transform ideas into impactful solutions for the betterment of society.
- ICIISTM-2026 envisions becoming a catalyst for advancing knowledge, fostering collaborative research, and encouraging sustainable, technology-driven development across diverse domains.

OUR MISSION

- **To Foster Innovation:**
  Encourage the exploration of new ideas, emerging technologies, and inventive methodologies across science, engineering, technology, and management.
- **To Promote Interdisciplinary Collaboration:**
  Bring together experts, researchers, and students from multiple disciplines to address complex challenges through collective insight and research synergy.
- **To Strengthen Academia–Industry Engagement:**
  Facilitate knowledge exchange and meaningful interaction between academic institutions, research bodies, and industry professionals, supporting practical and scalable outcomes.
- **To Empower Young Researchers:**
  Provide a supportive platform for budding scholars to present their work, gain feedback, and engage with the global research community.
- **To Support Sustainable and Impactful Research:**
  Encourage studies and innovations that contribute to societal well-being, environmental sustainability, and technological progress.

Through these mission objectives, ICIISTM-2026 continues to uphold the commitment of Shri Chhatrapati Shivaji Maharaj College of Engineering to quality education, research excellence, and the advancement of knowledge.

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PATRONS

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- Hon. Shri. Dr. Vivek Prabhakar Bhapkar
- Hon. Adv. Vishwasrao Dattatraya Athare
- Hon. Shri. Mukesh Madhavrao Mulay
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Track-1: Science

SCI01: Synthesis, Characterization and Photo-catalytic Activity of Spinel Ferrite Nano-composites for Efficient Dye Degradation

Amol Kardile[1], Shashikant Kuchekar[2], Satish Jadhav[1*]

[1]Department of Chemistry, R. B. Attal Arts, Science and Commerce College Georai Dist-Beed

[2]Ex-Principal, PMT's, Arts, Commerce and Science College Shevgaon, Dist-Ahilyanagr.

[*]Corresponding Author Email ID: orgchem.jadhav79@gmail.com

Abstract

Spinel ferrite nano-composites have emerged as appealing multifunctional materials due to their unique magnetic, structural and optical properties making them suitable for remediation of the environment. We have successfully synthesized spinel ferrite nano-composites using a simple and cost-effective sol-gel method. The sol-gel method enabled controlled nucleation and uniform particle distribution, resulting in the synthesis of nanostructured materials with enhanced surface activity. The structural characteristics of the synthesized nano-composites were studied by X-ray diffraction (XRD), XRD confirming the formation of a single-phase cubic spinel structure with high crystallinity. Fourier Transform Infrared Spectroscopy (FT-IR) which was used to perform functional group analysis shows distinctive metal-oxygen stretching vibrations that corresponded to the spinel lattice tetrahedral and octahedral sites. UV-Visible Diffuse Reflectance Spectroscopy (UV-Vis DRS) was used to analyse optical properties. The calculated band gap energy shows improve visible-light absorption capability, which is helpful for photo-catalytic applications. The degradation of organic dye under light radiation was used to evaluate the produced nano-composites photo-catalytic activity. Because of their strong surface reactivity, enhanced charge separation and narrow band gap. The nano-composites shows notable degrading efficiency. The study shows that spinel ferrite nano-composites made from sol-gel method are effective, stable and reusable photo-catalysts for wastewater treatment applications.

Keywords: *Spinel Ferrite, Nano-composites, Sol-gel, Photo-catalytic, Degradation.*

SCI02: Integrated Photocatalytic Efficiency Analysis Model (IPEAM): A Theoretical Framework for Photocatalytic Reactor Design and Hydrogen Production

Dr. Shashikant R. Kuchekar[1], Kabir B. Inamdar[2], Dr. Ismail B. Shaikh[3], Dr. Dilip S. Aute[4], Sachin V. Balsane[5]

[1]Former Principal, PMT'S Arts, Commerce and Science College, Shevgaon;
Former Head, Department of Chemistry, P. V. P. College, Pravaranagar,
Dist. Ahilyanagar, Maharashtra, India

[2]Assistant Lecturer in Chemistry, Pravara Medical Trust's A.C.S. College, Shevgaon,
Dist. Ahilyanagar, Maharashtra, India.

[3]Assistant Professor in Chemistry, Shri Anand College, Pathardi,
Dist. Ahilyanagar, Maharashtra, India

[4]Assistant Professor in Chemistry, Padmashri Vikhe Patil College of Arts, Science and Commerce, Pravaranagar, Tal. Rahata, Dist. Ahilyanagar, Maharashtra, India.

[5]Head, Department of Chemistry, ACS College, Alkuti, Parner,
Dist. Ahilyanagar, Maharashtra, India.
[*]Corresponding Author Email ID: kabir3131@gmail.com

Abstract: The present work develops a theoretical modeling framework for photocatalytic reactor efficiency analysis. This study introduces the Integrated Photocatalytic Efficiency Analysis Model (IPEAM), a unified two-dimensional theoretical framework for predicting hydrogen production in photocatalytic slurry reactors. The model combines three fundamental principles of photocatalysis: photon absorption based on the photoelectric effect, surface reaction kinetics described by Langmuir–Hinshelwood theory, and depth-dependent light attenuation governed by the Beer–Lambert law. The model starts from a unified microphysical rate expression that includes photon threshold effects, adsorption equilibrium, and intensity-dependent saturation behavior. The local reaction kinetics is extended to reactor-scale performance by integrating the reaction rate across reactor depth. The model identifies three characteristic optical thickness regimes: optically thin thin ($\tau < 0.3$), optically ($0.3 < \tau < 2$), and optically thick ($\tau > 3$). Efficient reactor operation is predicted within the regime $\tau \leq 0.5$, where photon utilization efficiency remains reasonably high and optical efficiency satisfies $\eta opt \geq 0.44$. Dimensional analysis confirms the physical consistency of the model with hydrogen production rates expressed in mol per second. The IPEAM framework provides a theoretical bridge between microscopic photocatalytic processes and macroscopic reactor-scale performance, offering rational guidelines for photocatalytic reactor design and scale-up. This purely analytical model establishes a foundation for optimizing photocatalytic hydrogen production systems prior to experimental implementation.

***Keywords**: Photocatalysis, 2D modeling, Hydrogen production, Reactor design, Langmuir-Hinshelwood kinetics, Beer-Lambert law, Optical thickness.*

SCI03: Fluorescent Chemosensors: Evolution, Current Advances, and Future Perspectives

Borawake G.A.[1]
[1]Jijamata College of Science & Art's Bhende
[*]Corresponding Author Email ID: ganesh.borawake77@gmail.com

Abstract: Fluorescent chemosensors are molecular systems designed to detect specific chemical species through measurable fluorescence signals. Over the past several decades, these sensors have become powerful analytical tools in chemistry, biology, environmental science, and medical diagnostics. Their effectiveness arises from the combination of a recognition unit, which selectively binds to a target analyte, and a signaling unit, which produces a fluorescence response upon interaction. The development of fluorescent chemosensors has progressed significantly since the early studies of ion-selective probes. Modern research focuses on improving sensitivity, selectivity, and real-time detection of various targets such as metal ions, anions, reactive oxygen species, and biological molecules. Advances in supramolecular chemistry, photo physics, and nanotechnology have further expanded the design strategies for these sensors. Fluorescent chemosensors are now widely applied in cell imaging, environmental monitoring, clinical diagnostics, and chemical sensing. Despite these advancements, challenges remain in developing sensors with higher stability, better

biocompatibility, and improved performance in complex biological systems. Future research is expected to integrate new materials, nanostructures, and advanced optical techniques to enhance the capabilities of fluorescent sensing systems.

***Keywords**: Fluorescent Chemosensors, Chemical sensing, Fluorescence detection, Metal ion detection, Nanotechnology.*

SCI04: Water Quality Analysis: Assessment of Physicochemical Parameters and Environmental Implications

Bhushan Govardhan Wable[1]

[1]Jijamata College of Science & Art's Bhende

[*]Corresponding Author Email ID: bhushanwable3151@gmail.com

Abstract: The present work develops a theoretical modeling framework for photocatalytic reactor efficiency analysis. This study introduces the Integrated Photocatalytic Efficiency Analysis Model (IPEAM), a unified two-dimensional theoretical framework for predicting hydrogen production in photocatalytic slurry reactors. The model combines three fundamental principles of photocatalysis: photon absorption based on the photoelectric effect, surface reaction kinetics described by Langmuir–Hinshelwood theory, and depth-dependent light attenuation governed by the Beer–Lambert law. The model starts from a unified microphysical rate expression that includes photon threshold effects, adsorption equilibrium, and intensity-dependent saturation behavior. The local reaction kinetics is extended to reactor-scale performance by integrating the reaction rate across reactor depth. The model identifies three characteristic optical thickness regimes: optically thin thin ($\tau < 0.3$), optically ($0.3 < \tau < 2$), and optically thick ($\tau > 3$). Efficient reactor operation is predicted within the regime $\tau \leq 0.5$, where photon utilization efficiency remains reasonably high and optical efficiency satisfies $\eta opt \geq 0.44$. Dimensional analysis confirms the physical consistency of the model with hydrogen production rates expressed in mol per second. The IPEAM framework provides a theoretical bridge between microscopic photocatalytic processes and macroscopic reactor-scale performance, offering rational guidelines for photocatalytic reactor design and scale-up. This purely analytical model establishes a foundation for optimizing photocatalytic hydrogen production systems prior to experimental implementation.

***Keywords**: Photocatalysis, 2D modeling, Hydrogen production, Reactor design, Langmuir-Hinshelwood kinetics, Beer-Lambert law, Optical thickness.*

SCI05: A Machine Learning Approach for Predicting Graph Isomorphism Using Structural Features

M.S.Shaikh[1] , Dr. M.K.Bhosale[2]

[1,2]Shri. Chhatrapati Shivaji Maharaj College of Engineering, Nepti, Ahilyanagar, Maharashtra.

[*]Corresponding Author Email ID: smohammadmoin25@gmail.com

Abstract: Graph isomorphism is a fundamental problem in graph theory and computer science that involves determining whether two graphs are structurally identical. Although polynomial-time solutions exist for specific graph classes, a general efficient algorithm is still an open question. This paper proposes a novel approach that integrates machine learning techniques to

predict graph isomorphism based on extracted structural features such as degree distribution, eigenvalues, clustering coefficients, and adjacency matrix patterns. Experimental results demonstrate that supervised models can effectively classify isomorphic and non-isomorphic graph pairs with high accuracy, reducing computational time compared to traditional combinatorial algorithms.

Keywords: *Graph isomorphism, Matrix*

SCI06: Fractional Differential Equations: From Mathematical Foundations to Real-World Applications

Dr. M.K.Bhosale[1], Poonam Jagtap[2], M.S.Shaikh[3]

[1,3]Shri. Chhatrapati Shivaji Maharaj College of Engineering, Nepti, Ahilyanagar, Maharashtra.

[2]Pamashri Vithalrao Vikhe Patil College of Engineering, Nepti, Ahilyanagar,Maharashtra

[*]Corresponding Author Email ID: smohammadmoin25@gmail.com

Abstract: Fractional differential equations (FDEs) have emerged as a powerful tool for modeling complex phenomena across various fields of science and engineering. These equations generalize traditional integer-order differential equations by introducing fractional derivatives, which provide a more flexible and accurate representation of anomalous behavior in natural systems. This paper explores the mathematical foundations of fractional differential equations, including the concept of fractional calculus and various definitions of fractional derivatives. We then discuss a range of real-world applications in physics, biology, engineering, and economics, highlighting the advantages of fractional models in capturing memory, non-local interactions, and anomalous diffusion. We conclude by addressing the challenges in solving FDEs and the ongoing developments in numerical methods, analytical techniques, and the broader applicability of fractional models in emerging scientific fields.

Keywords: *Fractional Differential Equations, Fractional Calculus, Anomalous Diffusion, Viscoelasticity, Memory Effect, Non-local Operators, Caputo Derivative, Riemann-Liouville Derivative.*

SCI07: QSAR- Quantitative Structure -Activity Relationship

Ashvini kisan Khetmalis[1], Vidhate Akshada Arjun[2], Shital A. Sherkar[3]

[1,3]Shri. Chhatrapati Shivaji Maharaj College of Engineering, Nepti, Ahilyanagar,Maharashtra

[2]New arts, commerce and science college Ahilyanagar,Maharashtra

[*]Corresponding Author Email ID: kulangeashvini@gmail.com

Abstract: This study focuses on understanding the relationship between the chemical structure of drug molecules and their biological activity using the QSAR approach. QSAR (Quantitative Structure–Activity Relationship) is an important method in medicinal chemistry that helps to predict drug behaviour based on molecular properties. In this study, molecular descriptors such as molecular weight, Log P (lipophilicity), hydrogen bond donors, hydrogen bond acceptors, and topological polar surface area are used to represent the structural characteristics of compounds. These descriptors influence the absorption, distribution, and interaction of drugs within the biological system. The biological activity of drug molecules is evaluated using IC50 values, which indicate the concentration required to inhibit a biological process. A lower IC50

value represents higher drug potency. By analyzing these values, the effectiveness of different compounds can be compared scientifically. In addition to chemical properties, clinical factors such as age, dosage, gender, and treatment duration are considered to study drug effectiveness under real-life conditions. This integration of chemical and clinical data provides a more realistic and comprehensive understanding of drug response in patients. The dataset is analyzed using statistical methods such as descriptive statistics, correlation analysis, and predictive modelling techniques. A suitable model, such as logistic regression, is applied to establish the relationship between molecular descriptors and biological response. The model is then used to predict whether a drug will be effective or not. The results of the study show that chemical structure plays a significant role in determining drug activity. Certain molecular properties, especially lipophilicity (Log P) and IC50, strongly influence drug effectiveness. The study also highlights that combining QSAR with clinical data improves prediction accuracy compared to using chemical data alone. Overall, this approach is highly useful in medicinal chemistry and drug design, as it helps in identifying promising drug candidates before laboratory testing. It reduces time, cost, and experimental effort, and supports the development of safer and more effective drugs. This study demonstrates the importance of integrating chemistry with data analysis for advanced pharmaceutical research.

Keywords: *QSAR, Molecular Descriptors, IC50, Drug Effectiveness, Chemical Structure, Drug Activity*

SCI08: Improving the Detection sensitivity in laser spectroscopy for simultaneous water vapour and temperature measurement

S.N.Pawar[1]

[1]Assistant Professor, Department of Engineering Science, Shri Chhatrapati Shivaji Maharaj College of Engineering, Ahilyanagar

[*]Corresponding Author Email ID: pawarsn@gmail.com

Abstract: Tunable Diode Laser Absorption Spectroscopy (TDLAS) is a highly precise gas measurement technology that determines gas concentrations by analyzing the absorption characteristics of gases at specific laser wavelengths. It offers significant advantages, including real-time monitoring, non-contact measurement, high sensitivity, and strong selectivity. This paper reviews recent research progress in the application of TDLAS for water vapor concentration measurement and explores various methods for enhancing detection sensitivity. Key technical advancements are discussed, including improvements in laser sources, gas absorption cell design, signal demodulation techniques, and background interference suppression. Additionally, environmental adaptability and measurement limits are evaluated. The effectiveness of different sensitivity enhancement strategies is analyzed, emphasizing the critical roles of temperature and pressure compensation, as well as noise suppression, in improving detection limits and overall sensitivity. The paper also summarizes the application of TDLAS-based water vapor measurement in complex and harsh environments. Finally, future development trends of TDLAS technology in environmental monitoring and industrial applications are outlined.

Keywords: *TDLAS*

SCI09: Nanotechnology: Applications and Future Directions

Chine S. S[1], K.D. Thete[2]
[1]Department of Electronics and Computer Engineering(Chemistry), Sanjivani College of Engineering, Kopargaon, Savitribai Phule Pune University Pune, Maharashtra, India
[2]Department of Zoology, PVP Art, Science and Commerce College, Pravaranagar, Loni(KD), Savitribai Phule Pune University Pune, Maharashtra, India
[*]Corresponding Author Email ID: pawarsn@gmail.com

Abstract: Nanotechnology has emerged as one of the most transformative scientific disciplines of the 21st century, fundamentally altering our approach to materials science, medicine, electronics, and environmental remediation. This review provides a comprehensive overview of the current state of nanotechnology, encompassing the synthesis and characterization of nanomaterials, their diverse applications across various sectors, and the challenges that remain to be addressed. We examine the fundamental principles governing nanoscale phenomena, recent advances in fabrication techniques, and the rapidly expanding array of applications in healthcare, electronics, energy, and environmental science. Furthermore, we discuss the ethical considerations, safety concerns, and regulatory frameworks that accompany the proliferation of nanomaterials. By synthesizing findings from seminal research publications and authoritative reviews, this paper aims to provide a holistic understanding of nanotechnology's trajectory and its potential to address pressing global challenges.

Keywords: *Nanotechnology, Nanomaterials, Nanofabrication, Nanomedicine, Nanoelectronics, Environmental Nanotechnology, Nanosafety*

SCI10: Eco Friendly Fabrication of Copper Oxide Thin Films by Spray Pyrolysis using Cassia Tora Leaf Extract

Sonali C. Magar[1], Mahendra D. Shinde[2]
[1]Research Scholar, School of Science, Sandip University, Nashik, India.
[2]Professor, School of Science, Sandip University, Nashik, India.
*Corresponding author mail: magarsonali94@gmail.com

Abstract: Copper oxide (CuO) thin films were successfully prepared through an ecofriendly green synthesis route utilizing spray pyrolysis techniques and employing cassia tora leaf extract as a natural reducing and stabilizing agent. Solutions of copper acetate precursors with concentrations of 0.1 M, 0.2 M and 0.3 M were prepared to examine their efficacy in the formation of thin films using a sustainable method. The plant's aqueous extract allowed for the controlled formation and growth of CuO structures without needing extra chemical stabilizers. The precursor solutions were applied to a glass substrate preheated to 300 °C and subsequent annealing at 400 °C for one hour was performed to enhance film quality and eliminate any remaining organic substances. X-ray diffraction (XRD) analysis showed broad diffraction patterns characterized by a prominent hump within the 2-theta range of 20 °C to 30 °C which is indicative of nano-crystalline CuO thin films. Furthermore, the observed increase in peak intensity with higher precursor concentrations implies enhanced film coverage and improved structural ordering.

Keywords: *Copper oxide, thin film, ecofriendly, green synthesis, sustainability*

Track-2: Computer Science & Information Technology

CSIT01: NUTRI-AI: An Intelligent Food Recognition and Personalized Diet Recommendation System Using Machine Learning

Ritik Singh[1], Hussain Vahora[2], Siddhi Dere[3], Simran Agrawal[4], Abhinav Singh[5]
[1,2,3,4,5]Vishwakarma nstitute of Technology ,Pune
[*]Corresponding Author Email ID: suraj.siddhi24@vit,edu

Abstract: As demand grows for automated nutrition assessment tools as well as customized meal plans, many people discover that they have difficulties keeping track of what they eat and following meal plans that are applicable to their specific health concerns or daily intake. NUTRI-AI will create user nutrition profiles using demographic, diet type, and medical condition information to help calculate the daily minimums of macro and micronutrients through a cloud-based artificial intelligence (AI) system. Once this information is captured, EfficientNet-V3 image recognition technology will be used to scan meal photos and obtain the corresponding nutritional information from an external food database in order to keep track of the amount of food that remains. Finally, the AI-derived nutritional information will allow NUTRI-AI's recommendation engine to suggest, based on user-defined preferences, the most balanced meals, adjustments to portion sizes, healthier options, and the ability to purchase food directly from the NUTRI-AI platform.

Keywords: *EfficientNet-V3, timm Library, Food Recognition, Deep Learning, Nutrient Estimation, Personalized Diet Recommendation Machine Learning, Nutrition Tracking, Medical Condition Aware Diet, AI Suggested Meals, Computer Vision, Health Monitoring E-commerce Integration, NUTRI-AI.*

CSIT02: Giving Patients the Keys - A Blockchain Architecture for Verifiable Consent Driven EHR Management

Amit shinde[1]
[1]Marathwada Mitra Mandal's Institute of Technology
[*]Corresponding Author Email ID: Amitushinde@gmail.com

Abstract: The swift digitalization of healthcare has been a major driver for the widespread adoption of Electronic Health Records (EHRs). Despite this, current EHR systems are still limited by their centralized architectures, fragmented data silos, lack of interoperability, and the absence of patient, centric consent mechanisms. These limitations have the effect of not only making clinical decision, making less efficient but also increasing the risk of data privacy, security, and regulatory compliance issues. In order to overcome these obstacles, this article presents a blockchain, based, patient, centric EHR management system concept which employs a hybrid architecture mixing a permissioned blockchain with secure off, chain storage. In the newly designed healthcare setup, blockchain technology is used as the control plane for managing patient consent, access policies, data provenance, and immutable audit trails whereas, to guarantee scalability and privacy, sensitive medical records are encrypted and saved off, chain. Smart contracts interpret patient consent as set by the patient into enforceable, granular access control mechanisms, thus allowing for data sharing that is time, limited and role, specific. The compatibility of FHIR, compliant APIs with existing healthcare infrastructures allows for integration to be completed without any disruption of the clinical

workflows. The sound framework not only eliminates the risk of single points of failure by separating data storage from access governance but also ensures compliance with healthcare regulations like HIPAA and GDPR. The designed system realizes the ideals of increased transparency, trust, and security as well as giving patients the real, time capability of controlling their health data which they can verify.

Keywords*: Blockchain; Electronic Health Records (EHR); Patient-Centric Consent; Healthcare Data Security; Interoperability; Smart Contracts*

CSIT03: A Data-Driven Approach to Predictive Maintenance for Early Fault Detection and Enhanced Operational Efficiency

Tanmay Sandip Benke[1],Shreyash Dipak Javalekar[2],Aditya Jaising Shinde[3],Prof. S. B. Bhosale[4],Prof.Abhisk P.Bangar[5]

[1,2,3,4,5]Jaihind College of Engineering, Kuran

[*]Corresponding Author Email ID: sachinbhosale@gmail.com

Abstract : The In modern industrial environments, unexpected equipment failures can lead to costly downtime, reduced productivity, and safety risks. To address these challenges, this study presents a data-driven approach to predictive maintenance aimed at enabling early fault detection and improving overall operational efficiency. The proposed framework leverages historical machine data, real-time sensor inputs, and advanced analytical techniques to identify patterns that indicate potential equipment degradation. By applying machine learning algorithms to structured and time-series data, the system can predict faults before they escalate into critical failures. Unlike traditional reactive or schedule-based maintenance strategies, the proposed approach focuses on condition-based monitoring, allowing maintenance activities to be performed only when necessary. This not only reduces maintenance costs but also extends equipment lifespan and minimizes unplanned downtime. Experimental evaluation demonstrates improved fault detection accuracy and more efficient resource utilization compared to conventional maintenance practices. The findings highlight the potential of data-driven predictive maintenance as a scalable and practical solution for enhancing reliability, productivity, and operational performance across industrial systems.

Keywords : *Predictive Maintenance, Data-Driven Modeling, Early Fault Detection, Machine Learning, Operational Efficiency.*

CSIT04: Big Data Analytics in Healthcare: Data Sources, Tools, Challenges, and Opportunities

Nawale Sanchita N[1] , Fatangare Snehal B[2], Ms. Nehe Ashwini R[3], Dighe Shamal G[4], Tribhuvan Rutuja .R[5]

[1,2,3,4,5]AVCOE, Sangamner

[*]Corresponding Author Email ID: sanchitanawale@gmail.com

Abstract : The rapid growth of healthcare data—arising from electronic health records (EHRs),medical imaging, wearable sensors, and patientgenerated content—offers unprecedented potential for improving patient outcomes, optimizing operations, and informing policy. However, extracting value from such large, heterogeneous data sets requires

sophisticated tools, robust analytical frameworks, and careful attention to challenges such as privacy, interoperability, realtime processing, and data quality. This paper synthesizes recent research to: (i) characterize major sources of healthcare big data; (ii) overview the tools, frameworks, and analytics types currently in use; (iii) propose a conceptual framework for realtime healthcare prediction; and (iv) discuss challenges and opportunities for future work. The survey is motivated by the increasing burden of chronic disease, the push for predictive and preventative medicine, and the demand for healthcare systems that can adapt in near real time.
***Keywords**: Big Data Analytics, Electronic Health Records (EHR), Machine Learning, Predictive Analytics; Precision Medicine*

CSIT05: Banking Security System Using Face and Liveness Detection Using Machine Learning and Image Processing

Sakshi jog[1], payal khambkar[2], gitanjali kamble[3], sandhya shinde[4]
[1,2,3,4]Rajiv Gandhi College of engineering, karjule harya ,Ahilyanagar
*Corresponding Author Email ID : sakshijog27@gmail.com

Abstract: With the rapid growth of digital banking services, secure and reliable authentication mechanisms have become essential to prevent identity theft, spoofing attacks, and unauthorized access. Traditional authentication methods such as passwords, PINs, and OTPs are increasingly vulnerable to phishing, replay attacks, and credential leakage. Although biometric authentication using face recognition provides convenience and contactless verification, conventional systems remain susceptible to spoofing through printed photographs, video replays, and 3D masks. This project proposes a Banking Security System using Face Recognition and Liveness Detection based on Machine Learning and Image Processing techniques. The system integrates a Convolutional Neural Network (CNN)-based face recognition model with a liveness detection module to ensure that the detected face belongs to a real, live person. The authentication pipeline includes image acquisition, preprocessing, face detection, feature extraction, spoof classification, and final authentication decision. The liveness detection module analyzes dynamic facial characteristics such as eye blinking, lip movement, texture variations, and illumination patterns to distinguish between real and fake representations. Extracted facial embeddings are compared with stored templates in a secure database using similarity threshold matching. Access is granted only when both recognition accuracy and liveness verification conditions are satisfied. Experimental evaluation demonstrates high recognition accuracy (above 95%) and effective detection of photo and video-based spoofing attempts with minimal latency. The proposed system enhances security for ATM machines, mobile banking applications, and online banking platforms by providing a scalable, real-time, and robust authentication solution.
***Keywords:** Face Recognition, Liveness Detection, Banking Security, CNN, Machine earning, Image processing, Anti-Spoofing.*

CSIT06: Real Time Disaster Alert System

Pratiksha Subhash Gadhave[1], Shabnam Sameer Pathan[2], Shruti Kumari[3],Gayatri Bharat Gite[4]
[1,2,3,4]Rajiv Gandhi College of Engineering
*Corresponding Author Email ID : pratikshagadhave23@gmail.com

Abstract : Natural disasters such as earthquakes, floods, and wildfires pose significant threats to human life, infrastructure, and economic stability. Rapid detection, monitoring, and dissemination of disaster information are essential for effective emergency response and mitigation strategies. This paper presents the design and implementation of a Unified Real-Time Disaster Intelligence Platform with Geospatial Mapping and Analytical Dashboard that integrates live disaster data streams, geospatial visualization, and analytical dashboards into a unified web-based platform. The proposed system collects real-time data from external disaster- reporting APIs, processes and classifies events based on type and severity, and visualizes them through an interactive map interface. A time-filtering mechanism enables users to analyze events across various temporal intervals. Additionally, a trend analytics dashboard provides insights into hourly disaster frequency distribution. The system also supports structured data export for extended analytics. The implementation demonstrates enhanced situational awareness and improved accessibility to disaster intelligence information.

Keywords *: Real-time monitoring, disaster management, geospatial visualization, data analytics, early warning systems, API integration.*

CSIT07: Stocksight AI-powered predictive analysis for stock and crypto market

Meet Jain[1], Keissa Pate[2], Gaurav Indorkar[3]
[1,2,3]Thakur College of Engineering & Technology
*Corresponding Author Email ID : indorkargaurav@gmail.com

Abstract : Financial markets are incredibly complex, they are constantly changing, unpredictable, and involve countless moving parts. Individual investors typically can access the powerful, integrated tools that big investment firms use, forcing them to piece together information from multiple disconnected sources.We are trying to fix that problem.We have built an all-in-one platform that brings together three essential AI-powered tools that work seamlessly together: Stox Sight predicts where asset prices might be heading. It combines several advanced neural network models (LSTM, GRU, and Transformers, think of these as different types of pattern-recognition engines) that each bring unique strengths to forecasting. We have also added an AI language model that can explain what's happening in plain English, turning numbers into insights you can actually understand. Algo Stox lets you test your trading strategies against historical data before risking real money. It is like a flight simulator for trading, you can see how your approach would have performed in the past. InvestMind acts as your knowledgeable research assistant, answering financial questions with reliable, sourced information rather than speculation. We designed the system using modern microservices architecture (essentially, independent modules that work together but can be updated separately) and built it with proven technologies like Python, FastAPI, Next.js, and Docker. When we tested it on the notoriously volatile cryptocurrency market, our ensemble

approach consistently outperformed traditional forecasting methods. What makes this work meaningful is that it brings everything together in one place, data analysis, price predictions, strategy testing, and educational resources, giving individual investors access to institutional-grade tools that were previously out of reach.

Keywords: *Financial Forecasting, Algorithmic Trading, Time-Series Analysis, Deep Learning, LSTM, GRU, Transformer, Retrieval-Augmented Generation (RAG), Decision Support Systems, Quantitative Finance*.

CSIT08: Bringing AI to Automatic Diagnosis of Diabetic Retinopathy using Image Processing and Machine Learning Technique

Ganesh P. Joshi[1], Prof. S.G.Tathe[2]

[1,2]Vishwabharati Academy's College of Engg, Ahilyanagar

*Corresponding Author Email ID : joshigp.scsmcoe@gmail.com

Abstract: Diabetic Retinopathy (DR) is a severe complication of diabetes that can lead to vision loss if not detected early. This project focuses on developing an AI-based system for automatic detection and multi-class diagnosis of Diabetic Retinopathy using image processing and machine learning techniques. By leveraging a Python-based web application, the system can process retinal images, extract meaningful features, and classify the severity of DR into multiple stages. The use of advanced Convolutional Neural Networks (CNNs) ensures high accuracy and reliability in identifying the disease at an early stage, making it possible to assist doctors in timely diagnosis. The system uses a Diabetic Retinopathy dataset for training and validation. Images are pre-processed to enhance features such as blood vessels, exudates, and microaneurysms, which are critical indicators of the disease. The CNN model is trained to recognize patterns corresponding to different stages of DR, ensuring efficient and automated diagnosis. This reduces the dependency on manual screening, which is time-consuming and prone to human error, providing a scalable solution for mass screening in healthcare systems. The proposed approach not only offers accurate classification of DR severity but also provides a user-friendly interface for medical professionals to upload images, visualize predictions, and monitor patient progress. By combining image processing and machine learning, this project aims to improve early detection rates, assist ophthalmologists in decision-making, and contribute to better patient outcomes in diabetic care.

Index Terms: *Diabetic Retinopathy (DR), Convolutional Neural Network (CNN), Multi- class Diagnosis, Feature Extraction, Automated Detection, Image processing & Machine Learning, Python web application, etc.*

CSIT09: Explainable Ensemble Machine Learning for Early Parkinson's Disease Detection Using Acoustic Voice Biomarkers

Mahie Anand Sabu[1], Tejaswini Patil[2], Shahebaz Shaikh[3], Danish Shaikh[4]

[1,2,3,4]Vishwakarma Institute of Technology, Pune

Corresponding Author Email ID : mahie.sabu24@vit.edu

Abstract: Parkinson's disease (PD) is a progressive neurodegenerative illness that results in the appearance of prodromal (earliest signs of disease) vocal changes before the onset of motor symptoms (symptoms of Parkinson's disease that involve movement) for many years before they are seen (many years before motor symptoms develop). The changes occur from a loss of functions in Dopaminergic system circuits (Consisting of Basal Ganglia and Motor Cortices), resulting in an unstable larynx/Vocal cord, resulting in variations in speech acoustics. This study presents a framework for creating an interpretable detection framework that uses perturbations and Non-linear extraction of acoustic biomarkers, along with a SMOTE-imbalance correction process and statistical validation. The Random Forest model achieved 98.3% accuracy, 96.55% sensitivity, and 100% specificity using AUC data from test data that had been held out (accumulated data that had been kept aside). The Jitter, Shimmer, and Entropy Measure features are the most indicative/important of the presence of vocal pathology associated with Parkinson's disease and provide a degree of interpretability that aligns with existing neuronal mechanisms. The proposed framework brings together elements of Neuroscience, Acoustic signal processing, and interpretable Machine Learning, demonstrating the potential of using VOICE biomarkers for large-scale, non-invasive screening of PD in Telehealth clinical environments.
Keywords : *Parkinson's disease, Acoustic biomarkers, Hypokinetic dysarthria, Explainable machine learning, Speech signal processing, Telemedicine screening, Non-invasive diagnostics*

CSIT10: SAKSHAM - An AI-Based Government Personalized Scheme Advisory System

Purva Gandhi[1], Shruti Todmal[2], Sneha Thete[3], Shrutika Chaudhary[4]
[1,2,3,4]Shri Chhatrapati Shivaji Maharaj college of engineering, Nepti
Corresponding Author Email ID : purvagandhi3105@gmail.com

Abstract : In India, there are many government welfare schemes made to help people, but citizens often face problems in finding correct information and knowing which schemes they are eligible for. Because of this, many people do not receive the benefits they deserve. To solve this problem, we have developed SAKSHAM (Smart Access to Knowledge of Schemes & Helpful Assessment for Merit) - a user-friendly web-based platform that helps citizens easily find suitable government schemes. Users just need to enter their basic personal and financial details, and SAKSHAM automatically checks their eligibility and recommends the right schemes. The system uses document analysis, rule-based checking, and recommendation logic to provide accurate and helpful results.The system utilizes Natural Language Processing (NLP) and Google Document AI / Vision API to extract and analyze data from user-uploaded documents. Based on attributes such as age, income, occupation, gender, and category, the system automatically evaluates the user eligibility and classifies schemes into Eligible and Ineligible categories, along with the reasons for ineligibility. It also provides details such as required documents and step-by-step application guidance for each scheme. The proposed system is developed using Next.js and TypeScript for the frontend, Node.js and Express.js for the backend, and Supabase (PostgreSQL) for secure data storage and authentication. This

centralized digital platform enhances transparency, efficiency, and accessibility, ensuring that citizens can easily discover and benefit from the schemes they truly deserve.

Keywords: *Document Analysis, Eligibility Checking, Recommendation Logic, NLP, Vision API, PostgreSQL*

CSIT11: Service Connect: Real-Time Service Booking Platform

Sanjana Vidhate[1], Payal Thokal[2], Renuka Tribhuvan[3], Manisha Wankhede[4]

[1,2,3,4]Shri Chhatrapati Shivaji Maharaj College of engineering Nepti

*Corresponding Author Email ID : sanjanavidhate2609@gmail.com

Abstract : Service Connect is a native Android application designed to bridge the gap between customers and verified service professionals through a comprehensive, real-time marketplace. The system addresses the challenges of finding reliable home services, such as plumbing, electrical work, and cleaning, by providing a centralized digital ecosystem. Developed using Java and the Firebase suite (Auth, Firestore, Storage, and FCM), the platform implements three distinct modules: a Customer interface for service discovery and booking, a Service Provider dashboard for task and earnings management, and an Administrator panel for system-wide oversight. The application features a dynamic, multi-state booking lifecycle—synchronized instantly across devices using Firestore real-time listeners. Key technical integrations include a secure digital payment gateway via the Razorpay SDK, incorporating HMAC-SHA256 signature verification to ensure transaction integrity. Direct communication is facilitated through a built-in, real-time messaging system, while Firebase Cloud Messaging (FCM) handles automated push notifications for booking confirmations and status updates. For financial transparency, the system employs an automated earnings split (90/10) within a provider- specific digital wallet. By centralizing verified service discovery and securing the transaction layer, ServiceConnect offers a scalable and efficient solution for professional service management in the modern gig economy.

Keywords *: Android Native Development, Service Marketplace, Real-time Synchronization, Firebase Integration, Cloud Firestore, Firebase Cloud Messaging (FCM), Role-Based Access Control (RBAC), Mobile Commerce (m-Commerce), Repository Pattern, In-App Chat,Digital Wallet.*

CSIT12: Sports Analytics by Data Science

[1,2,3,4]Dekhane Ankita Milind, Mahajan Gauri Ganesh, Jadhav Shruti Ramnath, Sayyed Suhani Kayyum

Shri Chatrapati Shivaji Maharaj college of Engineering, Nepti

Corresponding Author Email ID : dekhaneankita18214@gmail.com

Abstract : With the rapid growth and increasing influence of new technologies and devices in sports, intelligent assessment methods, efficient planning and automatic analysis of large amounts of data have gained great importance in recent years. One of the cutting-edge technologies revolutionizing numerous industries, including the sports industry, is artificial intelligence, along with its subfields of machine learning and deep learning. the use of data science in sports has significantly improved the way teams evaluate performance and make

strategic decisions. Sports analytics involves collecting and analyzing large amounts of data related to player performance, match statistics, and team strategies. This study focuses on applying data science techniques to analyze sports performance and extract meaningful insights that can help teams improve their results. The proposed approach includes collecting sports-related data such as player statistics, match outcomes, scoring patterns, and physical performance indicators. Using data analysis and visualization techniques, the collected data can be processed to identify patterns, evaluate player efficiency, and predict possible outcomes of matches. We use Tools such as Python, Excel, and data visualization platforms to represent the data through charts and graphs for better understanding. This platform can help coaches and teams make real time data-driven decisions regarding player selection, game strategies, and performance improvement. Sports analytics can also help in predicting match results and identifying strengths and weaknesses of players and teams. This study highlights the importance of data science in modern sports and demonstrates how analytical techniques can enhance decision-making and performance evaluation in competitive sports environments. This paper aims to conduct a systematic literature review on the application and impact of artificial intelligence, machine learning and deep learning on sports performance, focusing on various techniques and algorithms used for handling data and their benefits in achieving superior ports results.

***KEYWORDS**: Sports Analysis, Data Science, Artificial Intelligence, Machine Learning, Deep Learning, Player Performance Analysis, Data Visualization, Match Prediction, Performance Evaluations, Sports Data Analytics.*

CSIT13: Tree Data Structures in Modern Computing: Evolution and Significance

Ashwini sopan shidore[1], Dipali ankush kamble[2], Ganesh joshi[3], Kanchan raikar[4]
[1,2,3,4]SHRI CHHATRAPATI SHIVAJI MAHARAJ COE, NEPTI, A.NAGAR
*Corresponding Author Email ID : ashwiniakolkar@yahoo.com

Abstract: Tree data structures have played a pivotal role in the evolution of modern computing, offering efficient solutions for organizing, storing, and retrieving hierarchical data. Originating from mathematical concepts in the 19th century, trees were first introduced in computer science in the 1950s, with early applications in compiler design and hierarchical data representation. Over the decades, various types of trees, such as binary search trees, AVL trees, and B-trees, have been developed to optimize operations like searching, insertion, and deletion. These structures have significantly influenced areas including database indexing, file systems, and network routing. This paper traces the historical development of tree data structures, examines their diverse applications, and highlights their enduring impact on modern computing systems.

***Keywords**: Tree Data Structures, Binary Trees, Binary Search Trees, AVL trees, Tree traversal technique.*

CSIT14: Transitioning Programming Paradigms: A Comparative Study of Procedural and Object-Oriented Approaches

Dipali ankush kamble[1], Ashwini sopan shidore[2], Ganesh p. joshi[3], Kanchan k. Raikar[4]

[1,2,3,4]SHRI CHHATRAPATI SHIVAJ MAHARAJ COE,NEPTI,AHILYANAGAR
*Corresponding Author Email ID : dipali.kamble4@gmail.com

Abstract: This Procedure-Oriented Programming (POP) and Object-Oriented Programming (OOP) are two fundamental programming paradigms that differ in their approach to software development. POP follows a structured, function- driven methodology, focusing on a step-by-step execution of procedures, while OOP is centered around objects that encapsulate data and behavior, promoting modularity and reusability. This study provides a comparative analysis of these paradigms based on key factors such as code structure, data handling, reusability, security, maintainability, and scalability. The paper also explores the advantages and limitations of each approach, highlighting their applicability in various real-world scenarios. By understanding the strengths and weaknesses of POP and OOP, developers and organizations can make informed decisions when selecting the appropriate paradigm for software development.

***Keywords**: Procedure-Oriented Programming (POP), Object- Oriented Programming (OOP), Class and Object, Encapsulation, Inheritance, polymorphism, data abstraction, dynamic binding*

CSIT15: Federated Learning Framework for Privacy-Preserving Artificial Intelligence

Kanchan K. Raikar[1], Dipali A. Kamble[2] , Ashwini S. Shidore[3] ,Ganesh P. Joshi[4]
[1,2,3,4]Shri Chatrapati Shivaji Maharaj College of Engineering, Nepti A'Nagar
*Corresponding Author Email ID : kanchan.raikar@scoea.org

Abstract: Artificial Intelligence (AI) systems require large volumes of data for training accurate models. However, sharing sensitive data such as healthcare, financial, or personal information raises serious privacy concerns. Federated Learning (FL) has emerged as a promising approach to enable collaborative machine learning while preserving data privacy. In this paper, we propose a Federated Learning framework for privacy-preserving artificial intelligence where multiple clients collaboratively train a global model without sharing their raw data. The proposed framework uses decentralized training, secure model aggregation, and differential privacy techniques to ensure data confidentiality. Experimental evaluation demonstrates that the proposed approach maintains high model accuracy while significantly improving privacy protection. The results highlight the effectiveness of federated learning in enabling secure and scalable AI systems.

***Keywords** : Federated Learning, Privacy Preservation, Artificial Intelligence, Distributed Machine Learning, Data Security.*

CSIT16: AI-Driven Vision-Guided Robotic Arm For Real Time Defect Detection and Quality Control in Smart Manufacturing

Aparna Prakash Bairagi[1]
[1]Amruthvahini College of engineering, Sangamner
*Corresponding Author Email ID : aparnasbairagi03@gmail.com

Abstract : Quality inspection plays a critical role in modern manufacturing industries to ensure product reliability and maintain production efficiency. Conventional inspection methods rely heavily on manual visual inspection, which is often slow, inconsistent, and susceptible to human error. With the emergence of Industry 4.0, intelligent automation using artificial intelligence and robotics has become an effective solution for improving manufacturing quality control. This paper presents the design and development of an AI-driven vision- guided robotic arm system for real-time defect detection and automated sorting of industrial components. The proposed system integrates computer vision with deep learning techniques, including You Only Look Once (YOLO) for real-time object detection, Convolutional Neural Networks (CNN) for feature extraction, and Convolutional Fuzzy Neural Networks (CFNN) for accurate defect classification. Industrial component images are captured using an industrial camera and processed through image preprocessing, object detection, feature extraction, and classification stages. Based on the classification results, a robotic arm performs automated pick-and-place operations to separate defective and acceptable components. The proposed framework improves inspection accuracy, reduces manual intervention, and enhances overall production efficiency. The system demonstrates the potential of integrating artificial intelligence, machine vision, and robotic automation to support reliable and intelligent quality control in smart manufacturing environments.
Keywords: *AI, YOLO, CNN, CFNN.*

CSIT17: AI-Based Resume Generator with ATS Analysis and Job Integration System

Mohalkar Akash Yuvraj[1] , Pangavhane Sandip Navnath[2] , Khendake Pradip Goraksh[3] , Paudmal Nikhil Namdev[4]

[1,2,3,4]Pravara Rural Engineering College, Loni

*Corresponding Author Email ID : akashmohalkar34@gmail.com

Abstract: The AI-Based Resume Generator with ATS Analysis and Job Integration System is a web-based platform designed to assist job seekers in creating professional, optimized resumes while improving their chances of passing Applicant Tracking Systems (ATS) used by modern recruiters. The system integrates AI- driven resume generation, ATS compatibility analysis, and job opportunity integration into a single unified platform. It enables users to generate customized resumes by entering their personal, educational, and professional details, while intelligent algorithms analyze the resume content to ensure it meets ATS standards and industry requirements. The platform consists of multiple modules where users initially log in through a secure authentication interface and gain access to resume creation, editing, ATS scoring, and job search functionalities. The AI Resume Generator module helps users automatically structure resumes based on predefined templates and best practices. The ATS Analysis module evaluates resumes for keyword optimization, formatting compatibility, and recruiter readability, providing suggestions for improvement. The Job Integration module allows users to explore relevant job opportunities and align their resumes with job descriptions. Developed using React.js for the frontend and Python/Java-based backend services with RESTful APIs, the system ensures scalability, efficient processing, and secure data handling. By combining artificial intelligence with automated resume optimization and job integration, the platform

simplifies the resume creation process, enhances job application success rates, and provides an intelligent career support tool for job seekers.

***Keywords**: Web-Based Application; AI Resume Generator; ATS Analysis; Job Integration System; React.js; RESTful APIs; Resume Optimization; Career Support Platform*.

CSIT18: Design and Implementation of Vehicle Theft Detection and Remote Engine Locking System

Akanksha Pokale[1], Monali Ghogare[2], Krushna Mengade[3], Abhishek Ghogare[4]

[1,2,3,4]Pravara Rural Engineering College Loni

Corresponding Autor Email ID : akankshapokale4@gmail.com

Abstract: Vehicle theft has become a serious issue in both urban and rural regions, especially with the rapid increase in the number of vehicles. Traditional security systems such as mechanical locks, alarms, and central locking mechanisms provide only basic protection and often fail to prevent theft effectively. In many cases, these systems only alert nearby people and do not allow the owner to monitor or control the vehicle remotely. To address these limitations, this research proposes an IoT-based vehicle theft detection and remote engine locking system. The system uses an ESP32 microcontroller along with sensors to monitor the vehicle's condition and detect unauthorized access or movement. Once suspicious activity is detected, the system immediately sends a notification to the vehicle owner through the Telegram messaging platform. The owner can then remotely lock the vehicle engine using a command through the Telegram bot. The proposed system also integrates GPS technology to provide real-time location information of the vehicle. By combining tracking, remote monitoring, and engine immobilization features, the system offers a more reliable and efficient solution for preventing vehicle theft. The implementation demonstrates that IoT-based communication can significantly improve vehicle security while remaining cost-effective and user friendly.

***Keywords:** IoT, Vehicle Theft Detection, ESP32, GPS Tracking, Remote Engine Locking, Telegram Notification System*

CSIT19: MAATIBHASHA AI – MARATHI DIALECT TRANSLATOR WITH VOICE, CHAT AND GOVME INTEGRATION

Vidya Gagare[1] , Renuka Patil[2], Bhakti Sase[3], Prajwal Deshmukh[4]

[1,2,3,4]Shri Chhatrapati Shivaji Maharaj College Of Engineering, Nepti , Ahilyanagar

*Corresponding Autor Email ID : vidyagagare2224@gmail.com

Abstract: MaatiBhasha AI – Marathi Dialect Translator with Voice, Chat, and GovMe Integration is a multilingual, dialect-aware system designed to bridge the linguistic gap for Marathi speakers across Maharashtra, including underrepresented tribal and regional dialects. The system focuses on enabling seamless communication through translation between Marathi dialects, standard Marathi, and English, while integrating voice input/output, conversational chatbot support, and government service accessibility via GovMe APIs. The primary objective is to provide accurate dialect-to-standard translation with ≥85% semantic fidelity, robust speech-to-text for noisy real-world audio, natural Marathi text- to-speech output, and a chatbot

capable of handling FAQs and government queries.Unlike existing systems (e.g., Google Translate, AI4Bharat, CommonVoice), MaatiBhasha AI emphasizes dialect coverage, code-mixing handling, and end-to-end integration of ASR,NMT,chatbot and GovMe services. The system leverages state-of-the-art algorithms such as Whisper and Wav2Vec2.0 for ASR, Tacotron2/WaveGlow for TTS, IndicTrans and mBART for translation, and Rasa for chatbot intent handling. Datasets like CommonVoice, AIKosh, L3Cube-MahaNLP, and code-switched corpora form the backbone of training. Methodology includes preprocessing (normalization, annotation), fine-tuning models, and deploying on scalable cloud platforms (AWS/GCP) with GovMe API integration. Key innovations lie in dialect normalization, handling of Hindi–English code-switching, emotion detection for adaptive chatbot responses, and government service integration tailored for rural and regional Marathi speakers. This work aims to enhance inclusivity by providing a practical, AI-driven solution for linguistic accessibility, ensuring citizens can interact effectively with digital and governance platforms in their mother tongue.

Keywords: *Marathi Dialects, Speech-to-Text, Text-to-Speech, Neural MachineTranslation, Chatbot, GovMe Integration,Code-Mixing.*

CSIT20: A Comprehensive Review on Deep Learning-Based Diabetic Retinopathy Detection and Severity Grading Using ResNet-50 and Fundus Imaging

Ms.Jyoti Tulshiram Vairal [1]

[1]Amrutvahini college of Engineering, Sangamner

*Corresponding Author Email ID : jyotivairal1992@gmail.com

Abstract : Diabetic Retinopathy (DR) is still one of the most common causes of blindness that may be avoided. This makes it very important to have automated screening technologies that can find early signs of disease in retinal fundus images. Recent progress in deep learning, especially Convolutional Neural Networks (CNNs), has made it possible to very accurately classify the severity levels of DR. This review looks at the best ways to find DR, with a focus on the ResNet-50 and Retinopathy Severity Grading (RSG-Net) designs. The paper shows how preparing images, adding more data, and using transfer learning can make DR grading more reliable across different datasets, including the APTOS 2019 Blindness Detection dataset. To find out how well modern CNN models can diagnose, several stages of DR, from No DR to Proliferative DR, are looked at. The paper also looks at the pros and cons of binary and multi-class classification systems in more detail, as well as how useful they are in the clinic. This research gives a detailed look into CNN-based methods for finding DR early and talks about how they could help with large-scale, real-time eye screening.

Keywords*: Diabetic Retinopathy, ResNet-50, RSG-Net, Fundus Images, Deep Learning, Convolutional Neural Network, DR Classification, APTOS 2019, Medical Image Analysis, Automated Screening.*

CSIT21: Next-Gen Industrial AI Driven Wearable Equipment Detection and Alert System

Sahil Bodke[1], Devika More[2], Samruddhi Pansare[3]

[1,2,3]Jaihind College of Engineering
*Corresponding Author Email ID : moredevika44@gmail.com

Abstract : Ensuring worker safety in industrial settings is a persistent challenge due to hazardous environments, equipment failure, and limited human supervision. This paper introduces a Smart Industrial Safety Wearable Device integrated with Artificial Intelligence (AI) and Internet of Things (IoT) technologies to proactively identify risks, monitor conditions, and generate real-time alerts. The proposed wearable system captures physiological parameters such as heart rate and body temperature, along with environmental data like gas levels, humidity, and ambient temperature. An AI-based predictive model analyzes these readings to forecast potential hazards and automatically notify both workers and control centers. The system enhances workplace safety, reduces human errors, and promotes data-driven preventive actions.

Keywords*: Artificial Intelligence, IoT, Industrial Safety, Wearable Technology, Predictive Analytics, Worker Protection*

CSIT22: Smart VR-Integrated 3D Visualization System for Real Estate and Interior Design using Machine Learning

Mr. Todmal Chaitanya Bharat[1], Mr. Athare Krishna Ashok[2] , Mr. Wable Rohan Sharad[3] , Mr. Wani Girish Gorkshanath[4] , Prof.Surekha.S.Nimase[5]
[1,2,3,4,5]Shri Chhatrapati Shivaji Maharaj College of Engineering, Nepti, Ahilyanagar
*Corresponding Author Email ID : chaitanyatodmal13@gmail.com

Abstract: Real estate and interior design industries increasingly demand smarter, faster, and more immersive design solutions. Traditional 2D and static 3D design methods often fail to communicate spatial depth and functional flow effectively. This paper presents a Smart VR-Integrated 3D Visualization System that combines Virtual Reality (VR) with Machine Learning (ML) to enhance user experience and decision-making in spatial design. The system enables users to generate 3D layouts, visualize them in immersive VR environments, and receive ML based layout recommendations that consider spatial harmony, ergonomics, and aesthetic balance. The proposed system reduces design time, improves visualization accuracy, and enhances user satisfaction compared to traditional tools. Experimental results show smooth rendering at 55–60 FPS and layout recommendation accuracy exceeding 90%, demonstrating the effectiveness of VR– ML integration for real estate and interior applications.

Keywords*: Virtual Reality, Machine Learning, 3D Visualization, Real Estate Technology, Layout Optimization, Neural Networks, Deep Learning, Human-Computer Interaction*

CSIT23: Deepfake Image Detection using CNN and LSTM

Abhishek Subhash Dhanapune[1]
[1]Amrutvahini collage of engineering sangamner
*Corresponding Author Email ID : abhisheksubhashdhanapune@gmail.com

Abstract: Deepfake technology utilizes Generative Adversarial Networks (GANs), a breakthrough of machine learning technology, and has received significant attention as it allows for the generation of fake, yet extremely realistic images and videos. The risks of these media are significant in terms of politics, cyber security and personal privacy, to name a few. This paper is a large-scale survey on the current state of deepfake image detection. The paper reviews state-of-the-art methods for detecting deepfake images including their development, effectiveness, and limitations. We consider classic image forensics, machine learning-based approaches, and modern deep learning methods and consider their performance on major datasets. The paper also outlines the challenges facing detection approaches, including the rapidly mutating space of deepfake generation, dataset bias, as well as real-time detection. Finally, this paper discusses potential future research directions with an emphasis on the development of detection methods that are more robust, more generalizable, and more explainable.
***Keywords**: Deepfake, Image detection, Generative adversarial networks, Machine learning, Classical image forensics, Datasets, Robustness, Explainability.*

CSIT25: Implementation of a Digital Daily Pass and E-Ticketing Mobile Application for Local Public Transport

Wathore Shivaji Pandit[1], Dr.Wakchaure M.A.[2]
[1,2]Amrutvahini collage of engineering Sangamner
*Corresponding Author Email ID : shivajiwathore220@gmail.com

Abstract: The current public transportation system primarily relies on traditional paper-based ticketing and manual fare collection methods, which lead to operational inefficiencies, revenue leakage, increased boarding time, and inconvenience for passengers and transport authorities. This dissertation presents the design and implementation of a Digital Daily Pass and E-Ticketing Mobile Application aimed at modernizing local public transport fare management. The proposed system replaces conventional ticketing with a secure, mobile-first solution that enables instant purchase, digital storage, and real-time validation of single-journey e-tickets and digital daily passes. The system follows a three-tier architecture consisting of a customer mobile application, a conductor mobile validation application, and a cloud-based backend infrastructure. A key security feature of the system is the use of dynamic QR codes generated using a Time-Based One-Time Password (TOTP) mechanism. This ensures that each digital pass produces a time sensitive encrypted token, preventing screenshot fraud and unauthorized sharing. The conductor application validates tickets in real-time through a secure Validation API, ensuring that responses are delivered within a minimal time frame to reduce passenger dwell time. The backend infrastructure manages transaction processing, pass generation, audit logging, and real-time reporting for administrative oversight. The system enhances operational transparency, reduces paper waste, eliminates cash handling complexities, and enables data-driven decision-making through accurate transaction records. By integrating mobile technology, secure tokenization, and scalable cloud architecture, the proposed solution contributes to improved efficiency, enhanced passenger convenience, reduced fraud, and the advancement of smart, sustainable public transportation systems.
***Keywords:** Digital Daily Pass, E-Ticketing System, Mobile Application, Public Transport,*

Dynamic QR Code, TOTP, Validation API, Cloud Architecture, Secure Payment Gateway, Fraud Prevention, Real-time Validation, Microservices, Operational Transparency, Smart Mobility, Sustainable Transport.

CSIT26: PRICISION AGRICULTURE USING IOT AND ML

Vikas P. Gaikwad[1], Swapnil R.Gaikwad[2], Omkar P.Maid[3], Ajinkya S.Mhaske[4], Prof. Mutkule Prasad Raghunath[5]

[1,2,3,4]Shri Chatrapati Shivaji Maharaj College of Engineering, Nepti Ahilyanagar

*Corresponding Author Email ID : vikasgaikwad04166@gmail.com

Abstract: India's agricultural sector, which accounts for 58% of employees and produces 18% of GDP, is in crucial danger due to climate extremes, resource scarcity, and inefficient farming practices, as evidenced by the 2023 Punjab floods (₹5,000 crore damage on 1.2 lakh hectares) and Maharashtra (₹2,200 crore on 2.5 lakh hectares), and the need for resilient agriculture. Conventional methods yield poor results from generalized advice that does not consider local variations, resulting in wastage of resources and losses during harvesting. This project aims to have an integrated precision farming system with IoT sensors for measurement of soil moisture, temperature, humidity, and rainfall levels in real-time using machine learning (ML) models for predictive irrigation planning, organic vs. chemical fertilizer guidance, and disease risk forecasting. Intended for smallholder farmers, the system offers personalized notifications via a multilingual (Hindi, Marathi, English) WhatsApp and SMS bot backed by voice support for low literacy users and explainable AI (XAI) to ensure trust. There is an admin console for data management and feedback-based ML fine-tuning using the Random Forest, SVM, CNNs, and LSTMs algorithms on diverse datasets like Kaggle, IMD APIs, and satellite imagery. By lowering water/fertilizer use by 30- 50% and improving yields by 20-30%, the affordable, scalable platform promotes sustainable practice, improves climate resilience, and supports SDGs on food security

***Keywords**: Precision agriculture, Internet of Things (IoT), Machine learning, Sustainable farming, Climate resilience, Flood impacts, Multilingual chatbot, Explainable AI, Smallholder farmers, Irrigation optimization.*

CSIT27: Gyaan Yatra - An AI-Powered Immersive Learning Using AR/VR

Suraj Chauhan[1], Mitesh Kanojiya[2], Sanika Shinde[3]

[1,2,3]Thakur College of Engg & Technology

*Corresponding Author Email ID : 1032231241@tcetmumbai.in

Abstract: This paper introduces GyaanYatra, an innovative educational ecosystem that synergizes Artificial Intelligence with Augmented and Virtual Reality to reimagine experiential learning for underserved communities [11], [18]. Unlike conventional digital learning tools, GyaanYatra embeds culturally resonant 3D environments within a pedagogically grounded framework rooted in Kolbs experiential learning cycle and Vygotsky social constructivism [29], [30]. The platform adaptive AI engine dynamically tailors content delivery based on real-time learner engagement patterns and multimodal preferences, ensuring inclusive access across

diverse cognitive profiles. Implementation across three educational institutions in Maharashtra, India, involving 120 learners, revealed transformative.

Keywords: *Artificial Intelligence (AI), Augmented Reality (AR), Virtual Reality (VR), Immersive Learning, Experiential Education, Personalized Learning, Educational Technology, Adaptive Learning, Cultural Localization, Digital Divide, WebXR, Low-Cost Education, India Education, AI in Education, VR in Classroom.*

CSIT28: Real-Time Bidirectional Indian Sign Language Recognition and Translation to Multiple Regional Languages using CNN-LSTM & YOLOv8 with Voice Output

Kalyan Vishnu Giram [1], Arya Deepak Dhumne [2]

[1,2]Shri Chhatrapati Shivaji Maharaj College Of Engineering , Nepti Ahilyanagar

*Corresponding Autor Email ID : aryadhumne@gmail.com

Abstract: Communication remains a fundamental barrier for India's 6.3 million people with speech and hearing disabilities who rely on Indian Sign Language (ISL). Unlike Western nations with unified sign languages, ISL varies regionally across Hindi, Marathi, Tamil, and Telugu communities. Existing solutions focus on ASL or BSL with English-only output, supporting fewer than 16 sign classes, requiring desktop installation, and offering no two-way communication — making them impractical for rural hospitals, schools, and government offices. This paper proposes a real-time, bidirectional ISL recognition and translation web platform using YOLOv8 deep learning and MediaPipe, delivering regional language voice output to bridge communication between deaf-mute individuals and the general public.

Keywords: *Indian Sign Language, Real-Time Gesture Recognition, YOLOv8, Regional Language Translation, Bidirectional Communication.*

CSIT29: Smart Surveillance System for Real-Time Armed Person Detection using YOLO, AI-ML

Om Satish Ingale[1], Pranav Santosh Jagdale[2],

Mukund Sandesh Ugale[3], Ankit Pandurang Gavkhare[4]

[1,2,3,4]Shri Chatrapati Shivaji Maharaj college of engineering, Ahilayanagar, nepti

*Corresponding Autor Email ID : Omsingale8998@gmail.com

Abstract: Ensuring public safety has become a critical challenge due to the increasing occurrence of security threats involving armed individuals in public and private spaces. This study proposes a smart surveillance system for real-time armed person detection using advanced Artificial Intelligence (AI) and Machine Learning (ML) techniques. The proposed framework uses the YOLOv8n object detection model to automatically detect firearms and identify suspicious human activities from live CCTV video streams. The system is designed to operate in real-time, enabling continuous monitoring and rapid alert generation to assist security personnel in preventing potential incidents. The model was trained and evaluated on a curated dataset of weapon-related images and video frames to improve detection capability under various environmental conditions. Experimental results demonstrate that the proposed

system achieves an overall detection accuracy of approximately 70%, while maintaining efficient inference speed suitable for real-time deployment. The results indicate that the proposed approach can significantly enhance modern surveillance infrastructure by providing intelligent threat detection and proactive security monitoring. This research highlights the potential of integrating deep learning–based object detection models into surveillance systems for improving situational awareness and public safety.

Keywords: *Smart Surveillance System, Armed Person Detection, YOLOv8, Artificial Intelligence, Machine Learning, Real-Time Monitoring, Computer Vision.*

CSIT30: AI POWERED WEB UI GENERATOR FROM TEXT PROMPTS

Humbe Sanket Pramod[1], Khatik Tushar Ashok [2], Gursali Atharv Mahadev[3]

[1,2,3]Sharadchandra Pawar college of Engineering and Technology,Someshwarnagar

*Corresponding Autor Email ID : atharvgursali6267@gmail.com

ABSTRACT: This project, titled "AI Powered Web UI Generator from Text Prompt", aims to develop an AI-powered system capable of generating fully functional web user interfaces (UIs) from simple natural language prompts. By integrating Natural Language Processing (NLP) with code generation models such as transformer-based architectures, the system converts user descriptions into responsive HTML, CSS, and JavaScript code. The approach reduces front-end design time and allows non-technical users to create professional websites through conversational input. The project demonstrates the potential of AI in automating software development and improving productivity in UI/UX design.

Keywords: *AI, NLP*

CSIT31: TeleTrack:- AI powered drug trafficking identifier

Bhusari Shruti[1], Nimase Bharati[2], Wable Snehal[3],Wagh Akansha[4]

[1,2,3,4]Shri Chatrapati Shivaji Maharaj college of engineering Ahilyanagar nepti

*Corresponding Autor Email ID : bharatinimase2004@gmail.com

ABSTRACT: Telegram is one of the fastest-growing messaging services to gain global popularity. Its main selling points have been the privacy and encryption features. Ironically,the same features are protecting drug traffickers who adopted hidden groups, bots, and coded language to evade law enforcement. This project proposes an AI-powered Multimodal Detection Framework—TeleTrack that combines Natural Language Processing(NLP), Computer Vision, and Graph Neural Networks (GNNs) for the identification and tracking of illegal drug-trading activities on Telegram. Textual, visual, and metadata information is extracted from groups and channels and then processed by Transformer-based NLP models like BERT, the visual models like CLIP and YOLO, and graph-based analysis through Node2Vec to uncover the network of dealers hidden in the layers of the graph.The modality that connects these different modalities is the Fusion Enginehere to achieve the classification

of suspicious users and groups thus empowering law enforcement to intervene locally and in real-time at the very scene of the crime.To enable the officers to better understand the overall situation, a law enforcement dashboard is made available to them for plotting the results of their surveillance. It helps in the determination of the location, the time, and the suspect to take the necessary steps in response.One of the goals is to increase the system's performance, reduce the number of false alarms, and offer a scalable cloud solution that can adapt to the changes in communication trends and slang.The first version of multimodal AI usage in combination has a lot of detection capabilities that single-modal systems hardly have, as it is demonstrated by prototype results.

Keywords: *Telegram, Drug Trafficking Detection, Multimodal Artificial Intelligent, Natural Language Processing, Computer Vision, Graph Neural Networks, Cyber security, Real-time Monitoring*

CSIT32: Food Hope - Community Food Donation App

Chaitanya Kaldante[1],Aditya Kadam[2],Yash Mhaske[3],Sachin Vidhate[4]

[1,2,3,4]Shri Chatrapati Shivaji Maharaj college of engineering Ahilyanagar nepti

*Corresponding Autor Email ID : kaldantechaitanya13@gmail.com

Abstract: Food wastage remains one of the major globalchallenges, while millions of people continue to face food insecurity. Addressing this gap requires intelligent systems that can efficiently connect surplus food sources with individuals and organizations in need. Food Hope is a cloud-based mobile and web platform designed to reduce food wastage by enabling real-time collaboration between food donors, volunteers, and receivers.The application leverages modern web and mobile technologies, including Angular, Ionic Framework, and Google Firebase, to provide a scalable and responsive platform for food redistribution. Donors can quickly publish surplus food details along with location and expiry information, allowing nearby receivers to identify and collect donations before spoilage occurs. Real-time synchronization ensures that updates are instantly reflected across the system, improving efficiency and reducing response time.To enhance user interaction and platform accessibility, the system incorporates an AI-assisted conversational support module that guides users, resolves queries, and improves the overall donation workflow. The platform also integrates an intelligent food lifecycle management approach that automatically monitors donation validity and handles expired food responsibly. Instead of allowing food to go to waste, the system redirects unsuitable items toward sustainable alternatives such as composting, biogas generation, and other environmentally beneficial reuse processes.Developed using an agile methodology, the platform prioritizes usability, safety, and scalability while encouraging community participation in food redistribution. By combining real-time technology, intelligent assistance, and sustainable waste-handling strategies, Food Hope demonstrates how digital innovation can contribute to reducing food wastage and promoting a more sustainable and socially responsible food ecosystem.

Keywords: *Intelligent Food Redistribution Systems, AIPowered Conversational Agents, Real-Time Cloud Computing, Sustainable Food Lifecycle Management, Circular Economy for Food Systems, Smart Donation Infrastructure*

CSIT33: Garage Dhundo : Roadside and home mechanic

Gaikwad kedar Shankar [1], Jagtap Rohit Badrinath [2] , Kadam Rupesh Ramrao[3]
[1,2,3]Sharadchandra Pawar college of Engineering and Technology,Someshwarnagar
*Corresponding Autor Email ID : jagtaprohit2005@gmail.com

Abstract: The Garage Dhundo web-based application intended to monitor all garage operations, is presented in this study. Users may schedule deliveries, manage inventory, and look up repair estimates using the GMS. It keeps track of the time spent by mechanics and keeps a thorough record of the vehicle service history. Additionally, it keeps track of the inventory of auto parts and automatically reminds customers to get their cars serviced based on previous dates. Roles (administrator, supervisor, and receptionist) limit access, guaranteeing methodical control. By automating key procedures, the application aims to eliminate manual work and provide a clever online tool that will improve customer service and garage efficiency in general.

Keywords: *GMS*

CSIT34: ContractX: AI Based Intelligent Tendering System

Prof.Sachin P. Vidhate[1], Yashwantrao Navnath Unde[2] ,
Jay Anil Repate[3] ,Yash Santosh Sonawane[4]
[1,2,3,4]Shri Chatrapati Shivaji Maharaj college of engineering Ahilyanagar nepti
*Corresponding Autor Email ID : sonawaneyasha955@gmail.com

Abstract: Traditional tendering systems rely heavily on manual processes involving excessive paperwork, physical bid submissions, and face-to-face evaluations, leading to inefficiency, lack of transparency, and susceptibility to corruption. This paper presents ContractX — an AI Based Intelligent Tendering System developed as a mobile application using the Ionic Angular framework and Firebase Firestore as the real-time cloud database.The proposed system digitizes the complete tendering lifecycle, from contract publication to bid evaluation and milestone based payment tracking, through a secure role-based platform accessible on Android devices. The system architecture follows a three-tier model comprising the Ionic Angular presentation layer, Firebase Firestore data layer, and a Node.js proxy server as the business logic and AI communication layer. Two distinct role-based dashboards are provided Contractor and Bidder each designed to handle specific workflow responsibilities within the procurement process. Firebase Phone OTP Authentication eliminates password-based vulnerabilities and ensures secure user access. A notable contribution of this work is the integration of an AI Assistant powered by the Anthropic Claude large language model, deployed via a secured proxy server hosted on Render.com. The AI module assists contractors with intelligent contract drafting, competitive bid analysis, and market-based recommendations, communicated in English to improve acces- sibility for Indian users. Experimental results demonstrate that the system successfully automates all core tendering operations with 100 test case pass rate across eight functional modules. The application is packaged as an Android APK using Capacitor and tested on physical mobile devices. ContractX addresses the core limitations of conventional procurement systems by providing a fully digital, transparent, and AI-assisted tendering

workflow. The system holds significant potential for deployment in government departments, private organizations, educational institutions, and NGOs seeking efficient and accountable pro- curement management.

Keywords: *E-Tendering, Intelligent Tendering System, Ionic Angular, Firebase Firestore, Anthropic Claude API, AI Assistant, Role-Based Access Control, Node.js Proxy, Mile stone Tracking, Digital Procurement, Android APK, Three Tier Architecture, OTP Authentication*

CSIT35: Emergency Vehicle Arrival Alert System for Traffic Clearance

Tejal Sunil Hande[1] , Kadambari Chandrakant ozarkar[2],
Mayuri santosh Ghangale[3], Pooja Mande[4]
[1,2,3,4]Jaihind college of Engineering
*Corresponding Autor Email ID : handetejal004@gmail.com

Abstract: Rapid and effective emergency response plays a crucial role in saving lives; however, many existing emergency management systems experience delays due to heavy traffic congestion, insufficient real-time coordination, and limited interaction between emergency vehicles and traffic control infrastructure. This paper presents the development and implementation of a cloud-based ambulance monitoring and alert system designed to enhance response time and improve operational efficiency during medical emergencies. The main objective of the proposed system is to provide continuous real-time tracking of ambulance locations and generate timely notifications to traffic signals and administrative authorities to support faster and safer vehicle movement. The system integrates mobile and cloud technologies to transmit live GPS location data of ambulances to a centralized cloud platform. A distance-based processing mechanism is used to detect nearby traffic signals and automatically generate alerts when the ambulance enters a predefined range, enabling traffic signal priority and reducing travel delays. In addition, a web-based administrative dashboard enables authorities to observe ambulance movement and monitor system activity in real time. The system also includes role-based access control and an alert management mechanism to avoid duplicate notifications. Experimental evaluation indicates that the proposed approach helps minimize emergency response time and improves coordination between ambulance services and traffic management systems, contributing to more efficient and reliable emergency transportation.

Keywords: *AI,ML*

CSIT36: Eye And Hand Gestures Recognition Based On Virtual Mouse

Shweta Shinde[1], Akshata Jagadale[2], Sujata Divekar[3]
[1,2,3]Sharadchandra Pawar college of Engineering and Technology,Someshwarnagar
*Corresponding Autor Email ID : shwetashinde1391@gmail.com

Abstract: This project focuses on creating a virtual mouse system that lets a computer be operated through eye movements and hand gestures instead of a physical mouse. A standard webcam captures real-time images of the user's eyes and hands, and computer vision techniques process these images to detect eye direction and hand shapes. Detected gestures are then converted into mouse actions such as moving the cursor, left or right clicking, and

scrolling, providing a completely touch-free way to interact with the computer. The system aims to make human–computer interaction faster, more natural, and more convenient by removing the need for external hardware like a traditional mouse or touchpad. It is especially helpful for people with physical disabilities who may find regular input devices difficult to use. In addition, it can be applied in gaming, virtual reality, smart home control, and other areas where hands-free operation is useful. Machine learning methods are used to improve detection accuracy and to adapt to different users and lighting conditions. By combining image processing with gesture recognition, the project demonstrates how simple eye and hand movements can provide an efficient, low-cost alternative to standard input devices, making technology more accessible and interactive for a wide range of users.

Keywords: *Virtual Mouse, Hand Gesture Recognition, Eye Tracking, Computer Vision, Human–Computer Interaction (HCI), Machine Learning, Webcam-Based Interaction.*

CSIT37: RAG-Lawyer: A Retrieval-Augmented Legal Advisor Using Generative AI

Prof. Nazirkar Swati B.[1],Dr. Shah Saloni N.[2],Barde Sakshi U.[3],
Bhapkar Sakshi Y.[4],Lonkar Monika D.[5]

[1,2,3,4,5]SHARADCHANDRA PAWAR COLLEGE OF ENGINEERING AND TECHNOLOGY, SOMESHWARNAGAR

*Corresponding Autor Email ID : monikalonkar2003@gmail.com

Abstract: RAG-Lawyer integrates Retrieval- Augmented Generation (RAG) with domain-specific legal knowledge bases to create a reliable and explainable AI legal advisor. The system retrieves the most relevant legal documents—such as statutes, case judgments, and regulatory texts—before generating responses grounded in verifiable sources. By combining a retrieval module with a fine-tuned generative model, RAG- Lawyer enhances factual accuracy, transparency, and interpretability in legal text generation. Legal judgment prediction (LJP) presents a formidable challenge in artificial intelligence, demanding intricate comprehension of legal texts, nuanced interpretation of statutes, and complex reasoning over multifaceted case elements prediction Our approach encompasses four key stages: 1) legal knowledge infusion, where we pre-train an LLM on a vast corpus of legal literature using contrastive learning techniques; 2) case-law retrieval, employing a graph neural network to identify relevant precedents and statutes; 3) multi-hop reasoning, utilizing a transformer- based architecture with a hierarchical attention mechanism to navigate complex legal arguments; and 4) judgment synthesis, where we employ a generative adversarial network to produce coherent and legally sound judgments. Despite its potential, challenges remain, including ensuring data reliability, addressing nuanced legal interpretations, and maintaining privacy and security of sensitive legal information. A RAG legal advisor acts as a powerful augmentation tool for human lawyers, streamlining routine tasks while preserving professional oversight and responsibility. Our work not only advances the field of automated legal reasoning but also offers a transparent and explainable system that could serve as a valuable tool for legal professionals. By bridging

the gap between AI and legal expertise, Legal Reasoner paves the way for more efficient, consistent, and air legal decision-making processes.

Keywords: *AI, Processing*

CSIT38: Smart AgriPet Care based on Android Application

Arya Deepak Dhumne[1] , Riddhi Rajendra Huljute[2] ,
Kshitija Prashant Nimase[3], Shreya Prashant Nimase[4]
[1,2,3,4]Shri Chatrapati Shivaji Maharaj college of engineering Ahilyanagar nepti
Corresponding Autor Email ID : aryadhumne@gmail.com

Abstract: In today's technologically advancing world, automation and intelligent systems are revolutionizing how we interact with our environment. The Smart AgriPet Care system is a comprehensive, dual-purpose mobile application and web platform that integrates smart pet care for domestic (love) pets and agricultural/breeding animals (AgriPets) under one unified solution. This project simplifies human effort, promotes sustainable practices, and ensures the health, wellbeing, and commerce of both pet communities and agricultural animals through modern mobile technology, cloud computing, and AI powered assistance. The application is divided into two primary domains: Pet Care (for domestic love pets like dogs, cats, etc.) and AgriPet Care (for agricultural breeding animals like cows, buffaloes, goats, etc.). Both domains offer rich features including pet profiles with breed information, health checkups, diet plans, vaccination tracking, a shopping marketplace, and a buy/sell/adopt listing platform. A unified AI-powered chatbot named AgriBot allows users to ask any agriculture or pet-related question and receive intelligent responses. The app supports multi-language functionality, secure user login with profile management via a sidebar, and a smooth logout mechanism. The web dashboard and mobile app together provide a seamless, feature-rich experience for farmers, pet owners, and animal breeders, making Smart AgriPet Care an all-in-one solution for modern animal and pet management.

KEYWORDS: *Smart AgriPet Care, Mobile Application, Milky Animals, Pet Health Monitoring.*

CSIT39: Chatbot for College Enquiry using Groq And Croma Db

Pooja Ramchandra Chandgude[1] , Sujal Santosh Jadhav[2] , Akshay Sambhaji Jadhav[3] ,
Pratik Dayanand Jagtap[4]
[1,2,3,4]Sharadchandra Pawar College of Engineering And Technology
*Corresponding Autor Email ID : sujaljadhav2732@gmail.com

Abstract: Artificial Intelligence (AI) has become a transformative force in multiple industries, including education, where it enables enhanced interaction, automation, and information management. Among its many applications, AI- driven chatbots have proven to be effective digital assistants capable of providing immediate responses to user queries. This paper presents a detailed survey of existing chatbot systems developed for college enquiry purposes.It consolidates findings from recent academic research to explore the transition from rule-based systems to intelligent models that integrate Natural Language Processing (NLP), Deep

Learning (DL), and Machine Learning (ML). The paper also discusses algorithmic design approaches, implementation architectures, development frameworks, and comparative evaluations of performance. The analysis concludes that combining NLP and deep learning models significantly enhances chatbot accuracy, adaptability, and user satisfaction.

Keywords: *Natural Language Processing, Chatbot, Artificial Intelligence, Voice Bot*

CSIT40: Text to image synthesis using modified GANs

Suryawanshi Shruti Santosh[1], Pokar Gauravkumar Vinod[2],
Taware Swanand Suryakant[3], Patil Nikhil Bhagwat[4]
SPCOET Someshwarnagar
*Corresponding Autor Email ID : shrutisuryawanshi7771@gmail.com

Abstract: The current period of artificial intelligence development needs automated systems which create innovative and intelligent digital content. The traditional methods of graphic design and manual lustration require designers to spend extended periods of time and use their skills for technical work. The Promtify system functions as a web-based SaaS platform which uses Java Spring Boot and RESTful APIs and AI-powered image generation models to create its Text-to-Image Generation System. Users can create high-quality realistic images through the system by entering text descriptions which will turn their ideas into digital artworks. The Promtify system functions through a credit-based system which permits users to produce images until they use up all their credits and they can buy additional credits when they require more. The backend system developed with Java Spring Boot handles various functions which include user authentication and prompt request management and API integration and credit tracking and generated image storage. The system establishes secure connections with third-party AI models through APIs which handle text prompts and produce images during the process. This system design enables users to interact with the system without difficulties while providing the ability to expand operations and maintain system efficiency.

Keywords: *AI, DeepL, API*

CSIT41: Smart Receipt OCR Expense Tracker using Machine Learning

Sawant Bhakti Shriram[1], Saswade Neelam Shekhar[2], Sanap Rahul Vilas[3]
[1,2,3]Pravara Rural Engineering College
*Corresponding Author Email ID : sawantbhakti2574@gmail.com

Abstract: This project presents a Smart Receipt Expense Tracker that automatically extracts and categorizes expense information from receipt images using machine learning techniques. The main aim of the system is to simplify personal expense management by reducing manual data entry and improving accuracy. The proposed system captures or uploads receipt images and processes them through an automated pipeline. Image preprocessing techniques are applied to enhance image quality before text extraction. The system uses Tesseract OCR with an LSTM-based recognition engine to convert receipt images into machine-readable text. A Convolutional Neural Network (CNN) is used for detecting text regions in receipt images, improving text detection accuracy. After text extraction, Regular Expressions (Regex) are

applied to identify important fields such as date, total amount, and merchant name. The extracted information is then processed using an LSTM-based classification model to categorize expenses into categories like food, travel, or shopping. All processed data is stored in a SQLite database for efficient storage and retrieval. The system provides a structured view of expenses and helps users analyze their spending patterns. The proposed solution improves automation, accuracy, and efficiency in expense tracking. This project demonstrates the practical use of machine learning, OCR, and database systems for intelligent financial management.

Keywords: *LSTM*

CSIT42: Automated Contract Review And Compliance Checker

Vaishnavi Sakunde[1] , Pallavi Nigade[2] ,Komal Patil[3]

[1,2,3]Shri. Someshwar Shikshan Prasarak Mandal's Sharadchandra Pawar College of Engineering And Technology Someshwarnagar

*Corresponding Author Email ID : vaishnavisakunde28@gmail.com

Abstract : In today's digital era, organizations handle numerous contracts and agreements daily, especially those involving data sharing and processing under privacy laws like the General Data Protection Regulation (GDPR). Manual review of these contracts is time-consuming, error-prone, and often leads to non-compliance risks. The proposed project, Automated Contract Review & Compliance Checker, aims to develop a web-based intelligent system using Python and Natural Language Processing (NLP) techniques to automatically review contracts and identify potential compliance violations. The system analyzes the content of Data Processing Agreements (DPAs) to verify whether they align with GDPR requirements and other relevant standards.This project also focuses on detecting fraudulent or invalid documents by assessing textual inconsistencies and compliance gaps. The tool generates a summary report that indicates the compliance percentage and highlights non-compliant clauses or suspicious patterns. By integrating advanced NLP models, machine learning, and document parsing techniques, the system enables faster, more accurate contract evaluation, thereby reducing legal and operational risks. The project not only streamlines legal document management but also empowers businesses to make informed decisions with transparency and reliability.

Keywords*: Contract Review, Compliance Checker, GDPR, NLP, Machine Learning, Document Fraud Detection, Data Privacy, etc.*

CSIT43: AgroBridge: Farmer-to-Farmer Equipment Sharing System

Prachi Rohidas Saste[1],Sakshi Anil Gaikwad[2],Vaishnavi Vijaysinh Kadam[3]

[1,2,3]SSPM's Sharadchandra Pawar College of Engineering and Technology, Someshwarnagar, Baramati-412306

*Corresponding Author Email ID : prachisaste29@gmail.com

Abstract: Agriculture plays a vital role in India's economy, yet many small and marginal farmers face difficulties accessing modern farming equipment due to high purchase costs, seasonal demand, and limited availability. Expensive machinery like tractors, harvesters, or

threshers often remains idle for most of the year, while other farmers depend on local brokers or informal sharing networks that lack transparency and reliability. This results in underutilization of equipment, financial losses for owners, and reduced productivity for those unable to access tools when needed. AgroBridge is developed as a farmer-to-farmer equipment sharing system that addresses these issues through a secure, transparent, and technology- driven platform. It allows equipment owners to list idle machinery and lets nearby farmers discover and book them based on location, availability, and pricing preferences. The system integrates modules for user verification (KYC), digital payments with escrow protection, and a rating-based reputation model to promote trust and accountability between users.The platform also includes features such as vernacular language support, offline-first access for rural connectivity, AI-assisted price guidance, and real-time logistics tracking. Through these functionalities, AgroBridge ensures inclusivity and ease of use even for users with limited technical experience. The system is built using scalable web and mobile frameworks, secure APIs, and a microservice architecture for efficient data handling and smooth user experience. By improving machinery utilization and enabling affordable access to advanced tools, AgroBridge helps reduce the capital burden on smallholders while improving productivity and profitability. It bridges the gap between technology and agriculture by promoting digital collaboration and sustainability within the farming community.

Keywords : *Agricultural Technology, Equipment Sharing, Digital Agriculture, Smart Farming, Rural Innovation, Farm Mechanization*

CSIT44: Smart Attendance System with QR Code & Location Tracking

Shirsath Shraddha Rajendra[1],Shirsath Akshara Anil[2],Salve Sandesh Ratnakar[3],Sonawane Sai Ashok[4]

[1,2,3,4] Pravra Rural Engineering College Loni

*Corresponding Author Email ID : saisonawane2004@gmail.com

Abstract: The Smart Attendance System with QR Code and Location Tracking is a web-based application designed to automate and improve the traditional attendance process. In many institutions and organizations, attendance is still recorded manually, which can lead to errors, proxy attendance, and time consumption. This system provides a secure and efficient way to manage attendance using modern technologies. The proposed system generates a unique QR code for each class or session, which students can scan using their mobile devices to mark their attendance. Along with QR code verification, the system also uses location tracking to ensure that the student is physically present in the designated location while marking attendance. This helps to prevent fake or proxy attendance. The system is developed using Java Full Stack technologies, where the backend is implemented using Spring Boot, the frontend is developed using React/Angular, and the database is managed using MySQL. The system allows administrators and teachers to create sessions, generate QR codes, monitor attendance records, and generate attendance reports. Overall, this project aims to provide a secure, efficient, and automated attendance management solution that reduces manual work, improves accuracy, and ensures transparency in attendance tracking.

Keywords: *MySQL, QR codes*

CSIT45 :Smart glasses for visually impaired using ESP32-CAM and AI based object detection

Arti Aute[1], Ishwari Farate[2], Samiksha Deshmukh[3]
Shri Chhatrapati Shivaji Maharaj college of engineering Nepti, Ahilyanagar
*Corresponding Author Email ID : autearti2004@gmail.com

Abstract— The advancement of artificial intelligence (AI), computer vision, and embedded computing has accelerated the development of assistive technologies for individuals with visual impairments. Despite substantial progress, existing smart glasses solutions often suffer from limitations such as high cost, reliance on cloud-based inference, and lack of portability. This review paper analyses contemporary research (2020–2025) on AI- enabled wearable assistive devices, focusing on object detection, optical character recognition (OCR), and real-time speech feedback. The paper synthesizes findings from ecent YOLO-based edge detection models, text detection frameworks (EAST, CRAFT), and embedded deployments on microcontrollers and Raspberry Pi platforms. Furthermore, it identifies the technical and usability challenges that constrain real-time offline performance, including computational limits, energy efficiency, latency, and environmental robustness. Based on this analysis, a conceptual model for Smart Glasses for the Visually Impaired is proposed, integrating lightweight YOLO for object awarene ss, OCR for text reading, and text-to-speech (TTS) for contextual audio feedback—entirely offline. The framework emphasizes cost-effectiveness, privacy, and accessibility through tactile control, local inference, and modular hardware design. Expected outcomes include enhanced autonomy, situational awareness, and confidence for visually impaired users, demonstrating the feasibility of combining AI and IoT for socially beneficial, low-cost assistive devices.

Keywords*: Smart Glasses, Visually Imaired, YOLO, Raspberry Pi, OCR, Object Detection, Text-to-Speech, Assistive Technology, Edge AI, IoT*

CSIT46: Text to image synthesis using modified GANs

Suryawanshi Shruti[1], Pokar Gauravkumar[2], Taware Swanand[3], Patil Nikhil[4]
[1,2,3,4]SPCOET Someshwarnagar
*Corresponding Author Email ID : shrutisuryawanshi8485@gmail.com

Abstract: The current period of artificial intelligence development needs automated systems which create innovative and intelligent digital content. The traditional methods of graphic design and manual illustration require designers to spend extended periods of time and use their skills for technical work. The Promtify system functions as a web-based SaaS platform which uses Java Spring Boot and RESTful APIs and AI-powered image generation models to create its Text-to-Image Generation System. Users can create high-quality realistic images through the system by entering text descriptions which will turn their ideas into digital artworks. The Promtify system functions through a credit-based system which permits users to produce images until they use up all their credits and they can buy additional credits when they require more. The backend system developed with Java Spring Boot handles various functions which include user authentication and prompt request management and API integration and credit tracking and generated image storage. The system establishes secure connections with third-

party AI models through APIs which handle text prompts and produce images during the process. This system design enables users to interact with the system without difficulties while providing the ability to expand operations and maintain system efficiency.

***Keywords**: API,IoT*

CSIT47: AI Driven Fuel Locator And Detector

Pawar Akanksha[1], Dandavate Sanika[2], Chaoudhari Tejasvi[3]
[1,2,3]Sharadchandra pawar college of engineering and technology
*Corresponding Author Email ID : pawarakanksha386@gmail.com

Abstract: An AI-driven fuel and charging station locator and detector is a smart system designed tohelp users find the nearest and most suitable fuel(gasoline, diesel) or electric vehicle (EV) charging stations based on real-time data, user preferences, and contextual factors. As the world shifts towards smarter mobility and cleaner energy, the demand for efficient, reliable, and intelligent navigation tools has never been higher. The AI-driven fuel and charging stations locator and detector is an innovative solution designed to help drivers—both of traditional fuel-powered vehicles and electric vehicles (EVs)— quickly and accurately find the nearest, most suitable fueling or charging options in real time. Powered by artificial intelligence, this system goes beyond basic GPS mapping. It leverages real-time data, machine learning algorithms, and predictive analytics to optimize route planning, reduce wait times, and enhance user convenience. Whether in an urban environment or a remote area, the system ensures always connected to the nearest and best-rated energy sources for your vehicle

CSIT48: Digital College Platform For Mutual Empowerment

Siddhi Tambe[1], Amrale Ekta2, Jagtap Sakshi[3]
[1,2,3]Sharadchradra pawar college of engg and technology someshwarenagar,Baramati
*Corresponding Author Email ID : siddhitambe0080@gmail.com

Abstract : The Digital College Platform for Mutual Empowerment is an innovative web-based system developed to strengthen the connection between students and alumni of an educational institution. The main objective of this platform is to create a collaborative environment where students can gain mentorship, career guidance, job opportunities, and academic support from experienced alumni. The system includes essential features such as user registration and login, student and alumni dashboards, messaging, and connection requests, which enable smooth communication and networking. The platform is developed using modern web technologies such as HTML, CSS, JavaScript, PHP, and MySQL. By improving interaction between students and alumni, this platform promotes knowledge sharing, professional networking, and career evelopment, ultimately contributing to a stronger and more connected academic community.

***Keywords :** Students, Alumni, Networking, Career Guidance, Web Application, Education Platform.*

CSIT49: Online Construction Resource Planning and Tracking Platform

Jadhav Pranali Sunil[1] , Khalate Vijay Shridhar[2], Pore Siddhant Dhananjay[3]
[1,2,3]SSPM's Sharadchandra Pawar College of Engineering and Technology, Someshwarnagar, Baramati – 412306
*Corresponding Author Email ID : jadhavpranali9922@gmail.com

Abstract: Construction projects often face challenges such as poor communication, unclear progress updates, and lack of reliable documentation. Traditional methods like verbal reports, paper- ased records, and messaging applications often lead to data loss, inefficiency, and delayed decision-making.This project introduces an online construction site monitoring and documentation system designed to simplify supervision and enhance project transparency. The platform allows supervisors to upload real-time images of ongoing site activities along with text notes for clarification. These records are stored securely in a centralized database for a minimum of six months, ensuring accountability and accessibility for future reference.Administrators can remotely access these updates via a secure login, review progress, and generate reports for performance analysis. The system is built with role-based access control, ensuring data privacy and reliability.Overall, the project enhances communication between on-site and management teams, reduces manual workload, and ensures efficient documentation of construction progress. Future enhancements may include IoT integration, material tracking, and workforce monitoring for complete project automation.

Keywords: *IoT,AIML*

CSIT50: AI Powered E-learning Video Summarizer

Sawant Sakshi Satish[1], Nikam Gayatri Vijay[2], Daund Dipali Shivaji[3]
[1,2,3]Sharadchandra Pawar collage of engineering and technology, someshwar nagar
*Corresponding Author Email ID : gayatrinikam80800@gmail.com

Abstract : In recent years, online learning platforms have become very popular, and students often need to atch long educational videos to understand the content. However, watching lengthy videos can be time-consuming and sometimes difficult to review quickly. The AI Powered E-Learning Video ummarizer is a system designed to automatically generate a short and meaningful summary of educational videos using Artificial Intelligence techniques. The proposed system converts the video into text using speech-to-text technology and then applies Natural Language Processing (NLP) techniques to analyze the content and extract important information. Based on this analysis, the system generates a concise summary that helps learners quickly understand the key concepts of the video without watching the entire content. This tool is useful for students, teachers, and online learners as it saves time and improves learning efficiency. It allows users to quickly review important points, revise topics easily, and enhance the overall learning experience. The system aims to make e-learning more effective by providing intelligent and automated video summarization.

Keywords*: E-learning, Video Summarization, Artificial Intelligence, Speech-to-Text, Natural Language Processing (NLP), Automatic Summarization, Online Learning*

CSIT51: Lifeline – AI Based Emergency Doctor Assistant with Ambulance Connectivity

Gauri Sunil Shinde[1],Shruti Maruti Thopate2,Shreya Anil Thopate3
[1,2,3]Shri Someshwar Shikshan Prasarak Mandal's Sharadchandra Pawar College of Engineering and Technology, Someshwarnagar
*Corresponding Author Email ID : gaurii7083@gmail.com

Abstract : The rapid advancement of artificial intelligence has significantly improved healthcare services, especially in emergency situations. This project, titled "Lifeline – AI Based Emergency Doctor Assistant with Ambulance Connectivity", aims to provide immediate medical assistance to users through an intelligent system. The proposed system integrates AI technologies such as Natural Language Processing (NLP), image analysis, and voice recognition to understand user symptoms through text, voice, or images. Based on the input, the AI assistant suggests reliminary medications and first-aid measures according to age groups. In critical conditions, the system can automatically connect to nearby ambulance services, ensuring timely medical support. dditionally, the application provides real-time communication, enabling users to interact with the AI assistant via calls or messages. The system is designed to reduce response time in mergencies and improve accessibility to healthcare services, especially in remote areas. Overall, this project aims to bridge the gap between patients and immediate medical care using advanced AI echniques, thereby potentially saving lives.

Keywords: *NLP*

CSIT52: RecipeBowl – Ingredient and Recipe Recommendation System

Priyanka Baravkar[1], Vaishnavi More[2], Priti Saste[3]
[1,2,3]Sharadchandra Pawar College of Engineering and Technology, Someshwarnagar
*Corresponding Author Email ID : priyanka66baravkar@gmail.com

Abstract: In today's fast-paced lifestyle, deciding what to cook with available ingredients is a common challenge. The RecipeBowl system is an intelligent web-based solution that uses Natural Language Processing (NLP) to analyze ingredients entered by the user and recommend suitable recipes, home remedies, and wellness tips. The system enhances user experience by providing step-by-step cooking instructions and integrating YouTube video tutorials for better understanding. It helps reduce food waste, saves time, and promotes a healthy lifestyle by offering personalized and efficient cooking assistance.

Keywords: *Ingredient-Based Recommendation, Recipe Suggestion, Natural Language Processing (NLP), Web Application, Java, Home Remedies, Wellness, Fitness, YouTube Integration, Personalized Cooking, etc.*

CSIT53: Phantom Ledgers: Detecting Hidden Manipulation in Public Budget Datasets Using Cryptographic Integrity, Temporal Edit Graph Analysis, Explainable Anomaly Triage, and Agentic Alerting

Sanika Sameer Tribhuvan[1]
[1]New Arts Commerce And Science College, Ahilyanagar
*Corresponding Author Email ID : tribhuvansanika@gmail.com

Abstract: Public budget data sets are the cornerstone of government transparency initiatives across the globe, yet they are susceptible to retroactive changes, coordinated insertion campaigns, and incremental value changes. The traditional auditing process typically evaluates only the current state of the data set, unable to detect changes reflected solely in the historical differences from the original data set. In this paper, the author introduce Phantom Ledgers, a monitoring framework that detects hidden data set manipulation based on the analysis of the data set's historical changes across periodic snapshots. The data records are hashed using SHA-256, grouped into Merkle trees, with the root hash of each snapshot anchored using the Bitcoin blockchain via OpenTimestamps for non-repudiable timestamping. The differences between the current and the previous snapshot are aggregated into a Temporal Edit Graph, which models the changes in the data set, including insertions, deletions, and modifications, across the snapshots. A three-tier Isolation Forest-inspired Triage Engine categorizes the changes as ROUTINE, SUSPICIOUS, CRITICAL, based on the weighted five-dimensional feature vector, with ROUTINE changes eliminated from the subsequent process. For the remaining changes, the SHAP-inspired Explainable AI Attribution Layer calculates the contribution of the normalised factors to the attribution score, providing the basis for the forensics process. The changes are grouped into campaigns using the union-find algorithm, with the Agentic Alerting Engine dispatching HTTP POST webhook notifications to the registered oversight bodies upon the verification of the campaigns at high confidence levels. The prototype was tested on an eighteen-snapshot synthetic data set with four campaigns, correctly identifying 22 suspicious changes across three campaigns with zero false positives, achieving a mean XAI attribution depth of 4.2 signals per event, with agentic alerts dispatched in under 200 ms upon the verification of the campaigns.

Keywords—*Budget data set integrity, Temporal Edit Graph, Isolation Forest Triage, Explainable AI, SHAP Attribution, Agentic Alerting, Merkle Tree, Blockchain Anchoring*

CSIT54: A Deep Learning Framework for Real-Time Deepfake Detection Across Social Media Platforms

Kurhe Pradnya[1], Dhere Rajashri[2], Kharat Mayuri[3], Wandhekar Prerana[4]
Shri Chhatrapati Shivaji Maharaj College of Engineering Nepti Ahilyanagar
*Corresponding Author Email ID : pradnyasambhaji5@gmail.com

Abstract: An increasing threat to digital authenticity has emerged due to deepfake technology, which uses artificial intelligence and deep neural networks to create realistic manipulated multimedia content. This research aims to develop a neural framework for real-time deepfake detection across ultimedia modalities such as image, audio, and video. The proposed model integrates Convolutional Neural Networks (CNN) for spatial feature learning, Recurrent Neural Networks (RNN) for temporal dependency analysis, and Transformer-based attention

layers for cross-modal feature fusion. This combination enhances the model's ability to detect inconsistencies between visual and auditory cues, enabling effective identification of manipulated content. The framework is designed to achieve high accuracy and scalability while ensuring interpretability and computational efficiency. It can be applied in fields like digital forensics, social media monitoring, and cybersecurity to preserve the integrity of digital communication. This study concludes that multimodal learning significantly improves deepfake detection performance and strengthens trust in digital media.

Keywords: *Deepfake Detection, Neural Networks, CNN, RNN, Transformer, Multimedia, Misinformation.*

CSIT55: AI Career Coach

Shinde Jaydeep[1], Onkar Deshmukh[2], Chaitanya Shelke[3]

[1,2,3]SPCOET, Someshwarnagar

*Corresponding Author Email ID : onkardeshmukh384@gmail.com

Abstract: The rapid expansion of career opportunities in the digital era has made it increasingly difficult for students and job seekers to identify suitable career paths, prepare professional resumes, and develop job- relevant skills. Traditional career guidance methods are often limited, time-consuming, and inconsistent due to the availability of multipleinformation sources and the lack of personalized mentorship. To address this challenge, the AI Career Coach project proposes an intelligent, automated guidance system that utilizes Artificial Intelligence to assist users in making informed career decisions. This system analyses the user's academic background, skills, interests, and professional preferences to recommend suitable job roles, required skill development paths, and resume improvement suggestions. The system is designed to support individuals at various stages of their academic or professional journey, ensuring continuous development and informed career planning. The proposed solution reduces dependency on human counsellors and ensures consistency, scalability, and cost-effectiveness. Thus, the AI Career Coach contributes to the empowerment of job seekers and students by providing a reliable, smart, and user-centric
platform for career development and planning.

Keywords: *Resume Builder, AI based Career guidance, Study Materials, Motivation.*

CSIT56: Privacy-Safe Data Sharing in Educational Technology: Frameworks, Challenges, and Emerging Solutions

kalyan Vishnu Giram [1], Arya Deepak Dhumne[2]

[1,2]Deogiri Institute of Engineering and Management Studies, Chhatrapati Sambhaji-Nagar

*Corresponding Author Email ID : kalyanvgiram@gmail.com

Abstract—The rapid integration of Educational Technology (EdTech) platforms into academic institutions has result to unmatched volumes of vulnerable student data. This raises major issues about privacy, data governance, and adherence to regulations. This paper survey's how to share data safely in the EdTech ecosystem. It analyze the clash between personalized learning analytics and the need to protect student privacy. We evaluate existing privacy-preserving techniques, including differential privacy, federated learning, k-anonymity, and

blockchain-based consent management, verifying their suitability in educational data environments. Additionally, we recognize major challenges such as crossborder data transfer, third-party vendor risks, and the boundaries of current regulations like FERPA, GDPR, and COPPA in addressing EdTech-specific issues. We recommend a layered privacy framework that blends technical controls, institutional policy, and user- centered consent models to promote responsible data exchange while maintaining analytical benefits.

Keywords—*EdTech; privacy-preserving data sharing; federated learning; differential privacy; GDPR; FERPA; learning analytics; student data protection; blockchain consent*.

CSIT57: Decentralized Ride-Sharing Platforms Using Blockchain

Adiraj A.Sanas[1], Dhanraj H.Natkale2, Aditya P.Saste[3], Vaibhav D.Salunkhe[4]

[1,2,3,4]Sahardchandra pawar college of engineering and technology someshwarnagar

Corresponding Author Email ID : dhanrajnatakle@gmail.com

Abstract : The recent growth and rise of ride-sharing services have had an incredible impact on rban mobility, allowing people to be transported in a more efficient and convenient manner, which has reduced traffic congestion. Ride-hailing services today are mostly centralized, creating challenges surrounding privacy, transparency, trust, and high commission rates. This led to researchers considering blockchain technology as a way to resolve the limitations associated with current bride-hailing services. This article is a survey paper that gives an in-depth overview of blockchain- based decentralized ride- hailing systems. This article reviews the previous literature pertaining to blockchain-based decentralized ride-hailing and evaluates the significant contributions made in the last few years by these technologies. It also compares the various existing frameworks related to decentralized ride-hailing systems, highlighting their respective advantages and disadvantages. Lastly,research challenges and future directions within the area of blockchain-based decentralized ride-hailing systems will also be addressed. The intention of this paper is to provide researchers with a better understanding of how blockchain technology can be utilized in the area of ride-hailing services.

Keywords: *Blockchain, Decentralized Ride-Sharing, Smart Contracts, Peer-to-Peer Transportation, Privacy Preservation, Smart Mobility*

CSIT59: Loksetu -From village to the World

Somnath Lavhate, Rima Kolate[1], Nisha Gite, Sahil Mutadak[2]

[1,2]Pravara Rural Engineering College Loni

*Corresponding Author Email ID : nishagite13@gmail.com

Abstract: Loksetu is a digital marketplace platform designed to empower rural women, artisans,nd small business owners by providing them a direct platform to buy and sell their products [1]. Many skilled artisans and women entrepreneurs lack digital exposure, marketing knowledge, and access to large customer bases [7]. Loksetu bridges this gap by connecting village-level producers directly to customers across regions using an e-commerce platform [2]. The system provides secure user authentication, product listing, order management, and payment integration using modern web application technologies [4], [5]. The proposed

platform focuses on promoting self-employment, rural economic development, and digital inclusion [6]. By leveraging web technologies and structured system design, Loksetu aims to create a sustainable and scalable e-commerce ecosystem for rural communities [3], [9].

***Keywords**: E-commerce, Rural Development, Women Empowerment, Digital Marketplace, Web Application, UML [8], [10].*

CSIT60: Creating Alert Messages Based on Wild Animal Activity Detection Using Hybrid Deep Neural Networks

Vikas Rajkumar Mandlik[1],Sapnali lambhate[2],Rushikesh Gaikwad[3]

[1,2,3]Sharad chandra pawar college of engineering and technology,someshwarnagar.

*Corresponding Author Email ID : vikasmandlik7229@gmail.com

Abstract: Human–wildlife conflicts continue to cause substantial economic loss and human injury across many regions worldwide. This paper presents a comprehensive, extended survey and implementation report of a complete system that performs realtime wild animal detection and generates alert messages using a Hybrid Deep Neural Network (HDNN) that combines Convolutional Neural Networks (CNN) for spatial feature extraction and Recurrent Neural Networks (RNN/LSTM) for temporal activity modelling. The system integrates IoT-based communication channels to deliver timely SMS and web notifications to stakeholders. We describe dataset preparation, preprocessing, HDNN architecture design, training strategies, evaluation metrics, deployment considerations, security and ethical concerns, and a thorough discussion of experimental results and comparative analysis against existing approaches. The extended content provides both theoretical background and practical guidance for deploying robust wildlife monitoring solutions in real-world conditions. Index Terms—Wild animal detection, hybrid deep neural networks, IoT, CNN, RNN, real-time alert, wildlife monitoring, deployment, security.

***Keywords:** CNN,RNN*

CSIT61: Chest X-Ray Outlier Detection Model Using Dimension Reduction and Edge Detection

Mujawar Sabaa Shafique[1], Baravkar Shruti Mohan[2], Jadhav Priti Vijay[3]

[1,2,3]SHARADCHANDRA PAWAR COLLEGE OF ENGINEERING AND TECHNOLOGY SOMESHWARNAGAR

*Corresponding Author Email ID : sabaamujawar13@gmail.com

Abstract: Early and correct diagnosis of diseases from chest X ray images is essential for efficient treatment and patient care. Here, we discuss the use of Liquid Neural Networks (LNNs) for improving diagnostic efficiency and accuracy in chest X- ray disease prediction. LNNs involve adaptive memory and dynamic fusion, providing a rich understanding of sequential and spatial information .We used a double-model strategy: one for discriminating ”Normal” vs. ”Pneumonia,” and another for discriminating ”Normal” vs. ”Lung Opacity.” Our web- based application using Flask combines these models for real- time disease identification .Experimental findings indicate that the ”Normal vs. Pneumonia” model attained 94accuracy with an AUC of 0.98, and the ”Normal vs. Lung Opacity” model attained 90 percent accuracy

with an AUC of 0.96. With a disease detection-prioritized methodology, our technique ensures any suspicion of disease outweighs a ”Normal” assignment .This study highlights the prospects of LNNs to bring radiological diagnostics into a new era, making disease detection easier and improving patients’ outcomes To convert the extracted edge, which is visual information, into a series of lines, it is computed in convolution with the detection lter that has a coefficient of 2n and the lines are divided into 16 types. By counting the converted data, a one-dimensional 16- size array per one segment-image is produced, and this reduced data is used as an input to the RNN-based learning model. In addition, the study conducted various experiments based on the COVID- chest X-ray dataset to evaluate the performance of the proposed model. According to the experiment results, the LFA- RNN showed the highest accuracy at 97.5 percent in the learning calculated through learning, followed by CRNN 96.1 percent, VGG 96.6 percent, AlexNet 94.1 percent, Conv1D 79.4 percent, and DNN 78.9 percent. In addition, LFA-RNN showed the lowest loss at about 0.0357. Index Terms—Computer-aided diagnosis (CAD) system, feature extraction, line feature analysis (LFA), RNN, deep learning.

***Keywords**: RNN,CNN,CAD*

CSIT62: AI Virtual Career Counselor

Pratiksha Shelke[1], Sakshi Jagtap[2], Tanvi Saste[3], Dhanshri Kale[4]

[1,2,3,4]Sharadchandra Pawar College of Engineering and Technology

*Corresponding Author Email ID : sakshijagtap1007@gmail.com

Abstract: In today’s world, people need to keep learning new skills throughout their lives. Because of this, career guidance has become more important than ever. However, providing easy and accessible guidance that connects education with real job opportunities is still a big challenge. So far, only a small amount of research has focused on how artificial intelligence (AI) can be used to improve career guidance for students and professionals. This paper explains the development and use of AI tools to support career counselling in higher education institutions. The study collected information through focus groups, scenario discussions, and real-life trials. It explored the needs, expectations, and possibilities of using AI for career guidance from the perspectives of students, career advisors, and educational institutions. The results show that AI can play an important role in helping students plan their careers, identify suitable job opportunities, and make informed learning decisions. It can also help guidance staff by saving time, analyzing data, and offering personalized advice. However, there are also challenges such as a lack of trust, limited data, technical barriers, and ethical concerns about privacy and fairness. Based on these findings, the authors describe different stages of how AI can be included in the guidance process — from simple support tools to advanced systems that can work alongside human advisors. The paper also highlights the need for future research on how AI canresponsibly share decision-making with humans, how to build a strong data-sharing ecosystem, and how to manage ethical issues in AI-based guidance systems.

***Keywords:** Artificial Intelligence, Career Guidance, Higher Education, Ethics, Data, Lifelong Learning*

CSIT63: E – Voting System with Aadhar Verification and Biometric Authentication

Kadam Pradip Pralhad[1], Kadam Abhijit Jalindar[2], Raut Ashlesha Dnyaneshwar[3]
[1,2,3]Sharadchandra pawar college of engineering and technology someshwar nagar, Baramati
*Corresponding Author Email ID : pradipkadam2323@gmail.com

Abstract: The proposed solution uses biometric-based authentication and Aadhaar to improve the security and transparency of the voting process. It utilizes the voter's unique fingerprint, which is cross-verified with their Aadhaar ID stored in a secure database, to eliminate fraudulent and duplicate voting. Each voter is allowed to cast only one vote, and any unauthorized access is quickly detected by the system. The voting data is securely transmitted to the Election Commission database using microcontroller and GSM modules. This system significantly reduces time, labor, and cost while streamlining the election process. By integrating biometric verification with Aadhaar identification, the system ensures accurate results and tamper-proof elections, thereby enhancing voter confidence and maintaining electoral integrity.

***Keywords**: GSM*

CSIT64: Crime Track Using Deep Learning

Prof.N.A.Nirmal[1] ,Omkar Dabhade[2] , Shriraj Gangale[3] ,Prajwal Bargal[4]
[1,2,3,4]Shri Chhatrapati Shivaji Maharaj Collage Of Engineering Nepti,Ahilyanagar
*Corresponding Author Email ID : omkardabhade2007@gmail.com

Abstract: This project introduces Crime Track, a centralized and intelligent digital ecosystem designed to transform traditional law enforcement workflows into a high-efficiency digital platform. By replacing manual, paper-based record-keeping with a unified digital database, the system enhances the accuracy, transparency, and speed of criminal investigations. The core of CrimeTrack- lows authorized personnel to register new cases, manage detailed suspect profiles, and track the entire investigation lifecycle. A key innovation is the Automated Complaint Logging System with Voice-to-Text conversion. For advanced suspect identification, the system leverages Deep Learning and image processing techniques like SSIM. Developed with a robust architecture, CrimeTrack minimizes human intervention and provides law enforcement with a powerful tool for data-driven decision-making.

***Keywords:** Deep Learning, CrimeTrack, Image Similaity (SSIM), Voice-to-Text Conversion, Criminal Investigation, Suspect Prediction, Automated Reporting, Law Enforcement Technology*

CSIT65: Secure Online Transaction System Using Cryptography

Kalyani Dattatray Gawali[1] , Sai Sushilkumar Andhale[2] , Priti Sandip Bhalsing[3] , Gauri Ramdas Japkar[4]
[1,2,3,4]Shri Chhtrapati Shivaji Maharaj College Of Engineering Nepti ,Ahilyanagar
*Corresponding Author Email ID : kalyanigawali98@gmail.com

Abstract: With the rapid growth of e-commerce and digital banking, ensuring secure online transactions has become a critical requirement. A secure online transaction system protects sensitive information such as user credentials, financial data, and transaction details from unauthorized access and cyber threats. Cryptography plays a vital role in securing these transactions by providing mechanisms for encryption, authentication, and data integrity. This paper presents a secure online transaction system based on cryptographic techniques to safeguard communication between users and servers. The system uses encryption algorithms to convert sensitive data into an unreadable format during transmission, preventing attackers from accessing confidential information. Additionally, authentication methods such as One-Time Password (OTP) verification and digital certificates help ensure that only authorized users can perform transactions. The proposed approach enhances security, maintains data confidentiality, and ensures the integrity of financial operations in online platforms. This system can be applied to banking, e- commerce, and digital payment applications to provide a reliable and secure transaction environment.
Keywords: *Secure Online Transaction, Cryptography, Encryption, Data Security, Authentication, One-Time Password (OTP), Digital Certificates, Cybersecurity, Secure Communication, E-commerce Security.*

CSIT66: Classification of Sign Language Characters by Applying a Deep Convolutional Neural Network

Sakshi Gorakh Babar[1] , Sakshi Appaso Kumbhar[2] ,Tejaswini Manik Gaikwad[3]
[1,2,3]Sharadchandra Pawar College Of Engineering And Technology ,Someshwarnagar,Baramati
*Corresponding Author Email ID : sakshibabar993@gmail.com

Abstract : Sign language is the only medium of communication for the speech-impaired community while the rest of the population communicate verbally. This project aims to bridge this communication gap by proposing a novel approach to interpret the static and dynamic signs in the Indian Sign Language and convert them to speech. A sensor glove, with flex sensors to detect the bending of each finger and an IMU to read the orientation of the hand, is used to collect data about the actions. This data is then wirelessly transmitted and classified into corresponding speech outputs. LSTM networks were studied and implemented for classification of gesture data because of their ability to learn long-term dependencies. The designed model could classify 26 gestures with an accuracy of 98%, showing the feasibility of using LSTM based neural networks for the purpose of sign language translation. To live in a society, it is very important to communicate with each other. But this poses a grave problem for people with hearing disabilities. As they can converse using only sign language, it becomes very difficult for others who don't know the sign language to understand them. So, the purpose of this paper is to create an interpreter which can convert american sign language into the Engilsh language. Through convolutional neural network we were able to create such an interpreter which can interpret the american sign language.
Keywords*: LSTM*

CSIT67: Thumb Gesture Controlled Mouse Movement Recognition System

Tanvi Dhananjay Mahajan[1], Sakshi Sunil Binawade[2], Sakshi Manohar Bhosale[3]
[1,2,3]Sharadchandra Pawar College of Engineering
*Corresponding Author Email ID : tanvidmahajan22@gmail.com

Abstract: Sign Language Recognition (SLR) targets interpreting the sign language into text or speech, to facilitate the communication between deaf-mute people and ordinary people. This task has a broad social impact but is still very challenging due to the complexity and large variations in hand actions. Existing methods for SLR use hand-crafted features to describe sign language motion and build classification models based on those features. However, it is difficult to design dependable features that can adapt to the wide range of hand gestures.
Keywords: *SLR*

CSIT68: Youshelt-online flat renting platform

Gaurav Borokar[1], Atharve Dhumal[2], Suresh Ghule[3]
[1,2,3]Sharadchandra pawar college of engineering and technology Someshwar
*Corresponding Author Email ID : Gauravborokar45@gmail.com

Abstract: This paper examines the transformative impact of digital platforms on rental housing through the lens of "digital informalisation." It argues that the rise of platform-based real estate and algorithm-driven systems is reshaping housing access, governance, and tenant-landlord relationships. By drawing on interdisciplinary scholarship on informal housing across the Global North and South, the study highlights how digital mediation introduces new, opaque mechanisms of control, particularly through risk management, tenant profiling, and predictive analytics. These technologies, often presented as neutral and efficient, enable practices such as algorithmic redlining, biased screening, and data-driven exclusion, which reinforce existing social and spatial inequalities. The paper further demonstrates how both private platforms and state institutions contribute to the expansion of informalisation by suspending or bypassing regulatory frameworks, thereby creating new grey areas in housing governance. Ultimately, the concept of digital informalisation is proposed as a critical framework to understand how technological innovation in housing markets reproduces discrimination, intensifies surveillance, and reshapes urban inequalities in contemporary rental systems.
Keywords : *Digital Informalisation, Rental Housing, Platform Real Estate, Algorithmic Redlining, Tenant Profiling, Risk Management, Housing Inequality, Digital Platforms, Urban Governance, Informal Housing*

CSIT69: Retina Authentication Security for Application

Surabh Gadekar[1], Vaibhav Kudal[2], Suyash Jadhav[3]
[1,2,3]SharadChandra Pawar College of engineering Someshwar Nagar Baramati
*Corresponding Author Email ID : jadhavsuyash86@gmail.com

Abstract: In today's digital era, the need for strong and reliable security systems is greater than ever. Traditional authentication methods such as passwords, PINs, or ID cards are vulnerable to theft, hacking, and duplication. To overcome these challenges, biometric

authentication provides a more secure and unique solution. This project focuses on developing a Retina Authentication Security Application, a web-based system built using Python, OpenCV, and machine learning techniques, that verifies users based on their eye retina patterns captured through a live camera. The retina, located at the back of the eye, contains a complex network of blood vessels that is unique to every individual, making it an ideal feature for identity verification. The proposed system captures a user's facial and eye images using a live webcam, isolates the retina region using image processing techniques, and then authenticates the user by matching it against stored patterns in the database. The system ensures high accuracy, real-time processing, and strong protection against unauthorized access. This approach leverages OpenCV for image capture and processing and machine learning algorithms for pattern recognition. By combining retina and facial recognition, the system enhances the reliability of user verification and prevents fraudulent activities. It can be implemented in sensitive areas such as secure banking systems, confidential data access, and defense security networks.

Keywords: *Retina Authentication, OpenCV, Biometric Security, Image Processing, Python, Machine Learning, etc.*

CSIT70: AI Based Person Detection and Alert System In Restricted Areas Using Raspberry pi

Nandini Shivaji Khunte[1], Kajal Mohan Nazirkar[2], Aniket Dhananjay Godage[3]
[1,2,3]Sharadchandra Pawar College of Engineering and Technology Someshwarnagar
*Corresponding Author Email ID : nandinikhunte22@gmail.com

Abstract : This project presents an intelligent, low-cost surveillance system that leverages Raspberry Pi 4 Model B and artificial intelligence for real-time person detection and alert generation. The system integrates a camera module with the Raspberry Pi to continuously monitor a designated area. Using computer vision techniques powered by OpenCV and deep learning models such as Convolutional Neural Networks, the system can accurately identify human presence in live video streams. A pre-trained object detection model, such as YOLO (You Only Look Once), is deployed to ensure fast and efficient detection even on resource-constrained hardware. Upon detecting a person, the system triggers an alert mechanism, which may include sending notifications via email, SMS, or mobile applications like Telegram. This enables users to receive instant updates about potential intrusions or activity in restricted zones. The system is designed to be scalable, energy-efficient, and suitale for applications such as home security, office monitoring, and smart surveillance. By combining edge computing with AI, the solution minimizes latency and reduces dependency on cloud infrastructure, ensuring privacy and reliability. Overall, this project demonstrates a practical implementation of embedded AI for real-time security monitoring.

Keywords: *Raspberry Pi, Person Detection, Artificial Intelligence, Computer Vision, OpenCV, YOLO, Convolutional Neural Networks, Surveillance System, Real-Time Monitoring, Alert System.*

CSIT71: PCOS Detection Using Machine Learning

Rutuja Rajendra Garje[1], Sumitra Nandkishor Deokar[2], Shravani Avinash Wable[3],
Vaishnavi Sanjay Chothe[4]
[1,2,3]Pravara Rural Engineering College Loni, Ahilyanagar
*Corresponding Author Email ID : rutujagarje8080@gmail.com

Abstract— Polycystic Ovary Syndrome (PCOS) is a common hormonal disorder affecting women of reproductive age, causing issues such as irregular menstrual cycles, weight gain, and infertility, making early detection very important for proper treatment and management. Traditional diagnostic methods are often time-consuming and require multiple medical tests, which can delay diagnosis. The aim of this project is to develop an intelligent system using machine learning techniques to detect PCOS at an early stage with better accuracy. This system uses patient data such as age, body mass index (BMI), hormonal levels, and other clinical parameters for prediction. The objective of the project is to collect and preprocess the dataset, apply different machine learning algorithms such as Logistic Regression, Decision Tree, and Random Forest, and compare their performance based on evaluation metrics like accuracy, precision, and recall. The system analyzes the input data and predicts whether a patient is likely to have PCOS or not. By comparing multiple algorithms, the most efficient model is selected for better prediction results. This approach helps in reducing manual effort and supports healthcare professionals in making faster and more reliable decisions. The proposed system is cost-effective, user-friendly, and can be used as a supportive diagnostic tool. Overall, the project demonstrates that machine learning can play a significant role in improving early detection and management of PCOS.

Keywords: *PCOS, Machine Learning, Disease Prediction, Healthcare Analytics, Logistic Regression, Decision Tree, Random Forest, Classification, Data Preprocessing, Early Detection,Medical Diagnosis, Predictive Modeling.*

CSIT72: Intelligent Tutorial System

Aishwarya Yadwadkar[1], Vedangi khedkar[2], Joya Shaikh[3]
[1,2,3]Shri Chatrapati Shivaji Maharaj college of engineering nepti Ahilyanagar
*Corresponding Author Email ID : yadwadkarash211@gmail.com

Abstract: An Intelligent Tutorial System (ITS) is an advanced computer-based learning system that utilizes Artificial Intelligence (AI) techniques to provide personalized and adaptive education to learners. The system aims to overcome the limitations of traditional teaching methods by offering individualized instruction based on the learner's knowledge level performance, and learning behavior. The proposed ITS analyzes student responses, tracks progress, and dynamically adjusts the learning content to suit the needs of each learner. It consists of key components such as the student model, domain model, pedagogical model, and user interface, which work together to create an interactive learning environment. The system provides real-time feedback, identifies weaknesses, and recommends appropriate learning paths to improve understanding and retention. Experimental results demonstrate that the ITS enhances student engagement, improves academic performance, and supports self-paced learning. The system is scalable, efficient, and applicable across various educational domains, making it a valuable tool in modern digital education.

Keyword: *Intelligent Tutorial System (ITS), Artificial Intelligence (AI), Adaptive Learning, Machine Learning, Personalized Education, Student Modeling, E-Learning Educational Technology, Learning Analytics.*

CSIT73: Smart Citizen Complaint and Resolution Platform using AI

Akash Sanas[1], Atharv Gade[2], Shivendra Phadtare[3], Shivraj Bhosle[4]
[1,2,3,4]sharadchandra pawar college of engineering & technology
*Corresponding Author Email ID : atharva.gade15@gmail.com

Abstract: The Smart Citizen Complaint & Resolution Platform using AI is designed to provide an efficient, transparent, and citizen-friendly solution for addressing public service grievances. Traditional manual complaint handling systems are often slow, poorly categorized, and lack effective tracking mechanisms, leading to delays and dissatisfaction among citizens. This platform leverages Artificial Intelligence (AI) and automation to streamline the entire process of complaint management. Complaints are automatically categorized and routed to the appropriate government or municipal departments, ensuring faster resolution. The system also provides real-time status tracking, enabling citizens to monitor the progress of their issues. Furthermore, built-in data analytics assists policymakers in identifying recurring problems and making informed decisions for long-term improvements in public services. By combining AI-driven complaint categorization, automated routing, and insightful analytics, the platform aims to improve government accountability, enhance citizen satisfaction, and contribute to the development of smar governance and sustainable communities.

Keywords*: AI*

CSIT74: AI-Based Beauty Advisor with Privacy Guardian for Intelligent Skin Condition Analysis

Anirudha Sanjay Sahane[1] , Nikita 74ishor Shende[2],
Rajgauri Rajendra Parkhe[3] , Yash Devidas Nawale[4]
[1,2,3,4]Pravara rural college of engineering,loni
*Corresponding Author Email ID : anirudhasahane2004@gmail.com

Abstract: The rapid growth of mobile skincare applications has increased the demand for intelligent and personalized skin health assessment systems. However, most existing solutions provide generic recommendations, lack accurate skin condition analysis, and raise serious concerns regarding the storage and misuse of sensitive facial data. To address these challenges, this paper presents an AI-Based Beauty Advisor with Privacy Guardian, an intelligent and privacy-preserving skincare advisory system. The proposed system employs deep learning–based facial image analysis using Convolutional Neural Networks (CNNs) with MobileNetV2 transfer learning to detect multiple skin conditions, including acne, wrinkles, pigmentation, dark spots, and redness. A multi-label classification approach is adopted to allow simultaneous detection of multiple skin issues, with confidence scores and severity levels generated for each condition. Based on the analysis results, the system provides personalized skincare recommendations, including morning and night routines, product guidance, and skincare best practices. A key contribution of this work is the integration of a Privacy Guardian layer, which ensures secure handling of user data by processing facial images locally or temporarily,

automatically deleting images after analysis, enforcing minimal data retention policies, and utilizing secure API communication. The system is implemented as a mobile-optimized application using Flutter for the rontend and a secure backend architecture for AI inference and request handling. Experimental results demonstrate that the proposed system effectively identifies multiple skin conditions with high confidence while maintaining strong privacy guarantees. This project highlights the potential of combining artificial intelligence, mobile computing, and privacy-aware design to deliver reliable and secure AI-powered skincare assistance.

Keywords: *Artificial Intelligence, Skin Condition Detection, Convolutional Neural Networks, Transfer Learning, MobileNetV2, Multi-Label Classification, Skincare Recommendation System, Privacy Preservation, Facial Image Analysis, Mobile Application*

CSIT75: Blockchain based Management for organ donation and tranplantati

Sangram Kale[1], Radha Deshmukh[2], Sneha Suryawanshi[3]

[1,2,3]Sharadchandra Pawar College of engineering and technology someshwarnagar

Abstract : Organ donation and transplantation systems face several challenges such as lack of transparency, security issues, data manipulation, and inefficient donor-recipient matching. To address these problems, this study proposes a blockchain-based management system using a private Ethereum network. The system ensures a decentralized, secure, and trustworthy environment for handling the entire organ donation process, including donor registration, patient waiting lists, matching, organ removal, transportation, and transplantation. Smart contracts are developed to automate processes, maintain data integrity, and provide traceability and auditability of all transactions. The proposed system enhances fairness, reduces illegal activities, and improves efficiency in organ allocation. Performance evaluation and security analysis demonstrate that the system is reliable, scalable, and resistant to common cyber threats, making it a promising solution for modern healthcare systems.

Keywords *: Blockchain, Organ Donation, Organ Transplantation, Ethereum, Smart Contracts, Traceability, Security, Decentralization, Healthcare System*

CSIT76: Intelligent E-Certificate Verification System using Blockchain And Ocr

Purva Suryawanshi[1] , Shruti Sangale[2]

Shrii Chhatrapati shivaji maharaj college of engineering nepti ahilyanagar

*Corresponding Author Email ID : purvasuryawanshi2908@gmail.com

Abstract : The rapid growth of digital education and online certifications has made it crucial to have secure and reliable methods to verify if a certificate is genuine. Traditional ways of checking certificates often rely on manual processes. These methods are time-consuming and can be deceived by fake documents. This paper presents Cert Verify AI, a complete system for verifying certificates that utilizes Optical Character Recognition (OCR), fraud detection, and blockchain technology to secure records. Users can upload certificates in image or PDF format. The system uses OCR to read the text, compares it with the submitted information, and checks for fraud using a scoring system. Once verified, each certificate is stored in a secure, tamper-

proof record similar to a blockchain. The system includes separate interfaces for users and administrators to manage certificates, verify them, reject them, and send notifications. The technology stack includes HTML, CSS, JavaScript, Node.js, Express.js, MongoDB, Tesseract.js, and pdf-parse. The results indicate that this system can automatically perform most verification tasks, reduce manual labor, and provide a clear method for securely validating digital certificates.

Keywords: *AIML*

CSIT77: MeetMatch - AI-Powered Event Suggestions

Sahil pisal[1], Aditya pawar[2], Pratik holkar[3]

[1,2,3]Sharadchandra Pawar College of Engineering and Technology, Someshwar nagar Baramati

*Corresponding Author Email ID :sahilpisal06@gmail.com

Abstract : With the rapid growth of digital platforms, a vast number of events such as workshops, conferences, seminars, and social gatherings are organized daily, making it increasingly difficult for users to identify relevant opportunities efficiently. This paper presents "MeetMatch," an AI-Powered Event Suggestions designed to provide personalized event suggestions based on user preferences, location, and historical interactions. The proposed system leverages Machine Learning techniques, including collaborative filtering and content-based filtering, to analyze user behavior and generate accurate recommendations. In addition, Natural Language Processing (NLP) is incorporated to process event descriptions and improve contextual understanding. A hybrid recommendation approach is employed to overcome the limitations of individual methods, such as cold-start problems and lack of diversity. The system is implemented using modern web technologies, ensuring scalability, real-time data processing, and a user-friendly interface. Furthermore, a feedback mechanism is integrated to continuously refine recommendations and enhance system performance. Experimental evaluation demonstrates that the proposed system significantly improves recommendation accuracy, user engagement, and efficiency in event discovery compared to traditional methods.

Keywords: *Artificial Intelligence, Machine Learning, Recommendation System, Event Management, Collaborative Filtering, Content-Based Filtering, Natural Language Processing (NLP)*

CSIT78: RockSense: An IoT-Based Rockfall Prediction and Alert System Using Machine Learning for Open Pit Mine Safety

Komal Sanjay Anap[1], Arti Balasaheb Mane[2], Sagar Ramnath Ekhande[3], Vikrant Sachin Doke[4]

[1,2,3]Pravara Rural Engineering College

*Corresponding Author Email ID : VIKRANTDOKE08@GMAIL.COM

Abstract—Rockfall incidents in open pit mining environments pose a significant threat to human life, equipment, and operational continuity. Early detection of unstable slope conditions is essential to reduce accidents and improve mine safety. This paper presents RockSense, an

Internet of Things (IoT)-based rockfall prediction and alert system designed for real-time monitoring of slope stability in open pit mines. The system integrates multiple sensors, including vibration, rain, tilt, and ultrasonic distance sensors, interfaced with an Arduino microcontroller to continuously monitor environmental and geotechnical parameters. Sensor data is transmitted in real time via Bluetooth to a backend server, where it is stored and analyzed using a Random Forest machine learning model to predict rockfall risk levels. Based on the predicted risk category—Safe, Warning, or Danger—alerts are displayed on a live monitoring dashboard, and critical warnings are sent to nearby personnel through SMS notifications. The proposed system enables early detection of hazardous conditions, supports timely preventive action, and contributes to improved safety and risk management in mining operations.

Keywords—*Rockfall prediction; IoT; Open pit mining; Random Forest; Sensor networks; Safety monitoring; Machine learning; Arduino.*

CSIT79: Skin-Sense: An AI-Based Skin Disease Detection System .

Akshada Ganesh Jadhav[1], Sakshi Mahesh Gore[2], Narendra Nandu Ghuge[3], Avdhoot Vijay Kode[4]

[1,2,3]Pravara Rural Engineering College, Loni

*Corresponding Author Email ID : jadhavakshada414@gmail.com

Abstract—Skin diseases have become increasingly common due to environmental changes, lifestyle factors, and limited availability of dermatological services. Early identification of skin disorders is essential to prevent complications and to support timely treatment. This paper presents the implementation of an artificial intelligence-based skin disease detection and management system that analyzes skin images to assist users in identifying possible skin conditions. The system enables users to upload or capture images of affected skin areas and optionally provide visible symptom information to improve prediction of reliability. A deep learning-based image classification model is employed to process the input images and generate predicted disease results along with confidence scores. In addition to disease detection, the platform provides preventive guidance, basic care recommendations, automated report generation, and healthcare support features such as chatbot assistance and doctor appointment facilitation. The implemented system aims to improve accessibility to preliminary assessment, reduce dependence on immediate physical consultations, and promote early awareness and management of skin-related health issues.

Keywords—*Skin Disease Detection, Deep Learning, CNN, Computer Vision, Healthcare System, Image Processing, AI in Healthcare.*

CSIT80: AI-Based Nutrition Web Application for Personalized Diet Recommendation Using BMI Analysis and Image Based Calorie Prediction

Prof. P.L. Rahinj[1], Mr. Aher Dhiraj Ramchandra[2], Ms. Bhise Dhanashri Dnyaneshwar[3], Ms. Adhav Gauri Janardhan[4], Mr. Tejas Bhabad Dattatray[5].
[1,2,3,4,5]RGCOE
*Corresponding Author Email ID : pj583167@gmail.com

Abstract: In recent years, maintaining a healthy lifestyle has become increasingly important due to the rise in lifestyle-related diseases. Proper nutrition plays a vital role in improving overall health; however, many individuals lack personalized guidance and accurate tools to monitor their dietary habits. This research presents an AI-based nutrition web application designed to provide personalized diet recommendations using Body Mass Index (BMI) analysis and image-based calorie prediction. The proposed system integrates multiple modules, including user authentication, BMI calculation, diet recommendation, daily dietary history tracking, and an intelligent image processing module. The application uses Convolutional Neural Networks (CNN) to analyze food images and estimate their calorie content, enabling users to track their intake more accurately. Additionally, a database system is implemented to store user information and historical data, allowing long-term monitoring and analysis. The system is developed using modern web technologies and machine learning frameworks, ensuring scalability and efficiency. Experimental results indicate that the application provides reliable calorie predictions and improves the accuracy of diet recommendations compared to traditional methods. The proposed solution offers a user-friendly and intelligent platform that promotes healthier lifestyle choices and enhances awareness about nutrition management.

Keywords: *CNN*

CSIT81: AI-Based Real-Time Personal Finance Dashboard

Prof. Unde S.P.[1], Mr. Ajitkumar Bhagwan Ghule[2], Mr. Raj Sunil Jaware[3], Mr. Sopan Namdev Kanawade[4], Mr. Yogesh Kishor Koli[5].
[1,2,3,4,5]RGCOE
*Corresponding Author Email ID : pj583167@gmail.com

Abstract—The rapid growth of digital payment systems such as UPI, mobile wallets, credit cards, and online subscriptions has significantly increased the complexity of personal financial management. Individuals often struggle with fragmented financial data, lack of real-time insights, and ineffective traditional tools that primarily focus on manual tracking rather than intelligent decision-making. To address these challenges, this project proposes an AI-Based Real-Time Personal Finance Dashboard that functions as a smart financial assistant. The system integrates financial data from multiple sources and provides a unified, real-time view of income, expenses, and savings. Leveraging machine learning techniques, it automatically categorizes transactions, detects unusual spending patterns, and analyzes user behavior to generate personalized financial recommendations. The dashboard also includes features such as budget monitoring, overspending alerts, and goal-based savings automation, enabling users to manage their finances proactively. Interactive visualizations and analytics help users understand their financial habits and make informed decisions. By combining real-time data processing with artificial intelligence, the proposed system enhances financial awareness, promotes disciplined spending, and supports long-term financial planning. This approach

transforms traditional passive finance tools into an intelligent, adaptive, and user-centric financial management solution.

Keywords—*Artificial Intelligence (AI), Personal Finance Management, Real-Time Dashboard, Machine Learning, Expense Tracking, Budget Monitoring, Anomaly Detection, Financial Analytics, Smart Savings, Data Visualization*

CSIT82: RFID-Based Automated Attendance System Using IoT

Prof. P.L. Rahinj[1], Mr. Bharat Sampat Palaskar[2], Mr. Utkarsh Ramdas Patil[3], Mr. Aditya Vasant Randhavan[4], Mr. Irfan Dastagir Shaikh[5].

[1,2,3,4,5]RGCOE

*Corresponding Author Email ID : pj583167@gmail.com

Abstract—The traditional attendance management system used in educational institutions is often time-consuming, error-prone, and vulnerable to proxy attendance. To address these challenges, this paper presents an RFID-based automated attendance system integrated with IoT using the ESP32 microcontroller. In the proposed system, each student is assigned a unique RFID tag, which is scanned by an RFID reader to capture attendance data. The ESP32 processes the tag information and transmits it to a cloud-based database via Wi-Fi using HTTP or MQTT protocols. The system enables real-time attendance tracking, secure data storage, and easy access through a web or mobile interface. Compared to conventional methods, the proposed solution significantly reduces manual effort, improves accuracy, and minimizes chances of manipulation. Experimental results show enhanced efficiency, faster processing time, and reliable performance in real-time environments. This system contributes to the development of smart classrooms by providing an automated, scalable, and cost-effective attendance management solution.

Keywords: *AIML*

CSIT83: A Python-Based Intelligent Human-Computer Interaction System Integrating Eye Tracking and Voice Command Recognition for Accessibility Applications

Prof. P.L. Rahinj[1], Mr. Suyash Subhash Narawade[2], Mr. Sadik Samir Pathan[3], Mr. Shailesh Vilas Kadam[4], Mr. Dattatray Mininath Shirke[5].

[1,2,3,4,5]RGCOE

*Corresponding Author Email ID : pj583167@gmail.com

Abstract: The traditional attendance management system used in educational institutions is often time-consuming, error-prone, and vulnerable to proxy attendance. To address these challenges, this paper presents an RFID-based automated attendance system integrated with IoT using the ESP32 microcontroller. In the proposed system, each student is assigned a unique RFID tag, which is scanned by an RFID reader to capture attendance data. The ESP32 processes the tag information and transmits it to a cloud-based database via Wi-Fi using HTTP or MQTT protocols. The system enables real-time attendance tracking, secure data storage,

and easy access through a web or mobile interface. Compared to conventional methods, the proposed solution significantly reduces manual effort, improves accuracy, and minimizes chances of manipulation. Experimental results show enhanced efficiency, faster processing time, and reliable performance in real-time environments. This system contributes to the development of smart classrooms by providing an automated, scalable, and cost-effective attendance management solution.

***Keywords**: RFID*

CSIT84: Advanced Organ And Blood Donation Management System

Vishwajeet Satish Aher[1] , Kaustubh Prakash Gaike[2] , Anuj Jawahar Hotkar[3]

[1,2,3]Shri.Chhatrapati Shivaji Maharaj College Of Engineering , Nepti,Ahilyanagar

*Corresponding Author Email ID : anujhotkar77@gmail.com

Abstract— The increasing demand for blood and organ donation has created a need for a proper and efficient management system. The traditional methods are slow, unorganized, and sometimes lead to delays in emergency situations. To solve this problem, this project proposes an Online Advanced Organ and Blood Donation Management System. This system provides a digital platform where donors, receivers, and administrators can interact easily. Donors can register themselves, check their eligibility based on age and health conditions, and track their donation details. Receivers can submit requests by providing important information such as required organ, blood group, medical condition, and priority level. The system also includes advanced features like smart donor recommendation, emergency blood request handling, and patient behavior prediction. These features help in identifying urgent cases and suggesting suitable donors quickly. It improves decision-making and reduces the time required to match donors and receivers. An admin dashboard is provided to monitor all activities, approve requests, and manage the overall system. The system ensures data validation and secure handling of sensitive information. Overall, this project helps in improving the efficiency, speed, and reliability of the donation process. It plays an important role in saving lives by providing timely support to patients in need.

***Keywords:** AIDS,AI*

CSIT85: Dragon Fruit Disease Detection Using Convolutional Neural Networks

Dr. S. S. Lavhate[1], Niraj Nitin Jagdale[2], Aarti Jalindar Kandekar[3], Kiran Dadasaheb Kavade[4]

[1,2,34]Pravara Rural Engineering College Loni

*Corresponding Author Email ID : nishagite13@gmail.com

Abstract: Dragon fruit is a significant tropical crop cultivated as food and commercial products due to its nutritional and economical properties. Plant diseases are having a detrimental impact on the quality and quantity of dragon fruit production causing farmers monetary losses [6]. The early and accurate detection of these diseases play a vital role in the management of crop. In this work, we provide a CNN (Convolutional Neural Networks) based

disease detection system for dragon fruit plants which is a prominent deep learning technique in image level tasks. The dataset comprises a collection of dragon fruit images, both normal and diseased. Techniques like rescaling, normalization, and data augmentation are used before feeding to the model to enhance its performance and robustness. Important elements are automatically extracted from the photos by the CNN model, which then categorizes them into several disease groups. Accord- ing to experimental data, the suggested method can reliably detect infections at an early stage, allowing for prompt intervention and minimizing crop damage. By offering a quick, dependable, and automated way to track the health of dragon fruit plants and raise overall agricultural productivity, the technology can help farmers and agricultural professionals.
Keywords: *CNN, Deep Learning, Dragon Fruit, Disease Detection, Image Classification*

CSIT86: Brain Tumor and Types Detection

Ms. Ashwini Gunjal[1] , Prof. Prof. V. S. Dhongde[2]
SCSMCOE, A.Nagar
*Corresponding Author Email ID : ashwini13@gmail.com

Abstract: Brain tumors pose a major threat, so finding them early is key to better treatment and survival. Current ways to spot them involve radiologists checking MRI and CT scans by hand. This takes time, can be spotty, and might have mistakes. To fix these problems, this study shows off a system that uses deep learning to automatically find and sort brain tumors by checking medical images. This system puts image prep, cutting out parts, and sorting into one deep learning setup. First, MRI images are cleaned up to get rid of noise and look better. Then, a U-Net design is used to carefully cut out the tumor spots. After that, a CNN sorts the tumors into types like glioma, meningioma, and pituitary. The model is trained and checked using standard MRI datasets like BRATS. This makes sure it works well and is correct in both cutting and sorting jobs. Tests show this new method is over 97% correct. It cuts down on how long it takes to figure things out and lowers the amount of work needed by hand. The automatic setup makes diagnoses more trustworthy, shows where the tumor is, and helps radiologists make quicker, more right calls. By mixing AI with medical images, this work points out how deep learning can change healthcare diagnoses. This leads to early finds, smooth work, and better results for patients.
Keywords: *Brain Tumor Detection, Medical Imaging (MRI/CT), Deep Learning, U-Net, Convolutional Neural Network (CNN), Automated Diagnosis*

CSIT87: Data Science: Transforming Decision-Making in the Modern Era

Dr. V.V Patil[1], Janhavi Swapnil Kudal[2], Kalyani Satish Mhaske[3]
[1,2,3]Department of Artificial Intelligence and Data Science
Savitribai Phule University, Ahilyanagar
*Corresponding Author Email ID : vasudhapatil28@gmail.com

Abstract: Data Science has emerged as a transformative force across multiple industries, enabling organizations to leverage large and complex datasets for informed, data-driven decision-making. This paper investigates the methodologies, tools, and practical applications of data science in modern organizational contexts. By integrating predictive analytics, machine learning models, and big data platforms, organizations can improve operational efficiency, optimize strategic planning, and enhance decision accuracy. Several real-world examples are presented to illustrate the impact of data-driven approaches, including customer segmentation in marketing, predictive maintenance in manufacturing, and risk assessment in finance. The study also discusses key challenges in implementing data science solutions, such as data quality, model interpretability, privacy concerns, and integration with existing systems. Evaluation metrics for model performance are highlighted to ensure reliability and validity of predictions. Additionally, ethical considerations and governance frameworks are addressed to promote responsible use of artificial intelligence and analytics. By demonstrating how data science transforms decision-making, this research provides a comprehensive understanding of its practical benefits, limitations, and future directions. The findings suggest that organizations adopting data-driven strategies strategies gain competitive advantages, improve resource allocation, and support sustainable, evidence-based decisions.
Keywords: *Data Science, Machine Learning, Predictive Analytics, Big Data, Decision-Making*

CSIT88: Student Certificate Validation using Blockchain Technology

Nagude Vaishnavi Anil[1], Perane Tejaswini Sampat[2] , Uchit Shubhangi Sunil[3],
Umbarkar Ishwari Vitthal[4], Prof.Yogita.G.Suryavanshi[5]
Dr.Hemantkumar.B.Jadhav[6]
Department of Computer Engineering, Adsul technical Campus, Chas
*Corresponding Author Email ID : nagudevaishnavi29@gmail.com

Abstract: The present invention relates to a system and method for validating student certificates using blockchain technology to ensure secure, transparent, and tamper-proof academic credential management. The invention addresses the challenges of certificate forgery, manual verification delays, and dependence on centralized databases by utilizing a decentralized distributed ledger framework. In one embodiment, educational institutions generate digital student certificates and compute a unique cryptographic hash value for each certificate. The hash value, along with essential metadata such as institution identification and issuance timestamp, is recorded on a blockchain network through a secure transaction. The blockchain ledger ensures immutability, transparency, and traceability of issued credentials. The system further includes a verification module that enables authorized third parties, including employers and academic institutions, to validate certificates by uploading the digital document or scanning a QR code. The system recalculates the certificate's hash and compares it with the corresponding blockchain record. Any alteration in the certificate results in a hash mismatch, thereby detecting fraud or tampering. Smart contracts may be employed to automate certificate issuance, revocation, and status updates. Public-private key cryptography ensures secure access control and authentication of participating entities. By decentralizing certificate

validation, the proposed system enhances data integrity, reduces administrative overhead, and provides a reliable, scalable solution for global academic credential verification.

Key Words: *Research Paper,Technical Writing,Sciences,Engineering & Technology*

CSIT89: CloudShield: Enhancing Security for Direct File Uploads in Web Applications

ATHARV ACHYUT BHABARE[1] ,DHAKANE PRADIP BABASAHEB[2] ,SAMYAK SHASTRI KAMBLE[3], SAURABH BHIMA BHONDAVE[4], Prof.Yogita.G.Suryavanshi[5], Dr.Hemantkumar.B.Jadhav[6]

Department of Computer Engineering, Adsul technical Campus, Chas

*Corresponding Author Email ID : samyaa1412@gmail.com

Abstract: The present invention relates to a sophisticated security framework, CloudShield, designed to mitigate risks associated with direct file uploads in web applications. The system addresses critical vulnerabilities such as Remote Code Execution (RCE), Cross-Site Scripting (XSS), and malware injection by implementing a multi-layered, cloud-native validation architecture that operates independently of traditional, easily circumvented file-type checks. In one embodiment, CloudShield intercepts incoming file streams in a secure buffer zone before they reach the primary application server. The system performs Deep Packet and Content Inspection (DPCI), utilizing cryptographic hashing and signature analysis to verify the integrity of the file against known malicious patterns. By decoupling the upload process from the core storage layer, the framework ensures that unverified data remains isolated, preventing unauthorized execution of scripts or binaries. The system further incorporates an automated Metadata Sanitization and Transformation module. This component strips potentially harmful EXIF data and re-encodes files into secure, standardized formats to neutralize embedded threats. For high-risk file types, CloudShield utilizes a sandboxed detonation environment where the file is monitored for suspicious behavioral patterns. Integrated with Cloud-Edge Security protocols, the system provides real-time threat intelligence and logging, ensuring high availability and low latency. By employing this decentralized and proactive validation approach, CloudShield enhances the security posture of web platforms, reduces the burden of manual security audits, and provides a scalable, tamper-proof environment for user-generated content.keylogging. This adds an additional layer of authentication that is both user-friendly and resistant to data capture by malicious software.The integration of these two systems creates a multi-layered defense strategy. The dual-keypad mechanism reduces the risk of compromised keystrokes, while the visual authentication process ensures that even if keystrokes are captured, the authentication remains secure. The project aims to deliver a robust, secure, and intuitive authentication solution that enhances protection against keylogging and other cyber threats, providing a reliable means of securing sensitive information in various applications.

Keywords: *AIML, Cybercrime*

CSIT90: Advanced Development of Deep Learning-Based Information Abstraction Algorithms for Big Data Analysis: A Review

Vidhate S. P[1]

[2]SCSMCOE,A.Nagar

*Corresponding Author Email ID : 1412@gmail.com

Abstract: Deep Learning as a tool for information abstraction of Big Data. The rapid and widespread growth of big data across all sectors has created an imperative for developing highly sophisticated information abstraction algorithms that can extract useful and actionable insights from very large, complex, and heterogeneous data sets. Deep learning techniques are emerging as effective ways to enable the automatic extraction of features from big data; reduce dimensions; and perform semantic abstraction in the context of big data. This review will examine the latest developments in deep learning architecture and methods being used to abstract information from big data (i.e., autoencoders, CNNs, RNNs, Attention Mechanisms, Transformer-Based Models). Theoretical foundations, architectural advancements, training procedures, and various applications of each will be reviewed and discussed. In addition, challenges currently facing the use of deep learning to abstract information from big data such as scalability, interpretability, and computational efficiency will also be analyzed and promising areas for future research identified as this area continues to evolve at a rapid pace.

Keywords: *Deep learning, information abstraction, big data analysis, feature extraction, dimensionality reduction, neural networks.*

CSIT91: Scan To Park

Shruti Dongare[1] ,Smita Nat[2] ,Anjali Gohad[3],Gayatri Vairal[4]
T.P.Dhangekar[5]
[1,2,3,4,5]Shri Chhatrapati Shivaji Maharaj College of Engineering,Nepti,Ahilyanagar
Department Of Computer Engineering

Abstract: “Scan to Park” is a smart parking management system designed to simplify and automate vehicle entry, tracking, and parking operations using QR code technology. The system is developed with a Python-based backend and a web-based frontend using HTML and CSS, ensuring a lightweight and user-friendly interface. The system consists of three main modules: Entry, Exit, and Admin Dashboard. In the Entry module, the workflow begins when a user scans a QR code at the parking entry gate, which redirects them to a web-based homepage. The user is prompted to enter essential details such as owner name, vehicle number, and expected parking duration. Upon submission, the system generates a unique QR code that acts as a digital parking ticket containing a generated ticket ID. In the Exit module, the QR code is scanned again to validate the parking session, calculate the duration, and complete the exit process. The Admin Dashboard module displays all parking-related information, including vehicle details, entry time, exit time, and duration, allowing administrators to monitor and manage parking operations efficiently. The system reduces manual intervention, minimizes errors, and enhances efficiency in parking operations. It provides a scalable, secure, and contactless solution suitable for modern smart parking environments.

Keywords*: QR Code Technology,AWS,Ticket Generation, Web Application, Admin Dashboard , Vehicle Tracking,Smart Entry&Exit*.

CSIT91: AgroVision: Predictive Analytics for Onion Crop Yields In Maharashtra

Dale Anushka[1] , Walture Shravani[2] , Lawande Sakshi[3] , Patil Yash[4], Prof. S.P.Vidhate[5]
[1,2,3,4,5] Assistant Professor, Department of Computer Engineering, SCSMCOE, Ahilyanagar.
ABSTRACT: This paper proposes a machine learning framework integrating UAV multispectral indices, irrigation practice, weather data, and soil features to predict onion yield (t/ha) in Maharashtra, India. Using field records and UAV data collected from ICAR?DOGR research plots, the model achieved high accuracy in forecasting yields between 20–40?t/ha, validated against government?reported average yields (~15?t/ha) and farm?level observations.
Keywords: *Onion Yield Prediction ,Maharashtra Agriculture ,Precision Agriculture ,Machine Learning ,XGBoost Regression ,Remote Sensing ,UAV Multispectral Data, Irrigation, Management, Crop Modeling, Data-Driven Agriculture*

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Track-3: Civil Engineering

CIVIL01: AI in Irrigation: Enhancing Water Efficiency and Sustainable Agriculture

Nikita Rahul Kadu[1], Vaishali Somnath Mhaske[2]
[1]Lecturer, Civil Engineering Department, Padmashri Dr. Vitthalrao Vikhe Patil Institute of Technology & Engineering, Polytechnic, Loni
[2]Lecturer, Civil Engineering Department, Padmashri Dr. Vitthalrao Vikhe Patil Institute of Technology & Engineering, Polytechnic, Loni
[*]Corresponding Author Email ID: nikitayeole30@gmail.com

Abstract: Inefficient irrigation practices represent major barriers to sustainable agriculture, particularly in semi-arid regions. Traditional irrigation scheduling methods rely on manual observation or fixed time intervals, which often lead to over-irrigation, under-irrigation, and unnecessary consumption of energy and water resources. Artificial Intelligence (AI) has emerged as a transformative technology capable of improving irrigation efficiency through predictive analytics, machine learning algorithms, and intelligent automation. This study investigates the role of AI-based irrigation systems in optimizing water distribution and improving crop productivity. The paper proposes an integrated architecture combining soil moisture sensors, Internet of Things (IoT) devices, machine learning models, and automated irrigation controllers. The methodology involves data acquisition from environmental sensors, preprocessing of meteorological and soil parameters, prediction of crop water requirements using Artificial Neural Networks, and decision optimization through reinforcement learning-based irrigation scheduling. Experimental simulations demonstrate that AI-driven irrigation systems can reduce water usage by approximately 25–35% compared with conventional irrigation methods while improving crop yield by nearly 15%. The results also highlight the importance of predictive weather analysis and soil moisture monitoring for dynamic irrigation planning. Furthermore, the study evaluates economic feasibility, scalability, and challenges associated with deploying AI technologies in agricultural environments. The findings indicate that AI-enabled irrigation can significantly contribute to precision agriculture, climate-resilient farming, and long-term sustainability in global food production systems.

***Keywords**: AI, demonstrate, neural network*

CIVIL02: STRENGTHENING OF REINFORCED CONCRETE BEAM-COLUMN JOINTS USING FIBER-REINFORCED POLYMER (FRP) SHEETS

Mahesh .A. Nawale[1] , Dr. Parikshit Joshi[2]
[1]Research Scholar, Civil Department, Oriental University, Indore
[2]Professor, Civil Department, Oriental University, Indore
[*]Corresponding Author Email ID: nawale.mahesh123@gmail.com

Abstract: Strengthening existing reinforced concrete (RC) structures has become a vital necessity in India’s construction sector, primarily due to escalating load specifications and frequent updates to national design codes. Within moment-resisting frames, the beam-column

joint serves as the most critical zone for structural integrity. Despite the common design assumption of joint rigidity, these connections often experience intense shear forces during seismic events that exceed their capacity. This discrepancy frequently results in brittle shear failure—a catastrophic collapse mechanism that is unacceptable in seismic-prone regions. This study evaluates the enhancement of key structural parameters in RC joints through the application of Carbon Fiber Reinforced Polymer (CFRP) and Glass Fiber Reinforced Polymer (GFRP) sheets. By addressing the performance deficiencies highlighted by updated Indian code provisions, the research investigates how these fiber-reinforced polymers mitigate brittle failure modes and improve ductility. The findings provide critical insights into bolstering the seismic resilience and ultimate load capacity of existing structural frameworks using advanced composite materials.
***Keywords**: Beam-column joints, Carbon Fiber (CFRP), Glass Fiber (GFRP), Ductility, Fiber-reinforced polymers (FRP), Reinforced concrete (RC), Shear strength, Seismic retrofitting, Ultimate load.*

CIVIL03: INTEGRATED BIM IMPLEMENTATION FOR AUTOMATED CLASH DETECTION AND DYNAMIC COSTING ON A LIVE CONSTRUCTION PROJECT

Ms. Akanksha .P Athare[1]

[1]Department of Civil Engineering , Pimpri Chinchwad College Of Engineering Nigdi, Pune – 411044

[*]Corresponding Author Email ID: akankshaathare28@email.com

Abstract: The construction industry continues to struggle with design clashes, coordination gaps, cost overruns, and delays caused by fragmented 2D documentation and disconnected workflows. This study implements an integrated Building Information Modelling (BIM) approach on a live residential project to address these challenges. Architectural, Structural, and MEP models were developed in Autodesk Revit and integrated into a federated model for automated clash detection using Navisworks Manage. A total of 704 MEP vs MEP clashes and 489 Architecture vs MEP clashes were identified and resolved through iterative coordination. The study also links the BIM model with quantity extraction to support dynamic cost updates. Results show that BIM-based coordination reduces rework, enhances design accuracy, and improves multidisciplinary communication. The research confirms BIM as a practical tool for early clash avoidance, cost optimization, and informed project decision-making.
***Keywords**: BIM, Clash Detection, Navisworks, Revit, Dynamic Costing, MEP Coordination, Integrated BIM Workflow.*

CIVIL04: Parametric Evaluation of Seismic Performance of a Muti-storey Reinforced Concrete Building

C. Hudson Gnanadurai[1] , J.Revathy[1*]

[1]Department of Civil Engineering, B.S. Abdur Rahman Crescent Institute of Science and Technology, Chennai, Tamil Nadu, India

[*]Corresponding Author Email ID: hudson18.c@gmail.com

Abstract: This paper presents the analysis and design of a multi-storey reinforced concrete (RC) building using ETABS in accordance with the relevant Indian Standards. The main goal is to assess the structural behavior of the building under seismic loading and ensure the safety of its structural components. In this study, the seismic response of a 10-storey reinforced concrete building is investigated under three earthquake scenarios. The building is modelled as a Special Moment Resisting Frame (SMRF) system and analyzed under three different soil conditions to examine their influence on the seismic performance of the structure. The study uses both Time History Analysis (THA) and Response Spectrum Analysis (RSA) to evaluate the seismic behavior of the structure. THA is performed by applying real earthquake ground motion records, while RSA is carried out by considering different seismic zone factors and soil conditions as per the design code. The resulting structural responses, including storey drift, storey displacement, storey shear, and overturning moment, provide an assessment of the overall seismic stability of the structure. The results from both analyses are compared to provide a comprehensive understanding of the structural response under different seismic conditions.
***Keywords**: Seismic, Response spectrum analysis, Time history analysis, Storey displacement, Storey drift, Shear, Overturning moment*

CIVIL05: Performance Evaluation and GEP-Based Prediction of Hardwood Biochar Modified Bitumen for Sustainable Pavement Applications

S Mahalakshmi[1], Revathy Jayaseelan[1*]

[1]Department of Civil Engineering, B.S. Abdur Rahman Crescent Institute of Science & Technology, Chennai, Tamil Nadu, India

[*]Corresponding Author Email ID: revathyj@crescent.education

Abstract: The major challenge is to reduce the use of fossil resources and lower carbon emissions in road construction. Which are in greater need of sustainable pavement materials? This has inspired the exploration of bio-based modifiers for bitumen binders. In this study, hardwood biochar (HBC) derived from biomass through controlled laboratory pyrolysis was used as a modifier for VG30-grade bitumen. The biochar was incorporated into the binder at 5%, 10%, 15%, and 20% by weight of bitumen using a high-shear mixing process. Conventional binder tests, including penetration and softening point, were conducted to evaluate the influence of HBC on the physical properties of the binder. The microstructural characteristics of HBC were analyzed using scanning electron microscopy (SEM) to examine its surface morphology and interaction potential with the binder. Furthermore, Gene Expression Programming (GEP) was involved in developing predictive models for penetration and softening point values based on biochar content. The results show that the addition of HBC reduces penetration values and increases the softening point temperature, indicating improved stiffness and enhanced high-temperature resistance of the binder. This provides adaptable solutions for various climatic and loading conditions in road construction. The developed GEP models showed good predictive capability, and the findings suggest that HBC can serve as a sustainability and long-term viability of pavement construction.

***Keywords**: Biochar, Modified Bitumen, Sustainable Pavement, Gene Expression Programming, VG-30 Bitumen, Binder Modification*

CIVIL06: Quantifying the Influence of Roadside Parking on Urban Traffic Speeds Using Regression Modeling

Shishodiya Ghanshyam Singh[1], Challa Prathyusha[2]
[1]Department of Civil Engineering, MIT Art, Design & Technology University, Pune - 412201, Maharashtra, India
[2]Department of Civil Engineering, KSRM College of Engineering, Kadapa, YSR District-516005, Andhra Pradesh, India
[*]Corresponding Author Email ID: shishodiyaghanshyam@gmail.com

Abstract: Roadside parking is one of the major side friction factor on urban arterials that leads to reduction in the effective carriageway width, lowering speed, increased travel times thus ultimately leading to diminished capacity, efficiency and poor level of service. Quantification of roadside parking is carried out in this study which analyzes the vehicular speeds for different modes of vehicles plying on the Indian roads. Videographic based data collection was carried in Chikmagalur, which is a major tourist spot in Karnataka state. It is a four-lane undivided urban arterial road exhibiting varying levels of on-street parking throughout the day. The extracted data includes the vehicular speed, number of vehicles and parked vehicles for each five minutes interval, required for the development of traffic speed forecasting approach using the multivariable linear regression. The developed model with R^2 = 0.90 indicates that the approach exhibits strong robust explanatory power and precise forecasting with mean absolute percentage error of 2.8%. The findings of the study demonstrate that increased parking volume will substantially lower the effective carriageway width and operating pace of the vehicles. The results are highly pertinent to urban and traffic planners in optimizing curb space allocation and increasing vehicle movement efficiency on the congested high-density corridors.

***Keywords**: Roadside parking, Carriageway width, Vehicular Speed, Videographic data, Regression model*

CIVIL07: STUDY OF NATURAL AND COMPOSITE COAGULANTS, THEIR EFFECTIVE PRAPORTIONS FOR SUSTAINABLE SOLUTION IN TREATMENT OF WASTE WATER

Pradim Chandane[1] , Prof. (Dr.)Parikshit Joshi[2]
[1]Research scholar, Department of Civil Engineering, Oriental University, Indore,
[2]Professor Department of Civil Engineering, Oriental University, Indore
[*]Corresponding Author Email ID: pradeemchandane42161@gmail.com

Abstract: The treatment of wastewater has become an increasingly important global challenge due to growing industrialization, urbanization, and the associated environmental pollution. Among various treatment methods, coagulation is a widely adopted technique for the removal of suspended solids, colloidal particles, and dissolved contaminants. However, traditional coagulation agents often exhibit limitations in terms of efficiency, cost, and environmental impact. Recent studies have focused on the development and application of composite coagulating agents, which combine multiple materials to enhance coagulation performance and

sustainability. This study aims to evaluate the efficacy of composite coagulants in wastewater treatment, focusing on their ability to remove contaminants, improve treatment efficiency, and reduce environmental impact. A range of composite coagulants, including combinations of inorganic salts (such as alum or ferric chloride) with natural or synthetic polymers, bio-based materials, and nanoparticles, were synthesized and tested in laboratory-scale experiments. Key performance indicators, including turbidity reduction, chemical oxygen demand (COD) removal, and pH stability, were assessed under varying operational conditions.

***Keywords**: Coagulants, Composition*

CIVIL08: Smart Traffic Management System Using IoT and Artificial Intelligence

Gaurav Pradip Walhekar[1], Rahul Dattatray Vikhe[2], Sandip Kacharu Gorde[3]

[1,2]Lecturer, P. Dr. V. Vikhe Patil Institute of Technology Engineering (Polytechnic) Loni

[3]HOD CE, P. Dr. V. Vikhe Patil Institute of Technology Engineering (Polytechnic) Loni

[*]Corresponding Author Email ID: gauravpwalhekar@gmail.com

Abstract: Rapid urbanization and the exponential growth in vehicle ownership have led to severe traffic congestion, increased travel time, and higher environmental pollution in metropolitan areas. Traditional traffic management systems often rely on fixed-time signal controls, which are inefficient under dynamic traffic conditions. This paper proposes a smart traffic management system that integrates Internet of Things (IoT) devices and Artificial Intelligence (AI) techniques to optimize traffic flow in real time. The proposed system utilizes IoT-enabled sensors, cameras, and connected devices to continuously collect traffic data such as vehicle density, speed, and congestion levels at intersections. This data is transmitted to a centralized processing unit where AI algorithms, including machine learning models, analyze traffic patterns and predict congestion trends. Based on the analysis, traffic signal timings are dynamically adjusted to minimize delays and improve traffic efficiency. Additionally, the system incorporates features such as emergency vehicle prioritization, incident detection, and adaptive rerouting to further enhance road safety and reduce congestion. A simulation-based evaluation demonstrates that the proposed approach significantly reduces average waiting time, fuel consumption, and vehicular emissions compared to conventional systems. The study highlights the potential of integrating IoT and AI technologies in developing intelligent transportation systems for sustainable and efficient urban mobility..

***Keywords**: Smart Traffic Management, Internet of Things (IoT), Artificial Intelligence (AI), Intelligent Transportation Systems (ITS), Traffic Congestion, Machine Learning, Real-Time Monitoring, Adaptive Traffic Signals, Urban Mobility, Sustainable Transportation*

CIVIL09: A study on influence of steel slag in concrete block

Akshay Nimase[1], Siddhi Todmal[2], Supriya Holkar[3] Tejswini Kulthed[4], Kartika Dhase[5], Kausar Shaikh[6]

[1]Assistant Professor, Department of Civil Engineering, SCSMCOE, Ahilyanagar, Maharashtra, India

[2]Department of Civil Engineering SCSMCOE, Ahilyanagar, Maharashtra, India

[*]Corresponding Author Email ID: holkarsupriya364@gmail.com

Abstract: The rapid growth of the construction industry has increased the demand for natural aggregates used in concrete production. At the same time, large quantities of industrial waste such as steel slag are generated from steel manufacturing plants, creating environmental and disposal challenges. This study investigates the potential use of steel slag as a partial replacement for natural aggregates in concrete block production. Concrete specimens were prepared by replacing natural aggregate with steel slag at different percentages of 0%, 5%, 10%, 15%, and 20%. The performance of the concrete blocks was evaluated by conducting compressive strength tests after curing periods of 3 days and 7 days. The experimental results indicate that the compressive strength of concrete blocks varies with the percentage of steel slag replacement. At 3 days of curing, the compressive strength values ranged approximately from 13.87 MPa to 18.16 MPa for different replacement levels. After 7 days of curing, the strength values increased significantly, ranging from about 16.27 MPa to 26.85 MPa. The results demonstrate that the inclusion of steel slag improves the mechanical properties of concrete blocks due to the angular texture and higher density of slag particles, which enhance the bonding within the concrete matrix. The result concludes that steel slag can be effectively utilized as a partial replacement material in concrete blocks without compromising strength. Proper utilization of steel slag not only improves concrete performance but also contributes to sustainable construction by reducing industrial waste and conserving natural resources. Therefore, the use of steel slag in concrete production presents an environmentally friendly and economically beneficial alternative for modern construction practices.
***Keywords**: Steel Slag, Concrete Block, Compressive Strength*

CIVIL10: Performance of concrete containing GGBFS and RCA at Elevated Temperature

Undare R.S.[1], Jadhav S.S.[2], Agre S.V.[3], Ghodake A.R.[4], Dr. Palaskar S.M.[5]
[1,2,3,4,5]Department of Civil Engineering, SCSMCOE, Ahilyanagar, Maharashtra, India
[*]Corresponding Author Email ID: ashutoshghodake143@gmail.com

Abstract: The construction sector consumes vast natural resources and generates significant demolition waste, while cement production contributes heavily to CO2 emissions. This study explores sustainable concrete by replacing coarse aggregate with recycled concrete aggregate (RCA) and cement with Ground Granulated Blast Furnace Slag (GGBFS). Experimental investigation assesses workability, compressive strength, and durability. The result assesses the residual strength, thermal stability, and degradation behavior of concrete. Scanning Electron Microscopy (SEM) analysis is conducted to examine microstructural changes, including surface morphology and crack development, providing insight into the effects of elevated temperature on material properties. This study aims to promote sustainable and thermally resistant concrete for modern construction applications.
***Keywords**: Ground Granulated Blast Furnace Slag (GGBFS), Recycled Concrete Aggregate (RCA), SEM (Scanning Electron Microscopy)*

CIVIL11: 3D BIM model of multistorey green building

Girme S.D[1], Rajale S.S[2], Darode S.B[3], Chaure O.S[4], Yewale V.[5]

[1,2,3,4,5]Department of Civil Engineering, Shri Chhatrapati Shivaji Maharaj college of engineering Nepti , Ahilyanagar, Maharashtra, India
[*]Corresponding Author Email ID: sharvarigirme1234@gmail.com
Abstract: This project focuses on the development of a 3D Building Information Modeling (BIM) model of a Multistorey green building, integrating sustainability principles with modern design and construction workflows. The 3D BIM model not only enhances design accuracy but also facilitates cost estimation using schedules in Revit software, Energy optimization, reducing construction waste and improving project efficiency. By merging BIM technology with sustainable design strategies, this project demonstrates an innovative approach to planning and executing modern multistorey green building that align with global sustainability goals.
Keywords*: 3D Modeling, Multistorey Building, Green Building, Sustainable Design, Energy Efficiency, Ecofriendly materials, Cost optimization.*

CIVIL12: Use of Crushed Granite Stone as Partial Replacement of Coarse Aggregate in Concrete

Kadam A.M[1] , Kadam S.M[2] , Sayyad N.S[3] , Pawar S.A[4]
[1,2,3,4]Department of Civil Engineering, Shri Chhatrapati Shivaji Maharaj college of engineering Nepti , Ahilyanagar, Maharashtra, India
[*]Corresponding Author Email ID: abhishekkadam7@gmail.com
Abstract: Concrete is one of the most widely used construction materials in civil engineering, in which aggregates constitute a major portion of the total volume. Due to rapid urbanization and continuous infrastructure development, the demand for natural aggregates has increased significantly, leading to depletion of natural resources. Hence, there is a need to explore alternative materials which can partially replace natural aggregates without affecting the performance of concrete. In the present study, an attempt has been made to utilize crushed granite stone as a partial replacement of coarse aggregate in concrete. The granite material used is generally obtained as waste from quarrying and stone cutting operations, and its disposal is a major concern. The use of such waste material in concrete can help in reducing environmental impact as well as construction cost. Concrete specimens were prepared by replacing natural coarse aggregate with crushed granite stone at 0%, 10%, 20%, 30%, and 40%. Standard mixing, casting, and curing procedures were followed. The specimens were tested for compressive strength after 7 and 28 days of curing. During casting, no major segregation or bleeding was observed, although workability was found to decrease slightly with increase in granite content due to its angular texture. From the test results, it was observed that compressive strength increased gradually up to 20–30% replacement, after which a slight reduction was noticed at 40% replacement. However, the strength values at higher replacement levels were still within acceptable limits for practical use. The results clearly indicate that crushed granite stone can be effectively used as a partial replacement of coarse aggregate in concrete. This approach not only helps in conserving natural resources but also promotes sustainable and economical construction practices.
Keywords: *Concrete*

CIVIL13: Non-Destructive Testing and Chemical Treatments for Corrosion Protection Using Epoxy and Anti-rust Agents

Avinash Ghogare[1], Jitendra Gite[2], Aditya Kale[3], Mohit Unavane[4]
[1,2,3,4]Department of Civil Engineering Pravara rural Engineering College, Loni
[*]Corresponding Author Email ID: ghogareau@pravaraengg.org.in

Abstract: Corrosion of steel reinforcement is a primary cause of structural failure in reinforced concrete (RC) elements, leading to significant economic losses and safety hazards. This study investigates the effectiveness of epoxy coatings and antirust agents in enhancing the durability and bond strength of RC structures. M30 grade concrete specimens reinforced with Fe550 steel bars (16mm and 20mm diameters) were subjected to three environmental conditions: normal water curing, accelerated acid attack (3.5% HCl), and elevated temperature exposure (400°C). The structural integrity was assessed using Non-Destructive Testing (NDT) specifically Ultrasonic Pulse Velocity (UPV) and Rebound Hammer tests and destructive Pull-Out tests to determine bond strength. Results indicate that epoxy-coated bars exhibited superior bond strength (up to 12.87 N/mm² for 16mm bars) compared to uncoated bars (7.76 N/mm²) under normal conditions. While acid attack and thermal stress significantly reduced the UPV values and bond strength across all specimens, epoxy-coated samples consistently maintained higher residual strength than uncoated specimens. This study confirms that epoxy coatings provide a robust barrier against aggressive environmental factors, significantly extending the service life of civil infrastructure.

Keywords: *Corrosion Protection, Epoxy Coating, Bond Strength, Non-Destructive Testing (NDT), Acid Attack, Elevated Temperature.*

CIVIL14: Transfer of Microplastics from River Systems to Underground Water Resources: Mechanisms, Impacts, and Research Needs

Prasad G. Nikam[1], Dr. Parokshit Joshi[2], Dr. Chaitanya Mishra[3]
[1,2,3]Civil Engineering Department, Oriental University, Indore, India

[*]Corresponding Author Email ID: prasad.nikam999@gmail.com

Abstract: This review examines the pathways through which microplastics travel from rivers into groundwater. Microplastics originate from the breakdown of larger plastic materials as well as various human activities. These particles persist in the environment and can migrate through water, soil, and rock layers into underground water sources. Several mechanisms contribute to microplastics entering groundwater. They can move through the pores between soil and rock particles, enter when river water seeps through riverbanks, or during the natural replenishment of groundwater. The movement of microplastics depends on their characteristics, soil composition, water flow, and environmental conditions. Detecting microplastics in groundwater is essential to understanding the extent of contamination. However, sampling and analysis present significant challenges. The presence of microplastics in groundwater poses risks to both wildlife and human health. This review highlights the current gaps in knowledge regarding the transport and fate of microplastics in groundwater

systems. It emphasizes the need for collaborative research efforts and standardized methodologies to better understand microplastic pollution and develop strategies to protect our water resources. Addressing microplastic contamination is critical to safeguarding the quality of our groundwater.

Keywords: *Microplastics*

CIVIL15: To Determine Index Properties and Engineering Properties of Steel Slag Amended Soil

Akshay Rohidas Nimase[1]

[1]Assistant Professor, Department of Civil Engg, SCSMCOE, Ahilyanagar, Maharashtra, India

[*]Corresponding Author Email ID: akshay.nimase@scoea.org

Abstract: This study is carried out to determine the index properties and engineering properties of steel slag and its effect when mixed with landfill soil. Steel slag is an industrial by-product that can be reused in geotechnical applications due to its good strength and stability characteristics. In this work, different proportions of steel slag such as 5%, 10%, 15%, and 20% were mixed with soil to study the variation in its properties. The index properties like specific gravity, Atterberg limits, grain size distribution, and pH were determined through laboratory testing. The results showed that the specific gravity of soil increases with the addition of steel slag due to its heavier particles. At the same time, the liquid limit and plastic limit values decreased, indicating reduced plasticity and improved workability of the soil. Engineering properties were evaluated using standard Proctor test, direct shear test, and permeability test. The compaction results indicated that maximum dry density increased while optimum moisture content decreased as the percentage of steel slag increased. Shear strength parameters such as cohesion and angle of internal friction also improved, showing better resistance against shear failure. The permeability of the soil showed controlled variation, which is suitable for landfill applications. Overall, the results of this study indicate that steel slag significantly improves both index and engineering properties of soil. Hence, it can be effectively used as a stabilizing material in geotechnical works and landfill cover systems, providing strength, stability, and an eco-friendly solution for waste utilization.

Keywords: *Steel Slag; Index Properties; Engineering Properties*

CIVIL16: Utilization of plastic waste in rigid pavement

Garudkar Sanket[1] , Kardile Sayali[2] ,Nimse Shraddha[3], Shaikh Parvin[4]

[1,2,3,4]Department of Civil Engineering, Shri Chhatrapati Shivaji Maharaj college of engineering Nepti , Ahilyanagar, Maharashtra, India

[*]Corresponding Author Email ID: shraddhanimse8@gmail.com

Abstract: The quick rise in elastic unused generation positions a severe eco-friendly task universal. This project discovers an innovative and workable explanation by using recycled plastic unused as a partial replacement for coarse aggregate in M30 grade concrete suitable for rigid concrete structure. Using Ordinary Portland Cement (OPC), 20 mm coarse aggregate, and the given parameters (workability 75 mm slump, water-cement ratio 0.45, 0.5% admixture), the concrete mix is designed as per IS 10262:2019 guidelines. Plastic waste replaces coarse

aggregate by weight at four levels: 7%, 11%, 15%, and 20%. The primary objectives are to reduce natural aggregate consumption, promote circular economy principles, minimize landfill burden, and develop eco friendly rigid concretes with comparable or improved properties such as reduced self-weight and enhanced flexibility. The step-by-step mix design yields practical proportions for laboratory and field trials. This approach not only addresses plastic pollution but also supports workable infrastructure development in road construction, offering a cost-effective and environmentally responsible alternative without compromising the structural requirements of rigid pavements.

Key words: *Rigid pavement, plastic waste coarse aggregate, Ordinary Portland cement (OPC).*

CIVIL17: Sustainable Utilization of Processed Sugarcane Bagasse Ash as a Supplementary Cementitious Material: Effects of Calcination Condition and Mechanical Grinding on Pozzolanic Properties

Atul B. Jondhale[1], Dr. Sachin B. Mulay[2], Dr. Pramod K. Kolase[3]

[1] Research Scholar, Department of Civil Engineering, Sandip University, Nashik (MS), India.

[1] Assistant Professor, Department of Civil Engineering, PREC Loni, SPPU Pune (MS), India.

[2] Professor & Research Supervisor, Department of Civil Engineering, Sandip University, Nashik (MS), India.

[3] Associate Professor & Research Co-Supervisor, PREC Loni, SPPU Pune (MS), India.

[*]Corresponding Author Email ID: atul009jondhle@gmail.com

Abstract: The development of ecofriendly building material is currently become crucial in the evolution of sustainable infrastructure. This research work examines the valorization of sugarcane bagasse ash (SBA), an abundant agro industrial byproduct of sugar industry as a supplementary cementitious material (SCM). These raw SBA was processed by controlled calcination in electric furnace at temperature of 600 ^{0}C, 700 ^{0}C, 800 ^{0}C, 900 ^{0}C, 1000 ^{0}C for 2 hours and 4 hour's time intervals followed by mechanical grinding in ball mill for 2000 revolutions. The impact of these processing parameters on cementitious properties of SBA was studied through specific gravity, specific surface area, setting times and heat of hydration. XRD analysis confirmed the partial amorphization of silica at lower temperature for 2 hours' time intervals indicating improved pozzolanic activity. FESEM images shows irregular particle shapes promoting better cementitious interface. The result confirms that optimum calcination at 700 ^{0}C for 2 hours followed by mechanical grinding at 2000 revolution led to improved performance and early age hydration. In contrast, higher temperature and 4 hours calcination duration led to particle sintering and decreased reactivity. The results demonstrate a feasible and environmentally friendly way to turn waste bagasse ash into a useful partial cementitious material that fulfill the needs of modern infrastructures. Integrating the SBA material bring into line with the goals of a circular economy and reduces the environmental effects associated with production of cement. This study encourages sustainable innovation in civil engineering by providing an effective approach to decrease carbon emissions in cementitious systems.

Keywords*: Ecofriendly, Sustainable infrastructure, Sugarcane bagasse ash, Supplementary cementitious material, Calcination, circular economy.*

CIVIL18: Study on Structural Behavior of Self Compacting Concrete Confined With Ferrocement: A Review Paper

Nana M.Mulik [1] , Dr.Parikshit Joshi[2] Dr.Chaitanya Mishra[3]

[1]Department of Civil Engineering Research Scholar Oriental University Indore, India

[2,3]Department of Civil Engineering Oriental University Indore, M.P., India.

[*]Corresponding Author Email ID: nanamulik78@gmail.com

Abstract: This review paper provides a comprehensive analysis of the integration of ferrocement with self-compacting concrete (SCC) to enhance structural performance. The review highlights the advantages of SCC, such as improved workability and reduced segregation, and examines how the addition of ferrocement can further increase tensile strength, ductility, and durability. The paper summarizes various experimental studies, demonstrating how ferrocement confinement can effectively mitigate issues like cracking and enhance load-bearing capacity in SCC structures. It discusses the mechanics behind this improvement, including stress distribution and the role of ferrocement's mesh reinforcement. Furthermore, the review addresses design considerations, potential applications, and the economic aspects of utilizing this composite material in construction. Overall, the paper underscores the potential benefits of combining SCC with ferrocement, suggesting that this innovative approach could lead to more resilient and sustainable concrete structures. The findings aim to guide future research and practical applications in civil engineering.

Keywords: *SCC,Ferrocement, RCC, PCC*

CIVIL19: Experimental Investigation on Recycled Concrete aggregate with bituminous Mixture

Krishna Ambilwade[1], Ranjit Shelke[2],Shriram Ban[3],Harshal Suryawanshi[4], Girish Patil[5]

[1,2,3,4,5]Department of Civil Engineering, Shri Chhatrapati Shivaji Maharaj College of Engineering, Nepti

*Corresponding author mail: kirsh@gmail.com

ABSTRACT: Now a days increasing population in the world, then the constructions for living humans could be more like houses, bridges, apartments etc. Rapidly wastage of concrete aggregates is increasing. To minimize the wastage of concretes by utilizing properly with hot mix asphalt mixtures for construction of road pavements. The objective of this study is to understand the vitality of using RCA for the construction of bituminous pavements. Recycled concrete aggregates (RCA) materials are resulted from milling process. In this study samples of Recycled concrete aggregates (RCA) materials were collected and analysed for suitability of their usage in flexible pavements. Their characteristics including gradation, Aggregate Impact value, Aggregate Crushing value, Specific gravity, Flakiness Elongation Index, Loss Angles Abrasion value, and Water absorption were determined and compared to the standard specifications. The present study deals with evaluation of Marshall Stability, de- formation and the moisture damage resistance values of HMA which is mixed with RCA in varying proportions. The Marshall mix design method was adopted in this study to determine the optimum binder content (OBC) for the asphalt mixes containing five aggregate combination

with RCA contents of 0, 10, 20, 30 percent was found to be 5, 5.5, 6 percent which is optimum respectively.

Keywords: *Concrete, HMA*

CIVIL20 : Hybrid Road Energy Harvesting System Using Piezoelectric Sensors and Mechanical Plate Mechanism for Sustainable Power Generation

Tanuja Vyavhare[1], Rutuja Udage[2]

[1]Department of R&A, Shri Chhatrapati Shivaji Maharaj College of Engineering, Ahilyanagar

[2]Department of Civil Engg., Shri Chhatrapati Shivaji Maharaj College of Engineering, Ahilyanagar

Corresponding authors mail: Rutuja@gmail.com

ABSTRACT: The rapid increase in global energy demand and depletion of conventional energy sources has encouraged researchers to explore alternative renewable energy technologies. One of the potential sources of untapped energy is the mechanical pressure exerted by vehicles moving on roads. This paper proposes a hybrid road energy harvesting system that converts vehicular pressure into electrical energy using a combination of piezoelectric sensors and spring-based mechanical plates. When vehicles pass over the road surface, the applied load compresses the mechanical plate system which activates piezoelectric sensors to generate electric voltage. The generated electricity is rectified, stored in batteries, and utilized for low-power applications such as street lighting, traffic signals, and highway monitoring systems. The proposed design improves efficiency by integrating mechanical displacement amplification with piezoelectric energy conversion. The system provides an eco-friendly and sustainable method for harvesting wasted mechanical energy from traffic without affecting vehicle movement. The research evaluates system design, working principle, energy conversion process, and potential implementation in smart transportation infrastructure.

Keywords: *Piezoelectric Energy Harvesting, Road Energy Generation , Smart Transportation Systems, Renewable Energy Technology, Mechanical Pressure Conversion*

Track-4: Electronics & Electrical Engineering

EE01: Design and Implementation of a Cyber-Resilient Block chain Framework for Secure Healthcare Data Management

Mr.Pawar Bharat Ramdas[1], Dr. Sathyasri B.[2]

[1]Research Scholar, Department of Electronics and Communication Engineering, Vel Tech Rangarajan Dr. Sagunthala R&D Institute of Science and Technology, 400 feet Outer Ring Road, Avadi, Chennai – 600062, Tamil Nadu, India.

[2]Associate professor, Department of Electronics and Communication Engineering, Vel Tech Rangarajan Dr. Sagunthala R&D Institute of Science and Technology, 400 feet Outer Ring Road, Avadi, Chennai – 600062, Tamil Nadu, India.

[*]Corresponding Author Email ID: bharat.pawar14@gmail.com

Abstract: In order to handle the important issues of data security, privacy, and interoperability in healthcare systems, this research article suggests a blockchain-based architecture for handling Electronic Health Records (EHRs). Fragmentation, security breach risk, and limited patient control over personal health data define the present EHR architecture. This paper looks at how blockchain technology may provide patient-centric access control and offer a distributed, unchangeable, transparent answer for EHR administration. By means of thorough investigation of both primary and secondary data, this study shows that blockchain implementations may considerably lower data breaches by 67%, increase interoperability among healthcare providers by 78%, and raise patient satisfaction with data ownership by 84%. Smart contracts for automated access control, cryptographic methods for safe data storage, and consensus systems for verifying health record transactions comprise the suggested architecture. By building trust, increasing operational efficiency, and enabling patients with more control over their health information, the results imply that blockchain-based EHR systems may transform healthcare data management.

***Keywords**: Blockchain, Electronic Health Records (EHR), Healthcare Data Security, Interoperability, Smart Contracts, Patient Data Ownership, Decentralized Healthcare, Medical Data Privacy*

EE02: IoT Based HealthCare System For Paralysis Patient

Shrutika Shriniwas Battin[1], Janhvi Devendra Ballewar[2], Gayatri Sanjay Bodhe[3], Mr.S.A.Aher[4]

[1,2,3,4]Department of Electronics and Telecommunication Engineering Pravara Rural Engineering College, Loni

[*]Corresponding Author Email ID: shrutikabattin@gmail.com

Abstract: Continuous health monitoring and timely alerts are critical for paralysis patients who rely on caregivers. This paper presents an original IoT-enabled system using an ESP32 microcontroller that integrates multiple sensors to monitor key vitals in real time and automatically notify caregivers of emergencies. The system continuously measures physiological parameters – heart rate and SpO_2 via a MAX30102 pulse oximeter, body temperature via a non-contact MLX90614 sensor, and ambient temperature/humidity via a DHT11 – while an MPU6050 inertial sensor detects falls and gesture-based emergency signals. Data are transmitted wirelessly to a smartphone-based Blynk dashboard for real-time

visualization.. In an abnormal event (e.g., a fall or critical vital), a buzzer sounds immediately and a GSM SIM800L module (with GPS NEO-6M) transmits an emergency SMS with location to caregivers. Using IoT ensures that patient data is accessible remotely. Experimental evaluation shows that the device reliably tracks vitals within a few percent of medical devices and triggers alerts rapidly, demonstrating its potential to improve patient safety and reduce response time in emergencies.

***Keywords**: IoT, ESP32, Health monitoring, MAX30102, MLX90614, MPU6050, Fall detection, Blynk, GSM, GPS, Emergency alert*

EE03: Automatic Ration Dispensing System

Ranjeet Ashok Kadu[1], Samarth Dipak Gavali[2], Vedant Sudhir Badgujar[3], Milind Arvind Bhalerao[4]

[1,2,3,4]Department of Electronics and Telecommunication Engineering Pravara Rural Engineering College (PREC), SPPU Loni, Maharashtra, India

[*]Corresponding Author Email ID: ranjeetkadu181185@gmail.com

Abstract: Public Distribution Systems (PDS) are essential for maintaining food security by providing subsidized food grains and commodities to eligible citizens. However, traditional ration distribution methods frequently face problems such as human errors, limited transparency, fraudulent practices, and inefficient beneficiary identification. To overcome these limitations, this study proposes an Automatic Ration Dispensing System designed to improve the reliability and transparency of the distribution process. The developed system combines embedded system technology with digital authentication methods to automate ration delivery. Beneficiaries are verified using secure identification techniques such as RFID cards, biometric authentication, or smart card mechanisms, which helps prevent duplication and unauthorized access. After successful authentication, the system automatically dispenses predetermined quantities of ration items through electromechanical control mechanisms, ensuring accurate measurement and reducing dependence on manual handling. All transaction records are stored digitally and can be monitored in real time through a centralized database, enabling authorities to track distribution activities efficiently. Experimental evaluation indicates that the proposed approach decreases distribution time, reduces manual errors, and improves the efficiency of ration management compared to conventional systems. Furthermore, the solution provides a scalable and cost- effective approach for upgrading existing ration distribution infrastructure while supporting transparency, automation, and digital governance in public welfare systems.

***Keywords**: Automatic Ration Dispensing System, Public Distribution System (PDS), Embedded Technology, RFID Authentication, Smart Ration Card, Internet of Things (IoT), Transparency, Automation.*

EE04: Neo-Cradle: An IoT based Baby Monitoring System with AI power

Dr. P.P.Chitte[1], Ashish Nirmal[2],Vishal Thombare[3], Sagar Shingote[4]

[1,2,3,4]Department of Electronics and Communication Engineering, Pravara Rural Engineering College, Loni, Maharashtra, India.
[*]Corresponding Author Email ID: chittepp@pravaraengg.org.in
Abstract: Infant monitoring is a critical aspect of childcare, yet traditional systems often rely on simple audio streaming or cloud-based processing, which introduces latency and privacy concerns. This paper proposes Neo Cradle, a non- invasive, edge-computing-based baby monitoring system. The system utilizes a Raspberry Pi 4 as the central edge node to perform real-time Convolutional Neural Network (CNN) inference for baby cry detection without internet dependency. By leveraging Mel-Frequency Cepstral Coefficients (MFCCs) for feature extraction and TensorFlow Lite for on-device inference, the system distinguishes infant cries from background noise with high accuracy. Additionally, the system integrates environmental monitoring (temperature and hygiene) via IoT protocols to provide a holistic safety solution. Preliminary results demonstrate the efficacy of edge-AI in reducing notification latency compared to cloud-centric architectures.
***Keywords**: Edge AI, Internet of Things (IoT), Convolutional Neural Networks (CNN), Raspberry Pi, Audio Classification, MFCC*

EE05: Agriculture Based Crop Monitoring Drone

S. B. Lavhate[1], Karad Priti[2], Gudaghe Pratiksha[3], Kusalkar Vishal[4]
[1,2,3,4] Department of Electronics and Telecommunication Engineering
Pravara Rural Engineering College, Loni
[*]Corresponding Author Email ID: vishalkusalkar1111@gmail.com
Abstract: The increasing demand for sustainable agricultural practices has highlighted the need for accurate, real-time crop monitoring systems that can improve productivity while reducing resource wastage. This project introduces an advanced Agriculture-Based Crop Monitoring Drone targeted specifically at dragon fruit, plantations and maize. Fields, both of which present unique monitoring challenges. Dragon fruit cultivation requires continuous observation of stem health, flower development, fruit maturity, and susceptibility to fungal infections, especially during monsoon and high-humidity periods. Maize crops, on the other hand, demand vigilant monitoring of vegetative growth stages, leaf chlorophyll levels, pest attacks (such as fall armyworm), and moisture variations throughout the growing cycle. To address these complexities, the proposed system integrates an unmanned aerial vehicle (UAV) equipped with a high-resolution RGB/multispectral camera, GPS positioning, ultrasonic altitude sensors, and IoT-enabled communication modules. The drone captures aerial imagery and environmental parameters at scheduled intervals and transfers the data to a cloud-based processing unit. Machine learning–based image processing algorithms analyse crop health indicators such as NDVI (Normalized Difference Vegetation Index), canopy structure, colour segmentation, and disease pattern detection. The system generates actionable insights, including early detection of water stress, nutrient imbalances, fruit maturity levels in dragon fruit, and pest infestation hotspots in maize.
***Keywords**: Precision Agriculture, Crop Monitoring Drone (UAV), NDVI Analysis, Machine Learning in Agriculture, Smart Farming (IoT-based)*

EE06: Robotic arm for automated fruit sorting using AI and OpenCV

Shelke Shubham Dnyaneshwar[1], Maniyar Mohammedkaif Yusuf[2], Chaudhari Mrunal Sunil[3], Sutar Diptej Rajendra[4]
[1,2,3,4] Department of Electronics and Telecommunication Engineering
Pravara Rural Engineering College, Loni
[*]Corresponding Author Email ID: shubhshelke1011@gmail.com

Abstract: Sorting fruits by ripeness and visual quality strawberries in particular is tedious, slow, and prone to inconsistency when done by hand. Their soft flesh also makes strawberries a poor fit for conventional mechanical handling, where bruising and product loss are common. This paper presents an automated sorting platform that combines AI-based computer vision (OpenCV), and robotic pick-and-place to classify and sort strawberries by color and ripeness. A USB camera provides live imagery; Python handles detection and classification; and an ESP32-controlled, 3D-printed robotic arm moves the fruit. On a representative test sample, the system achieved 90% successful completion, pointing to real practical potential in agricultural quality control settings at substantially reduced labor cost.

Keywords*: Fruit sorting, Computer vision, OpenCV, Robotic arm, ESP32, Machine learning, Agricultural automation, Image processing*

EE07: Depression detection using machine learning

V.A.Aher[1], Chitra Agre[2], Pallavi Chavan[3], Rahul Sapte[4]
[1,2,3,4] Department of Electronics and Telecommunication Engineering
Pravara Rural Engineering College, Loni
[*]Corresponding Author Email ID: aherva@pravaraengg.org.in

Abstract: Depression is a serious psychological condition that affects emotional stability, thinking patterns, and daily functioning. Early identification plays an important role in preventing severe mental health consequences. Conventional diagnostic approaches mainly depend on self-assessment questionnaires and clinical interviews, which may introduce subjectivity and delay detection. This study proposes an automated depression detection framework using video-based analysis and deep learning techniques. Facial expressions, eye movement pat- terns, and head posture features are extracted from video frames through computer vision methods. A transfer learning approach based on the ResNet-50 architecture is employed for feature extraction and classification. The system is deployed as a web-based application integrating React.js for the user interface, Node.js for backend communication, and MySQL for secure data management. Experimental observations demonstrate that visual behavior analysis can support early-stage depression screening in a non-invasive and scalable manner.

Keywords*: Depression Detection, Machine Learning, Video Analysis, ResNet-50, Computer Vision.*

EE08: Leaf Disease Detection of Crops and Remedies for the Disease by Using Image Processing

Aarti Mitthu Shirsat[1], Sanika Manjabhapu Khemnar[2],Aniket Namdev Shinde[3], M.S.Anap[4]

[1,2,3,4]Department of Electronics and Telecommunication Engineering
Pravara Rural Engineering College, Loni
[*]Corresponding Author Email ID: artishirsath2003@gmail.com

Abstract: The agricultural sector plays a key role in supplying quality food and makes the greatest contribution to growing economies and populations. Plant disease may cause significant losses in food production and eradicate diversity in species. Early diagnosis of plant diseases using accurate or automatic detection techniques can enhance the quality of food production and minimize economic losses. In recent years, deep learning has brought tremendous improvements in the recognition accuracy of image classification and object detection systems. Hence, in this paper, we utilized convolutional neural network (CNN)-based pre-trained models for efficient plant disease identification. We focused on fine tuning the hyperparameters of popular pre-trained models, such as DenseNet-121, ResNet-50, VGG-16, and Inception V4. The experiments were carried out using the Plant Village dataset (54,305 images, 38 classes). The model was evaluated using accuracy, sensitivity, specificity, and F1 score. DenseNet-121 achieved 99.81% accuracy, outperforming other models.

***Keywords**: Deep Learning, Transfer Learning, CNN, Leaf Pathology, Leaf Disease*

EE09: Multi-Application Surveillance Robot for Border and Industrial Security

A.R.Mane[1], Sayali Devadhe[2], Harshada Dhaybar[3], Vaishnavi Khedkar[4]
[1,2,3,4]Department of Electronics and Telecommunication Engineering,
Shri Chhatrapati Shivaji Maharaj College of Engineering, Ahilyanagar, Maharashtra, India
[*]Corresponding Author Email ID: sayalidevadhe399@gmail.com

Abstract: The design and implementation of a multi-application surveillance robot capable of performing border, industrial, and hazardous-area monitoring using multiple sensors and wireless control. The proposed system integrates IR and ultrasonic sensors for obstacle detection, a temperature sensor for environmental monitoring, and a motor-driver module for robotic movement. Wireless communication via Wi-Fi or Bluetooth enables remote operation and live feedback. The robot minimizes human involvement in risky zones and enhances surveillance efficiency. Experimental testing demonstrates stable obstacle avoidance, reliable data transmission, and energy-efficient mobility. This project proposes the design and development of a multi application surveillance robot equipped with multiple sensors and wireless communication for monitoring and control. The robot is capable of detecting obstacles using IR and Ultrasonic sensors, monitoring environmental temperature with a temperature sensor, and navigating using a motor driver. Wireless modules such as Wi-Fi or Bluetooth enable remote monitoring and control. The system is designed for applications in border surveillance, hazardous environments, and industrial security, ensuring real-time monitoring, obstacle avoidance, and environmental sensing. Security and surveillance play a vital role in both border and industrial sectors. Manual surveillance often involves high risks, particularly in hazardous or remote environments. This project introduces an automated surveillance robot that integrates multiple sensors and wireless communication to perform tasks such as obstacle

detection, temperature monitoring, and controlled navigation. The proposed system minimizes human intervention and enhances operational safety.

Keywords *- Surveillance Robot, IoT, Wireless Monitoring, Obstacle Detection, Arduino, Industrial Security*

EE10: CONTEXT AWARE AUDIO GUIDANCE SYSTEM FOR VISUALLY IMPAIRED

Dr. S. M. Gulhane[1], M. R. Gaikar[2], Siddharth Pardeshi[3], Saideep Kulkarni[4], Mahesh Kanawade[5], Tejas Kale[6]

[1]Principal & Assistant Professor, Department of Electronics & Telecommunication, Pravara Rural Engineering College, Loni India

[2,3,4,5,6]Department of Electronics & Telecommunication, Pravara Rural Engineering College, Loni India

[*]Corresponding Author Email ID: siddharthpardeshi072@gmail.com

Abstract: Visually impaired individuals face significant challenges in safely navigating unfamiliar environments. Traditional assistive tools such as canes provide limited environmental awareness and lack contextual information. This paper presents a Context Aware Audio Guidance System for Visually Impaired that integrates real-time object detection and distance estimation to assist users during navigation. The proposed system employs a USB camera and a deep learning-based YOLO object detection model running on a Raspberry Pi to identify obstacles and their spatial direction. A multi-zone Time-of-Flight (ToF) sensor is used to measure absolute distance to nearby obstacles, enabling accurate and reliable distance estimation. The system provides intuitive audio feedback to the user through wired earphones, conveying obstacle type, direction, and proximity. Experimental results demonstrate that the system operates in real time with acceptable latency and distance accuracy, making it suitable for indoor and outdoor environments. The proposed solution enhances situational awareness and promotes safer, independent mobility for visually impaired users.

Keywords: *Visually Impaired Assistance, Object Detection, YOLO, Time-of-Flight Sensor, Raspberry Pi, Assistive Technology*

EE11: AI-Driven 6G-Enabled IoT Framework for Autonomous Unmanned Aerial Vehicle Networks

Archana Rajendra Mane[1]

[1]Assistant Professor, Department of Electronics and Telecommunication Engineering, Shri Chhatrapati Shivaji Maharaj College of Engineering, Ahilyanagar, Maharashtra, India

[*]Corresponding Author Email ID: mane.archana17@gmail.com

Abstract: Next-generation Internet of Things (IoT) systems are anticipated to benefit from ultra-reliable, low-latency, and intelligent connectivity made possible by the development of sixth-generation (6G) wireless communication. As mobile IoT nodes, unmanned aerial vehicles (UAVs) are essential to applications like precision agriculture, smart cities, disaster management, and surveillance. However, issues like spectrum limitations, limited onboard energy, and dynamic topology make it difficult for UAVs to coordinate and communicate effectively. For autonomous UAV networks, this paper suggests an AI-driven 6G-enabled IoT

framework that combines swarm coordination, edge intelligence, and adaptive resource allocation. The suggested architecture maximizes energy efficiency, trajectory planning, and communication reliability by utilizing deep learning and reinforcement learning. In comparison to traditional UAV communication methods, simulation results show increased mission completion rates, decreased latency, and increased network throughput.

Keywords*: 6G, UAV, Internet of Things, Artificial Intelligence, Edge Computing, Swarm Intelligence*

EE12: Color based object shorting and counting using deep learning

Sandip. S. Turane[1], Kajale Sanjyot[2], Rohan Dukre[3], Jagtap Sapana[4]

[1,2,3,4]Pravara rural engineering College Loni, Ahilyanagar

Abstract: The increasing demand for automation in industrial and pharmaceutical sectors has led to the development of intelligent systems capable of performing sorting and counting tasks with high accuracy and efficiency. This project presents a color-based object sorting and counting system using deep learning techniques. The system utilizes a conveyor belt mechanism integrated with a vision-based detection unit to identify objects in real time. A camera captures continuous images of moving objects, which are processed using computer vision techniques and a deep learning model such as YOLOv8 for accurate detection and color classification. The detected objects are categorized based on predefined color classes, and the system simultaneously maintains a count of each category. The classification results are transmittec to a microcontroller, which controls servo motors to physically separate objects into designated containers. This automated approach significantly reduces manual effort, minimizes errors, and enhances processing speed. The system is designed to be scalable and adaptable for various applications, including pharmaceutical pill sorting, food processing, and industrial product segregation. Overall, the proposed solution demonstrates the effective integration of artificial intelligence, embedded systems, and mechanical automation to achieve real-time sorting and counting with improved reliability and performance.

Keywords: *Color-Based Sorting, Deep Learning, YOLOv8, Object Detection, Computer Vision ESP32, Automation, Conveyor Belt System, Real-Time Counting, Smart Manufacturing*

EE13: Operating a Bell Using Arduino Board

Shubhangi Balte[1] , Mrunal Gholap[2]

[1,2]Department of Electronics and Telecommunication Engineering, Shri Chhatrapati Shivaji Maharaj College of Engineering, Ahilyanagar, Maharashtra, India

[*]Corresponding Author Email ID: balteshubhangi@gmail.com

Abstract: This project, "Operating Bell on Arduino Board," is designed to automatically control a bell using an Arduino microcontroller. The main objective of this project is to create a simple and efficient system that can ring a bell at specific times or based on user input. In this system, the Arduino board acts as the main controller, which processes the programmed instructions and activates the bell using a relay or buzzer. This project can be used in schools, colleges, or offices to automate bell ringing, reducing the need for manual operation. It helps in saving time and improves accuracy in maintaining schedules. The system is easy to design,

low-cost, and reliable. It also helps in understanding basic concepts of embedded systems, programming, and automation. Overall, this project demonstrates how Arduino can be used to develop simple automation systems for real-life applications..

Keywords: *Arduino UNO, Real Time Clock (RTC), Relay Module, Automatic Bell System, Arduino programmed, Transistor Switching*

EE14: IoT based Virtual Doctor Robot

Gauri Sanjay Gaikwad[1], Raut Snehal Milind[2], Abhang jayshri Baban[3]

[1,2,3]Pravara rural engineering College Loni Ahilyanagar, Maharashtra, India

[*]Corresponding Author Email ID: gauriigaikwad1507@gmail.com

Abstract: With the rapid growth of Internet of Things (IoT), healthcare systems are becoming smarter and more accessible. This paper presents the design and development of an IoT-based Virtual Doctor Robot that can monitor patients' vital parameters such as body temperature, heart rate, oxygen level, and blood pressure in real time. The system uses sensors integrated with a microcontroller to collect health data and transmit it to a cloud platform for remote monitoring by doctors.The proposed robot reduces the need for physical contact between patients and healthcare providers, which is especially useful during pandemics and in remote areas. It can also provide basic medical guidance and emergency alerts. The system is cost-effective, user-friendly, and improves healthcare accessibility. This research demonstrates how IoT technology can play a significant role in advancing modern healthcare systems.

Keywords: *Internet of Things (IoT), Virtual Doctor Robot, Healthcare Monitoring, Remote Patient Monitoring, Sensors, NodeMCU, Vital Signs, Smart Healthcare, Telemedicine, Automation, Embedded System*

EE15: Design and Implementation of Wireless Smart Roads Using IoT for Sustainable Transportation Systems

Markad Vivek Dadasaheb[1], Dalvi Atharv Dipak[2], Borude Aditya Harishchandra[3], Gangarde Rahul Dadasaheb[4], Bhole K. S.[5]

[1,2,3,4,5]Rajiv Gandhi College of Engineering, Ahilyanagar

[*]Corresponding Author Email ID: pj583167@gmail.com

Abstract: This project presents the design and implementation of an Eye Tracking Mouse system using Python, enabling users to control the computer cursor through eye movements without the need for a physical mouse. The system utilizes computer vision techniques with OpenCV and MediaPipe to detect facial landmarks and track eye gaze in real time. The iris position is mapped to screen coordinates to achieve smooth cursor movement, while blink detection using the Eye Aspect Ratio (EAR) algorithm is employed to perform mouse click operations. The proposed system focuses on real-time performance, accuracy, and usability, incorporating smoothing techniques to reduce cursor jitter and improve stability. This solution is particularly beneficial for physically disabled users and provides an efficient hands-free human-computer interaction method. The system demonstrates a low-cost and accessible approach to modern touchless computing technologies.

Keywords: *Eye Aspect Ratio (EAR), Tracking*

EE16: Underwater Wireless Communication Using Acoustic Waves for Marine IoT Systems

Bhoye Manisha Bhaurao[1], Daspute Pramod Sudhakar[2] , Ubale Rutuja Vijay[3], Chaudhari S. B.[4]

[1,2,3,4]Rajiv Gandhi College of Engineering, Ahilyanagar

(Same Abstact of EE18 and EE19)

[*]Corresponding Author Email ID: pj583167@gmail.com

Abstract: Underwater Wireless Communication (UWC) is a rapidly growing field essential for marine research, environmental monitoring, and underwater exploration. Conventional communication methods such as radio frequency and optical signals are ineffective underwater due to severe attenuation and scattering, making acoustic waves the most suitable medium for long-range communication. This Stage-I project focuses on the conceptual design and development of an acoustic-based communication system tailored for Marine IoT applications. The proposed system aims to enable reliable and energy-efficient data exchange between underwater sensor nodes, autonomous underwater vehicles, and surface monitoring stations. In this phase of the project, an extensive study of underwater acoustic channels, propagation characteristics, and major communication challenges such as low bandwidth, high latency, noise interference, and multipath effects has been conducted. A preliminary system architecture has been formulated, integrating underwater sensors, acoustic modems, microcontrollers, and cloud-based IoT platforms for real-time monitoring of parameters like temperature, salinity, and pollution levels. Block and circuit diagrams prepared in this stage provide a structural foundation for the transmitter, receiver, and signal processing units. The Stage-I work establishes the groundwork necessary for implementing a functional underwater communication network. The identification of hardware and software requirements, feasibility analysis, and understanding of system limitations will guide further development. Future stages will involve prototype implementation, modulation technique integration, channel testing, and system performance evaluation to achieve a robust and intelligent underwater Marine IoT communication system.

Keywords: *Underwater Wireless Communication (UWC), IoT*

EE17: IoT Home Automation & Weather Monitor Using NodeMCU with Cayenne

Dhotre Shreeraj Rajesh[1], Chindhe Akshada Bapusaheb[2], Chintamani Rohit Madhukar[3] Chaudhari S. B.[4]

[1,2,3,4]Rajiv Gandhi College of Engineering, Ahilyanagar

[*]Corresponding Author Email ID: pj583167@gmail.com

Abstract: Underwater Wireless Communication (UWC) is a rapidly growing field essential for marine research, environmental monitoring, and underwater exploration. Conventional communication methods such as radio frequency and optical signals are ineffective underwater due to severe attenuation and scattering, making acoustic waves the most suitable medium for long-range communication. This Stage-I project focuses on the conceptual design and development of an acoustic-based communication system tailored for Marine IoT applications. The proposed system aims to enable reliable and energy-efficient data exchange between

underwater sensor nodes, autonomous underwater vehicles, and surface monitoring stations. In this phase of the project, an extensive study of underwater acoustic channels, propagation characteristics, and major communication challenges such as low bandwidth, high latency, noise interference, and multipath effects has been conducted. A preliminary system architecture has been formulated, integrating underwater sensors, acoustic modems, microcontrollers, and cloud-based IoT platforms for real-time monitoring of parameters like temperature, salinity, and pollution levels. Block and circuit diagrams prepared in this stage provide a structural foundation for the transmitter, receiver, and signal processing units. The Stage-I work establishes the groundwork necessary for implementing a functional underwater communication network. The identification of hardware and software requirements, feasibility analysis, and understanding of system limitations will guide further development. Future stages will involve prototype implementation, modulation technique integration, channel testing, and system performance evaluation to achieve a robust and intelligent underwater Marine IoT communication system.

Keywords: *Underwater Wireless Communication (UWC), IoT*

EE18: Smart Classroom Energy Management System

A.R.Mane[1], Rutuja Agale[2], Falguni Dhadge[3], Poonam Mohite[4]

[1]Assistant Professor, Department of Electronics and Telecommunication Engineering Shri. Chhatrapati Shivaji Maharaj College of Engineering Ahilyanagar, Maharashtra, India

[2,3,4]Department of Electronics and Telecommunication Engineering Shri. Chhatrapati Shivaji Maharaj College of Engineering Ahilyanagar, Maharashtra, India

[*]Corresponding Author Email ID: rutujaagale2@gmail.com

Abstract: This paper aims to reduce electricity wastage in classrooms by using an automatic energy management system. The system automatically controls electrical devices like fans and light based on classroom conditions. A temperature level, the fan speed is automatically adjusted. The system can also include sensors to detect the presence of students, in the classroom. If no one present, the lights and fans automatically turned off to save energy. A microcontroller is used to process the sensor data and control the electrical devices. This smart system helps in saving electricity, reducing human efforts, and maintaining a comfortable classroom environment. The increasing demand for sustainable and energy-efficient solutions in educational institutions has highlighted the need for intelligent systems to optimize energy consumption in classrooms. Classrooms are significant contributors to electricity usage due to lighting, air conditioning, and the operation of electronic devices. This paper presents a Smart Classroom Energy Management System (SCEMS) that leverages Internet of Things (IoT) technology, real-time sensor networks, and automated control mechanisms to efficiently manage energy usage. The system incorporates occupancy sensors, ambient light sensors, and temperature sensors to monitor classroom conditions continuously. Based on the collected data, SCEMS automatically regulates lighting, ventilation, and other electrical devices, ensuring that energy is consumed only when necessary. This approach not only reduces unnecessary energy usage but also maintains optimal comfort levels for students and staff, promoting an environment conducive to learning.

Keywords: *Smart Classroom, Energy Management System (EMS), Internet of Things (IoT), Sensor Networks, Energy Monitoring, Automated control.*

EE19: Automated Temperature Sensing Fan

Sakshi Lokhande[1], Asmita Naik[2], Sunakshi Palve[3]
[1,2,3]Department of Electronics and Telecommunication Engineering Shri. Chhatrapati Shivaji Maharaj College of Engineering Ahilyanagar, Maharashtra, India
[*]Corresponding Author Email ID: sakshilokhande785@gmail.com

Abstract: Temperature control is important for maintaining comfort in homes, offices, and electronic systems. In many cases, fans are operated manually, which may lead to inefficient cooling and unnecessary power consumption. To overcome this problem, an automatic temperature sensing fan system can be used. This project is based on a temperature sensor that continuously monitors the surrounding temperature. The sensed temperature is processed by a microcontroller, which automatically controls the speed of the fan. When the temperature is low, the fan runs at low speed, and as the temperature increases, the speed of the fan increases accordingly. This allows efficient cooling without manual intervention. This automatic control mechanism helps in maintaining a comfortable environment while reducing unnecessary power consumption. The system is especially useful in places where temperature variations occur frequently, such as homes, offices, server rooms, and laboratories. It also helps in protecting electronic devices from overheating by ensuring proper cooling at the right time. The project is simple in construction, uses low-cost components, and does not require complex programming, making it suitable for beginners and academic mini projects. Through this project, students can gain practical knowledge of temperature sensing, microcontroller-based control, and automatic speed regulation of electrical devices. The proposed system is simple in design, low in cost, and easy to implement. It is suitable for basic temperature control applications and academic mini projects. This project helps in understanding the working of temperature sensors, automatic control systems, and speed control of DC fans.

Keyword*: Temperature Sensor, Automatic Fan, Microcontroller, Speed Control, Smart Cooling*

EE20: Ultrasonic IoT Communication Framework for Audio based Data Transfer

Pardeshi Rutuja Kishor[1], Kolhe Snehal Hanumant[2], Mulik Snehal Laxman[3], Shaikh T. N.[4]
[1,2,3,4]Rajiv Gandhi College of Engineering, Ahilyanagar
[*]Corresponding Author Email ID: pj583167@gmail.com

Abstract: Ultrasonic communication has emerged as a promising alternative medium for short-range, low-power data transmission in Internet of Things (IoT) applications. This project presents the design, development, and evaluation of an ultrasonic IoT communication framework capable of transmitting digital information using inaudible high-frequency sound waves (typically above 18 kHz). The primary objective is to establish a cost-effective, hardware-independent communication channel that operates using existing audio interfaces such as microphones and speakers, thereby eliminating the need for conventional wireless technologies like Wi-Fi, Bluetooth, or RF modules. The proposed system employs modulation

techniques such as Frequency Shift Keying (FSK) and Amplitude Shift Keying (ASK) to encode digital data into ultrasonic signals. These signals are transmitted through standard audio hardware and received by compatible devices, where demodulation and decoding processes reconstruct the original data. The architecture integrates microcontrollers, signal processing algorithms, and IoT interfaces to enable seamless device-to-device communication. In the implemented phase of the project, ultrasonic signal generation and modulation were successfully achieved, and experimental validation demonstrated reliable data transmission over short indoor distances. A basic receiver module was developed to perform signal filtering, decoding, and error detection. Performance evaluation indicates that ultrasonic communication offers advantages such as low energy consumption, enhanced security due to limited propagation range, and reduced interference compared to traditional RF-based systems. Furthermore, the system shows potential for applications in secure authentication, proximity-based data exchange, indoor localization, and smart environments. Despite challenges such as environmental noise sensitivity and limited bandwidth, the results confirm the feasibility of ultrasonic communication as an innovative and efficient solution for next-generation IoT networks.

Keywords: *RF, Ultrasonic Sound, IoT, Noise Sensitivity.*

EE21: Microwave-Based Wireless Battery Charging System

Shital Anil Gaikwad[1]

[1]S.P.I.T polytechnic, Kurund

[*]Corresponding Author Email ID: shitalagaikwad261991@gmail.com

Abstract: Wireless power transfer (WPT) using microwave technology has emerged as a promising solution for efficient and long-distance energy transmission without physical connectors. This paper presents the design and implementation of a microwave-based wireless battery charging system that utilizes electromagnetic radiation in the microwave frequency range to transfer energy from a transmitting antenna to a receiving rectenna. The system focuses on improving power transfer efficiency, minimizing energy loss, and ensuring safe operation. Key components include a microwave generator, transmitting antenna, rectifying antenna (rectenna), and power management circuit for battery charging. Experimental analysis demonstrates that the proposed system can effectively charge low-power devices over a moderate distance with acceptable efficiency. The study also addresses challenges such as alignment sensitivity, environmental interference, and safety concerns. This technology has potential applications in consumer electronics, electric vehicles, and remote power delivery systems.

Keywords: *Wireless Power Transfer (WPT), Microwave Charging, Rectenna, Electromagnetic Radiation, Battery Charging, Energy Efficiency, Wireless Energy Transmission, Antenna Systems*

EE22: IoT based smart plant health monitoring system

Snehal Ambade[1], Tejal Datir[2], Samrudhi Joshi[3]

[1,2,3]Department of Electronics and Telecommunication Engineering Shri. Chhatrapati Shivaji Maharaj College of Engineering Ahilyanagar, Maharashtra, India

*Corresponding Author Email ID: datirtejal123@gmail.com

Abstract: The Smart Agriculture Project, focusing on Crop and Plants Health Monitoring System combined with Air Quality Monitoring, aims to revolutionize modern farming practices by leveraging sensor-based technology for improved crop management and environmental monitoring. The system uses a combination of sensors and microcontroller technology to monitor crucial parameters such as soil moisture, temperature, humidity, and air quality. The core of the system is an Arduino UNO, which serves as the central control unit. Data is gathered from various sensors, including the MQ135 sensor for detecting air quality (particularly harmful gases), the DHT11 sensor for measuring temperature and humidity, and the soil moisture sensor to assess the water level in the soil. The system is equipped with a 16x2 LCD display to provide real-time feedback and readings for the farmer. The 10k preset and jumper wires are used to connect and adjust the sensors to the Arduino for accurate data acquisition, while the 9V battery and 3.7V rechargeable battery serve as the power sources. A power jack and switch are integrated for easy power management. The system is designed to provide instant feedback on the environmental conditions, allowing farmers to take timely actions to ensure the health of their crops, improve water efficiency, and monitor air quality, thus contributing to sustainable farming practices.

Keywords: *Internet of Things (IoT), Smart Agriculture, Crop Health Monitoring, Air Quality Monitoring, Arduino UNO, Soil Moisture Sensor.*

EE23: A Smart Women Protection System Using IoT and Open-Source

[1]G.Y. Mhaske, [2]Prajwal Zarekar, [3]Shrikant Zaware, [4]Bharath Handore

[1]Assistant Professor, Department of Electronics and Telecommunication Engineering,
Shri Chhatrapati Shivaji Maharaj College of Engineering, Ahilyanagar, Maharashtra, India

[2,3,4]Department of Electronics and Telecommunication Engineering,
Shri Chhatrapati Shivaji Maharaj College of Engineering, Ahilyanagar, Maharashtra, India

*Corresponding Author Email ID: zarekarprajwal@gmail.com

Abstract: This project presents the design and implementation of a SMART Women Protection System integrated with IoT (Internet of Things) and developed using open-source hardware and software platforms. The primary objective is to provide a wearable or portable safety device that offers an immediate response mechanism during emergencies. The system utilizes a microcontroller (such as an ESP32 or Arduino) interfaced with multiple sensors, including a GPS module for real-time location tracking and a GSM module or Wi-Fi connectivity for instant alerting. The device is triggered by a manual "Panic Button" or an automated sensor-based trigger (e.g., an accelerometer to detect sudden falls or forceful movements). Once activated, the system captures the user's precise coordinates and transmits an emergency SOS message to pre-defined contacts and the nearest police station via a dedicated mobile application or SMS. By utilizing open-source frameworks, the system ensures transparency, lower production costs, and the flexibility for future enhancements, such as audio recording or heart-rate monitoring for stress detection. The proposed solution aims to

bridge the gap between fear and safety, providing women with a reliable digital guardian in vulnerable situations.

***Keywords**: IoT (Internet of Things), Women's Safety, Open-Source Hardware, GPS Tracking, Emergency Alert System.*

EE24: Greenhouse Monitoring and Control System using IoT

A.R.Mane[1], Gangawane Priya[2], Kharad Yashwant[3], Gunjal Bhagyashri[4]

[1]Assistant Professor, Department of Electronics and Telecommunication Engineering,
Shri Chhatrapati Shivaji Maharaj College of Engineering, Ahilyanagar, Maharashtra, India

[2,3,4]Department of Electronics and Telecommunication Engineering,
Shri Chhatrapati Shivaji Maharaj College of Engineering, Ahilyanagar, Maharashtra, India

[*]Corresponding Author Email ID: gangawanepriya250@gmail.com

Abstract: Agriculture plays a vital role in the economy, and maintaining optimal environmental conditions inside a greenhouse is essential for improving crop yield and quality. Manual monitoring of temperature, humidity, soil moisture, and light intensity is inefficient and may lead to improper plant growth. This project presents the design and implementation of a Greenhouse Monitoring and Control System using Internet of Things (IoT) technology. The system continuously monitors key environmental parameters such as temperature, humidity, soil moisture, and light intensity using sensors like DHT11/DHT22, soil moisture sensor, and LDR. The collected data is processed using a microcontroller such as NodeMCU and transmitted to a cloud platform via Wi-Fi. Based on predefined threshold values, the system automatically controls actuators such as water pumps, exhaust fans, and artificial lighting through relay modules. Real-time data is displayed on a mobile application or web dashboard, allowing remote monitoring and control from anywhere. The proposed system helps in maintaining optimal greenhouse conditions, reducing water consumption, minimizing human intervention, and increasing crop productivity. The project is low-cost, energy-efficient, and suitable for smart agriculture applications. Future enhancements may include integration of advanced analytics and automation using Artificial Intelligence for predictive crop management.

***Keywords** – Internet of Things(IoT),Greenhouse Monitoring, Smart Agriculture, Environmental Parameters, Temperature and Humidity, Soil Moisture Sensor.*

EE25: Smart Energy Meter using IOT

Dr S.M.Walke[1], Vedant Yadav[2], Ajay Deshmukh[3], Huzaif Tamboli[4]

[1]HOD &Assistant Professor, Department of Electronics and Telecommunication Engineering,
Shri Chhatrapati Shivaji Maharaj College of Engineering, Ahilyanagar, Maharashtra, India

[2,3,4]Department of Electronics and Telecommunication Engineering,
Shri Chhatrapati Shivaji Maharaj College of Engineering, Ahilyanagar, Maharashtra, India

[*]Corresponding Author Email ID: vedantyadavk@gmail.com

Abstract: Collecting electricity utility meter readings manually can be a cumbersome and time-consuming process for electricity providers. Traditional methods require personnel to visit each location to record the readings, which may lead to errors, delays, and increased

operational costs. The integration of the Internet of Things (IoT) offers an effective and economical solution for wirelessly transmitting energy consumption data. By using IoT technology, electricity usage information can be monitored in real time and transmitted automatically through a communication network. The primary goal of this project is to measure the electricity usage of home appliances and automate bill generation through IoT technology. The proposed system continuously monitors energy consumption and sends the collected data to a cloud platform for storage and analysis. This allows both users and service providers to access energy usage information remotely at any time. As a result, consumers can easily track their electricity consumption and manage their power usage more efficiently. Our energy grid system is designed to function in a distributed topology that can adaptively manage various energy sources and devices connected to the network. This design improves the reliability and scalability of the system while supporting efficient communication between different components. The proposed approach is suitable for a range of smart grid applications, including distributed energy generation, consumption metering, energy monitoring, and intelligent energy demand management. By implementing this system, energy monitoring becomes more transparent, efficient, and convenient for both consumers and electricity providers.

Keywords: *Smart Grid, Energy Meter, Internet of Things.*

EE26: IoT BASED FIRE FIGHITING AND LIVE TRACKING ROBOT

Geeta Aghav*[1], Sanket Bangar *[2], Shraddha Pawale*[3],
Dr. S.M. Walke*[4]

*[1,2,3,] Student & *[4] . Professor, Electronics and Telecommunication Department, Ahilyanagar Jilha Maratha Vidya Prasarak Samaj's Shri. Chhatrapati Shivaji Maharaj College of Engineering, Nepti, Ahilyanagar, Maharashtra, India

Abstract: The Internet of Things (IoT) has opened new possibilities for developing intelligent systems that can respond to critical emergencies such as fire accidents. This paper presents the design and implementation of an IoT-Based Fire Fighting and Live Tracking Robot capable of detecting fire, extinguishing it, and sharing real-time location and status updates. The proposed system integrates flame sensors, temperature sensors, and a camera module for live monitoring. The robot is controlled remotely through IoT platforms, allowing users to observe and manage operations via the internet. A Wi-Fi-enabled microcontroller, such as Node MCU or Arduino with ESP8266, serves as the main control unit, ensuring seamless data transmission between sensors and the cloud. The system enhances safety in hazardous environments by reducing human intervention and providing immediate fire detection, suppression, and area surveillance. Experimental results demonstrate the robot's efficiency in detecting flames, navigating towards the source, and extinguishing small-scale fires effectively. This approach offers a cost-effective and reliable solution for industrial, residential, and institutional fire safety applications.

Keywords: *IoT, WIFI, Node-MCU,*

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Track-5: Mechanical & Production Engineering

MECH01: Modal Characterization and Experimental Validation of a Particle-Damped Spur Gearbox

Alina Baig[1], Hajare B.S.[2]
[1]Assistant Professor, Department Mechanical Engineering, Vishwabharti Academy College of Engineering, Ahilyanagar
[2]Head and Assistant Professor, Department Mechanical Engineering, Vishwabharti Academy College of Engineering, Ahilyanagar
[*]Corresponding Author Email ID: afbeg124@gmail.com

Abstract: This study presents a comprehensive modal characterization and experimental validation of a spur gearbox enhanced with particle damping. The primary objective is to quantify how internally contained granular media alter the gearbox dynamic properties and to validate numerical modal predictions with laboratory measurements. A finite element model of the gearbox assembly is developed and coupled with a discrete-element representation of the particle damper to predict shifts in natural frequencies, mode shapes, and modal damping ratios across operating-relevant boundary conditions. Experimental modal analysis is performed on an instrumented test rig using impact excitation and operational vibration measurements to extract modal parameters. Results show consistent reductions in peak response and measurable increases in modal damping for modes that engage particle–structure interaction, with frequency shifts dependent on particle fill level and confinement geometry. The combined numerical–experimental approach demonstrates that particle damping can be an effective passive strategy to mitigate resonant vibration in gearboxes without significant mass or stiffness penalties. Findings provide practical design guidance for integrating particle dampers into rotating powertrain components.

Keywords: *Particle Damping, Spur Gearbox, Experimental Modal Analysis, Passive Vibration Control*

MECH02: Recent Trends in Coating of sheet metal forming tools to reduce friction and wear

Solat Vijay Bhausaheb[1], Dr. Aher V.S[2]., Dr. Bajaj D. S.[3]
[1,2,3]Department Mechanical Engineering, Amrutvahini College of Engineering, Sangamner
[*]Corresponding Author Email ID: avcoecybercell@gmail.com

Abstract: This review paper provides a comprehensive evaluation of top most current and emerging coating technologies used to mitigate friction and adhesive wear (galling) in sheet metal forming (SMF) tooling. As the industry transitions toward advanced high-strength steels (AHSS) and aluminum alloys, traditional monolithic coatings often fail under extreme contact pressures (>1.5 GPa). This article synthesizes recent advancements in four critical areas of surface engineering: (a) the integration of Laser Surface Texturing (LST) with Ti3C2Tx (MXene)/PTFE hybrid solid lubricants; (b) nanolaminated ceramic-carbon architectures, such as AlTiSiN/a-C:H:Si; (c) TiCrN/DLC multilayer systems; and (d) functionalized halogen-terminated MXene (Ti3C2Tx-Cl) matrices. By comparing tribological data across these systems, this review identifies a consistent trend: the synergy between mechanical reinforcement (ceramic layers) and low-energy shear surfaces (2D nanomaterials and DLC) is essential for achieving ultra-low friction coefficients (0.05–0.08) and extending tool life in dry

forming applications. The paper concludes by discussing the economic and environmental benefits of these "self-lubricating" systems and proposes directions for future research into scalable Nano manufacturing for the automotive tooling sector.

Keywords: *sheet metal forming, galling, review, MXenes, diamond-like carbon, laser surface texturing, nanolaminate coatings, tribology*

MECH03: A Comprehensive Review of Multi-Stage Deep Drawing Processes for High Length-to-Diameter Ratio Components

Vishal Sahebrao Vadak[1] , Dr. P.A. Narwade[2]

[1]Dr. Vitthalrao Vikhe Patil College of Engineering, Ahilyanagar

[2]Assistant Professor, Dr. Vitthalrao Vikhe Patil College of Engineering, Ahilyanagar

[*]Corresponding Author Email ID: vadakvishal@gmail.com

Abstract: Deep drawing is one of the most widely used sheet metal forming processes for manufacturing cylindrical and hollow components in automotive, aerospace, and industrial applications. However, producing components with a high length-to-diameter (L/D) ratio presents significant challenges such as wrinkling, tearing, excessive thinning, and high forming loads when conventional single-stage drawing methods are used. To overcome these limitations, researchers and industries have adopted multi-stage deep drawing techniques and improved punch-die tooling designs. This paper presents a comprehensive review of the developments in multi-stage deep drawing processes for manufacturing high L/D ratio components. The study summarizes various research works related to punch and die design, blank holder pressure control, lubrication techniques, and process parameter optimization that influence the formability of deep drawn parts. Additionally, the review highlights different analytical, experimental, and numerical approaches used to analyze deep drawing performance, including forming load estimation, thickness distribution, and defect prevention. The findings from previous studies indicate that multi-stage drawing combined with optimized tooling design significantly improves material flow, reduces defects, and enhances dimensional accuracy. This review aims to provide researchers and industry professionals with a consolidated understanding of existing techniques and future research directions in the design and development of deep drawing tools for high aspect ratio components.

Keywords: *Deep Drawing, Multi-Stage Forming, Punch and Die Design, High Length-to-Diameter Ratio, Sheet Metal Forming.*

MECH04: Design and Optimization of Airplane Flooring Using Composite Materials

Sandeep Gundecha[1], Hajare B.S.[2]

[1]Department Mechanical Engineering, Vishwabharti Academy College of Engineering, Ahilyanagar

[2]Head and Assistant Professor, Department Mechanical Engineering, Vishwabharti Academy College of Engineering, Ahilyanagar

[*]Corresponding Author Email ID: afbeg124@gmail.com

Abstract: The increasing demand for lightweight, durable, and safe structures in the aerospace industry has driven significant advancements in material selection and structural design. This

project focuses on the design and optimization of airplane flooring using composite materials to enhance strength-to-weight ratio and overall performance. A sandwich structure is proposed, consisting of aluminium panels as the upper and bottom face sheets, and a honeycomb-structured composite core in between. The aluminium layers provide superior surface strength and resistance to impact, while the honeycomb composite ensures high stiffness, reduced weight, and efficient load distribution. Finite element analysis and optimization techniques will be employed to evaluate structural behavior under static and dynamic loading conditions, considering parameters such as strength, weight reduction, and fatigue resistance. The optimized flooring design aims to achieve significant weight savings, improve passenger safety, and contribute to enhanced fuel efficiency of the aircraft. This study highlights the potential of hybrid composite-aluminium sandwich structures as a sustainable solution for next-generation aerospace applications.

Keywords: *Punch and Die Design, High Length-to-Diameter Ratio, Sheet Metal Forming.*

MECH05: Particle Damping: A Comprehensive Review of Mechanisms, Modeling, and Applications in Mechanical Design

Baig A.F.[1,] Badade Rita[2]

[1,2]Department Mechanical Engineering, Vishwabharti Academy College of Engineering, Ahilyanagar

[*]Corresponding Author Email ID: afbeg124@gmail.com

Abstract: Particle damping is a passive vibration control technique that utilizes granular materials enclosed within cavities to dissipate vibrational energy through frictional sliding and inelastic collisions. Unlike conventional damping methods, particle damping offers broadband attenuation, robustness in harsh environments, and cost-effectiveness, making it suitable for aerospace, automotive, and civil engineering applications. This review synthesizes the fundamentals of particle damping, mechanisms of energy dissipation, modeling approaches including analytical, finite element, and discrete element methods, and experimental validation studies. Key design considerations such as particle size, density, cavity geometry, and fill ratio are discussed alongside advantages and limitations. The paper also highlights emerging research directions, including smart particle dampers, integration with meta materials, and additive manufacturing. The findings emphasize particle damping's potential as a versatile solution for next-generation mechanical systems requiring reliable vibration suppression.

Keywords: *Particle damping; vibration suppression; discrete element method (DEM); finite element method (FEM); nonlinear damping; mechanical design; energy dissipation; aerospace applications; automotive engineering; civil structures.*

MECH06: Wing Design of Airplane and Its Aspects

Mr. Gundecha S.D.[1,] Shaikh M.L.[2]

[1,2]Department Mechanical Engineering, Vishwabharti Academy College of Engineering, Ahilyanagar

[*]Corresponding Author Email ID: afbeg124@gmail.com

Abstract: Wing design plays a fundamental role in the performance, efficiency, and safety of an aircraft. As one of the most critical structural and aerodynamic components, the wing

directly influences an airplane's ability to generate lift, maintain stability, and achieve optimal fuel efficiency. This paper provides an in-depth exploration of the various aspects of wing design, beginning with the foundational aerodynamic principles that govern flight. It examines key geometrical parameters such as air foil shape, aspect ratio, sweep angle, and dihedral angle, each of which affects the lift-to-drag ratio and overall flight characteristics. Furthermore, the report delves into material selection for wing construction, highlighting the transition from traditional materials like aluminium alloys to modern composites such as carbon fibre-reinforced polymers (CFRP), which offer superior strength-to-weight ratios and fatigue resistance. Structural design considerations, including the internal arrangement of spars, ribs, and skin, are discussed with reference to the types of loads a wing must endure during various flight conditions. Advanced topics such as high-lift devices, control surfaces, and recent technological innovations like winglets and morphing wings are also analysed for their contribution to flight performance and control. Real-world case studies, such as the Boeing 787 and Airbus A350, are used to illustrate how modern engineering integrates aerodynamics, materials science, and computational analysis to design highly efficient wing structures. The report concludes with a discussion on current design challenges, including trade-offs between performance and manufacturability, and explores emerging trends such as smart materials, adaptive wing structures, and sustainable aviation.

***Keywords**: Wing design, aerodynamics, CFRP, structural analysis, morphing wings, high-lift devices.*

MECH07: Design And Development of EV Agricultural Sprayer

Tejas Dnyaneshwar Gunjal[1] ,Dattatray Baban Salunke[2],Tushar Balasaheb Bhawar[3],Krushna Sudhakar Kandekar[4]

[1,2,3,4]Department of Mechanical Engineering, Shri Chhtrapati Shivaji Maharaj Collage Of Engineering Nepti Ahilyanagar

[*]Corresponding Author Email ID: gunjaltejas92@gmail.com

Abstract: This project focuses on the design and development of an Electric Vehicle (EV)-based agricultural sprayer integrated with a grass cutting mechanism to address key challenges faced by Indian farmers. Traditional farming practices, particularly in spraying pesticides and cutting grass, are often labor-intensive, time-consuming, and expose farmers to harmful chemicals and physical strain. Existing tractor-operated sprayers and manual or fuel-powered grass cutters, while effective to some extent, lack efficiency, portability, and environmental sustainability. The proposed system aims to combine advanced electrical and mechanical technologies into a single, compact, and eco-friendly solution. The EV agricultural sprayer with grass cutter integrates an aero blast spraying mechanism with an efficient grass cutting unit, powered by an electric drive system. This design reduces dependency on fossil fuels, minimizes operational costs, and lowers environmental pollution. Additionally, the system enhances precision in spraying, ensuring uniform distribution of pesticides and reducing wastage. A key objective of this project is to improve the safety and comfort of farmers. By reducing direct exposure to chemicals and minimizing manual effort, the machine promotes better health and working conditions. The ergonomic and user-friendly design ensures ease of operation, making it suitable for small and medium-scale farmers. Furthermore, this project

reflects the application of classroom knowledge in real-world problem-solving, incorporating concepts such as electric mobility, power transmission, fluid mechanics, and machine design. It also considers how attitudes, beliefs, and values—particularly towards sustainability and innovation—play a role in engineering solutions. Overall, the EV agricultural sprayer with grass cutter represents a modern, efficient, and sustainable approach to farming, aiming to enhance productivity while supporting the well-being of farmers and the environment.

Keywords: *Power Transmission, Fluid Mechanics*

MECH08: USE OF POLYLACTIC ACID (PLA) MATERIAL FOR ASYMMETRIC SPUR GEAR MANUFACTURED BY 3-D PRINTING METHOD

Akshay Govardhan Dekhane[1]

[1]Assistant Professor, Department of Mechanical Engineering, Shri Chhatrapati Shivaji Maharaj college of Engineering, Nepti, Ahilyanagar, Maharashtra, India

[*]Corresponding Author Email ID: agdekhane@gmail.com

Abstract: When larger and smaller pressure angles are applied for the driving and coast side profile of the gear tooth, then such types of gears are categorizes as asymmetric gears. The two profiles of gear tooth are functionally different for most gear drives. The workload on one side of profile is significantly higher than other side of the gear. Design of gear with asymmetric teeth that enables to increase load capacity, reduced weight, size and vibration level. Design intent of asymmetric gear teeth is to improve performance of the primary drive profiles at the expense of the performance for the opposite coast profiles. The coast profiles are unloaded or lightly loaded during a relatively short work period. Asymmetric tooth profiles also make it possible to simultaneously increase the contact ratio and operating pressure angle beyond the conventional gears' limits. The main advantage of asymmetric gears is contact stress reduction on the drive flanks, resulting in higher torque density (load capacity per gear size). These benefits of the gears with asymmetric tooth profiles for unidirectional torque transmission are well known. In this work, asymmetric gear is designed, fabricated (by using 3-D printing method), and result obtained of analysis of the gear tooth (by using ANSYS software) and result obtained by experimentation have compared. By observing properties of different material I have selected the polylactic acid (PLA) for this work. From the comparison, it is concluded that the bending load carrying capacity of the spur gear is improved by using asymmetric teeth. Also it is noticed that the 3-D printing manufacturing method improves the strength of the gear.

Keywords: *Asymmetric Gear, Pressure angle, PLA, 3-D printing method, FEA*

MECH09: EFFECT OF EXHAUST GAS RECIRCULATION ON ENGINE PERFORMANCE AND EMISSION ON VARIABLE COMPRESSION RATIO ENGINE

Bharat Kharad[1]

[1]Assistant Professor, Department of Robotics & Automation Engg., Shri Chhatrapati Shivaji Maharaj college of Engineering, Nepti, Ahilyanagar, Maharashtra, India

[*]Corresponding Author Email ID: bnkvacoe@gmail.com

Abstract: Internal Combustion Engines are increased worldwide because internal combustion engine is having a better efficiency, maximum power with less maintenance cost. It is the most important power source for the automobile vehicles and transportation industry. Now a days emission norm becomes strict and very difficult for any I.C. Engine. The main pollutant are from Internal Combustion Engines are CO, CO2, HC, NOx, PM, soot, etc from which NOx are one of the most harmful component. Nitric Oxide affects the human and other living organisms also affect the environment. It is possible to limit the NOx in the atmosphere by various methods like exhaust gas recirculation (EGR), catalyst or SCR and water injection. The aim of this work is to find the effect of exhaust gas recirculation (EGR) to reduce the NOx emission from the I. C. Engine with measuring the all pollutants for different EGR rate with different load condition and find the optimum condition of emission and performance. It is found that adding EGR to the fresh air it is useful to reduce the NOx emission. Reductions in NOx emission are achieved by previous research with 5% to 30% EGR rate. However, EGR has other effects on combustion and emission production.
Keywords: *Exhaust Gas Recirculation (EGR), Nitric Oxide, I. C. Engine.*

MECH10: Performance, Combustion, and Emission Analysis of a Single-Cylinder Compression Ignition Engine Fueled with Diesel–Biodiesel–Ethanol Blends

Ganesh B. Salunke[1], Amol S. Kalhapure[2], Mangesh G. Kale[3]
[1,2,3]Assistant Professor, Department of Mechanical Engineering, Shri Chhatrapati Shivaji Maharaj college of Engineering, Nepti, Ahilyanagar, Maharashtra, India
[*]Corresponding Author Email ID: ganesh.salunke295@gmail.com

Abstract: Oil supplies keep shrinking while cleaner air rules grow stricter, pushing scientists to explore plant-based fuels for diesel motors. These new options carry oxygen inside them, which helps cut down harmful fumes when burned. Still, mixing ethanol straight into diesel fuel causes problems like separation, weaker ignition quality, less energy per drop. To fix that, researchers used biofuel as a bridge - helping everything stay mixed well in a three-part liquid mix: regular diesel, biofuel, plus ethanol. A single-cylinder, four-stroke, water-cooled diesel engine ran through an 8-mode steady-state cycle, left unmodified throughout testing. Using pure diesel as baseline, researchers examined three alternative blends instead. One mix combined 90% diesel with 10% biodiesel - labeled DB10. Another replaced 5% of the base fuel with ethanol, forming DBE10: 80% diesel, plus 10% biodiesel and 10% ethanol. The last variant, DBE20, reduced diesel content further - to 70%, holding biodiesel at 10% while increasing ethanol to 20%. Each blend went under identical conditions, ensuring fair comparison across performance measures. From the engine tests, brake power came through clearly alongside fuel use per unit of work. Measurements for how much heat got reused showed up in the data logs. Instead of grouping everything together, each efficiency point stood on its own. A device tracked gases like CO and CO_2 during operation. HC levels appeared lower when O_2 rose slightly. Though NOx behaved unpredictably at times, patterns emerged across runs. Each reading tied back to combustion behavior under load. Higher ethanol levels cut engine power because they carry less energy. Even so, fuel use climbs steadily as ethanol

rises. One mix, called DBE20, slashed harmful nitrogen oxides sharply - by as much as 56 percent - thanks to cooler burn temps. Then there is DB10, which burned cleaner with fewer carbon monoxide and hydrocarbon leftovers. That one, DBE10, held steady on both output and exhaust traits, standing out as a middle-ground option worth watching.

Keywords: *Diesel engine, Ethanol blending, Biodiesel, Alternative fuels, Emission control, Brake thermal efficiency.*

MECH11: Optimization of Suspension System for Minimization of Tyre Dynamic Force

Sachin Popat Jathar[1]

[1]Assistant Professor, Department of Mechanical Engineering, Shri Chhatrapati Shivaji Maharaj college of Engineering, Nepti, Ahilyanagar, Maharashtra, India

[*]Corresponding Author Email ID: sjathar48@gmail.com

Abstract: Suspension system has to perform complexity requirements, which includes road holding and equality, driving pleasure, riding comfort to occupant. Riding pleasure depends on vertical acceleration, with main objective to minimize vertical acceleration. The force transmitted by the deflection of tyres to the unsprung mass is known as the Tyre-Dynamic Force (TDF). The force should be considered while designing the suspension system. The TDF can cause vehicle instability and increase in the sprung mass acceleration. The objective of this research work is to minimize the TDF by optimum suspension design so that minimum vertical accelerations would be experienced by the passengers. To minimize the vertical acceleration, mathematical model of 2-DOF Quarter car model is considered for passive suspension system. Its equation of motion is formed with correlating objective function. Thus having objective function to, minimum weighted root mean square accelerations as per ISO 2631, viz. formulated for optimization. spring stiffness and damping coefficient are used as variables during optimization. The Genetic Algorithm is implemented as optimization tool with above mentioned objective function. The optimization results obtained were simulated and are compared with classical values. It is observed that the Seat acceleration, using optimized values, is reduced by 8.76% compared to the classical values. From experimental validation main objective function of Tyre dynamic force value reduced from 2300 N to 2100 N, it is decreased by 8.69 % and Seat acceleration reduced from 34.9176 m/s^2 to 32.0121 m/s^2 after replacing the conventional by optimized strut. Thus provides improved ride comfort to occupant/driver.

Key words: *Tyre Dynamic Force, Optimization of Suspension System*

MECH12: Shape Memory Alloy-Based Smart Composites: A Comprehensive Review of Properties, Fabrication, and Applications

Aditya B. Kale[1], Mohnesh D. Mandhre[2]

[1,2]Assistant Professor, Department of Mechanical Engineering, Shri Chhatrapati Shivaji Maharaj college of Engineering, Nepti, Ahilyanagar, Maharashtra, India

[*]Corresponding Author Email ID: adityakale26@gmail.com

Abstract: Shape memory alloys (SMAs) incorporated into smart composites represent a new class of advanced materials with the ability to recognize and respond to external stimuli. Because of reversible phase transformations between martensite and austenite structures, these materials exhibit unique properties like the shape memory effect and superelasticity. The fundamental concepts, characteristics, and production processes of shape memory alloys (SMAs), including vacuum melting, powder metallurgy, and thin-film techniques, are explained in this review. Nickel–titanium (NiTi), CuZnAl, and CuAlNi are among the alloys that are commonly studied because of their remarkable mechanical characteristics and functional performance. The study also examines the different applications of SMA-based smart composites in biomedical devices, aeronautical systems, and robotic actuators. By improving performance, reducing weight, and boosting efficiency, their use in aircraft morphing wings, artificial muscles, and bone plates, in particular, demonstrates that they have the potential to replace conventional systems. Despite its advantages, problems like high cost, short fatigue life, and difficult processing remain significant challenges. The primary objectives of current research are improving material performance and developing effective interfaces between active and passive components. All things considered, SMAs offer promising opportunities for the development of intelligent, adaptable, and multifunctional engineering systems.
Keywords: *SMA, Biomedical devices, Aeronautical systems*

MECH13: Optimization of Performance and Emission Parameters of Internal Combustion (IC) Engines : A Comprehensive Review

Pravin Ajinath Kekan[1], Bharat N. Kharad[2], Sonali N. Avhad[3]

[1]Assistant Professor, Department of Mechanical Engineering, Shri Chhatrapati Shivaji Maharaj college of Engineering, Nepti, Ahilyanagar, Maharashtra, India

[2,3]Assistant Professor, Department of Robotics and Automation Engineering, Shri Chhatrapati Shivaji Maharaj college of Engineering, Nepti, Ahilyanagar, Maharashtra, India

[*]Corresponding Author Email ID: pkekan6@gmail.com

***Abstract*:** Internal combustion (IC) engines are the heart of transportation industry. It is extensively used in transportation and industrial applications; however, they contribute significantly to environmental pollution. Optimization of engine performance parameters such as brake thermal efficiency (BTE), brake specific fuel consumption (BSFC), and emissions parameters including NO_x, HC, CO and particulate matter is essential to meet stringent regulatory standards. This review critically examines research progress in optimization strategies including biodiesel and ethanol blend usage, hydrogen enrichment, exhaust gas recirculation (EGR), and injection timing optimization. Methods such as Design of Experiments (DOE), Response Surface Methodology (RSM), Taguchi Technique, and artificial intelligence (AI) based modeling are investigated. Key findings, research gaps, and future directions are highlighted to guide further research and practical applications.
Keywords: *Internal combustion engine, emission optimization, performance, biodiesel, hydrogen enrichment, EGR, injection timing.*

MECH14: A Critical Review on Ternary Diesel-Biodiesel-Biodiesel blends with Nanoadditives for Diesel Engines

Nitin H Tambe[1], Dr. Y. R. Kharde[2]
[1]Research Scholar, PREC, Loni
[2]Principal, Shri Chhatrapati Shivaji Maharaj College of Engineering, Ahilyanagar
[*]Corresponding Author Email ID: nitin.tambe@avcoe.org

Abstract: The increasing energy demand globally and environmental concerns have accelerated research into sustainable alternatives to conventional diesel fuels. This paper is a comprehensive review of the possibilities and feasibilities of fueling diesel engines with ternary blends consisting of diesel, biodiesel, and biodiesel enriched with suitable nano additives. The study critically examines recent advancements in blend formation strategies, nanoadditive selection and their effects on fuel physicochemical properties, combustion characteristics, performance parameters, and emission profiles of engine. A comparative analysis of experimental findings from various studies is provided to highlight optimal blending ratios and the optimized role of nanoadditives in enhancing fuel stability, atomization, and thermal efficiency while managing NOx and particulate emissions. Further, the paper discusses challenges related to blend compatibility, production cost, and long-term engine durability. The review finally identifies research gaps and outlines future prospects for the commercial application of diesel–biodiesel–nanoadditive ternary blends as a viable pathway toward cleaner energy, healthier enviornment and more efficient diesel engines.

Keywords: *diesel engine, biodiesel, nanoadditives, ternary blends, combustion performance, emission reduction, fuel stability, renewable energy fuels*

MECH15: Recent Developments in Al-MMCs for Automotive Applications: A Critical Review of Reinforcements and Performance

Ganpat B. Jambukar[1], Dr. Y. R. Kharde[2]
[1]Research Scholar, PREC, Loni
[2]Principal, Shri Chhatrapati Shivaji Maharaj College of Engineering, Ahilyanagar
[*]Corresponding Author Email ID: gbjambo.me@gmail.com

Abstract: Aluminum Metal Matrix Composites (Al-MMCs) are increasingly being considered for automotive applications due to their ability to combine low density with enhanced mechanical and functional properties. This review presents a critical assessment of recent progress in Al-MMCs, with particular emphasis on the role of reinforcement strategies in determining material performance. Different categories of reinforcements, including ceramic particulates, fibers, and hybrid systems, are examined in terms of their interaction with the aluminum matrix, dispersion behavior, and contribution to property enhancement. The influence of commonly used fabrication methods—such as liquid-state processing, powder-based techniques, and emerging advanced manufacturing approaches—is discussed with respect to microstructural control and performance optimization. Relationships between processing conditions, reinforcement characteristics, and key properties such as strength, hardness, wear resistance, and fatigue life are analyzed. The application potential of Al-MMCs in automotive components requiring lightweight and high durability is also reviewed. Additionally, current limitations, including challenges in achieving uniform reinforcement

distribution, high production costs, and recycling concerns, are critically evaluated. The paper concludes by outlining future research opportunities focused on improving manufacturability, cost efficiency, and the development of innovative reinforcement systems for next-generation automotive materials.

Keywords: *Aluminum metal matrix composites (Al-MMCs); Automotive applications; Reinforcement; Mechanical properties; Wear resistance.*

MECH16: Synergistic Effects of Magnetic Field, Shake-Out Time, and Copper on the Microstructure and Mechanical Properties of As-Cast CI200

Mohnesh Mandhre[1], Sachin Ghalme[2]

[1]Department of Mechanical Engineering, Pravara Rural Engineering College, Loni, Affiliated to SPPU, Ahilyanagar, India

[2]Associate Professor, Department of Mechanical Engineering, Sandip Institute of Technology and Research Centre, Affiliated to SPPU, Nashik , India

[*]Corresponding Author Email ID: mdmscoea1@gmail.com

Abstract: CI200 cast iron is extensively utilized across industries due to its favorable casting characteristics, Properties, vibration damping, and economic viability. Nevertheless, the mechanical performance of alloy, especially its impact resistance and strength, is constrained by the inherent presence of graphite flakes within its microstructure. To address this limitation, the present study examines a combined processing approach that employs magnetic energy during solidification alongside adjustments in cooling schedule and chemical composition. The experimental investigation was structured using a Taguchi L9 orthogonal array to efficiently explore the effects of three variables: magnetic field strength (0, 3000, and 6000 mT), duration of mold retention (60, 90, and 120 minutes), and copper alloying content (0.5, 1.0, and 1.5 wt.%). Findings indicate that electromagnetic processing at the highest intensity (6000 mT) emerged as the predominant factor in impact resistance. This improvement is linked to the modification of solidification dynamics, leading to a more uniform dispersion of graphite and a homogenized metallic matrix. Pronounced synergistic interaction was identified when this high-intensity field was coupled with longest mold retention period (120 minutes), which promoted thorough and refined pearlitic transformation. The incorporation of 1.0 wt.% copper further optimized the microstructure by reinforcing the matrix, contributing to increased hardness.

Keywords: *Gray cast iron, SEM-EDS analysis, Impact toughness, Process optimization, Taguchi method*

MECH17: Review of the Functional Substitutes Derived from Industrial and Agricultural Process Wastes for Automobile Brakepad Manufacturing

Mr. Milind Gulve[1], Dr. Y. R. Kharde[2],

[1]PhD research scholar, Research Centre, Mechanical Engineering Dept. PREC Loni

[2]Research Guide, Research Centre, Mechanical Engineering Dept. PREC Loni

[*]Corresponding Author Email ID: principal@scoea.org

Abstract: Use of the process wastes finds a large number of applications in various product manufacturing industries. This not only helps in reducing the waste but also helps in reducing the overall manufacturing costs. The origin of such materials lies in the blast furnaces, cement manufacturing industries, agricultural and food processing industries,petroleum processing industry. There are numerous options available which can be substituted with the existing functional materials. The substitution of such materials may have a collateral damage but a careful selection of the substitute materials and their compositions in the product development would not sacrifice the performance of the replaced material in the main product. Flyash, slag, quarry dust, egg shells, agricultural wastes etc are classical examples which can be used in the manufacturing as substituting core ingredients. A conventional brakepad consists of Asbestos, metallic alloying elements such as copper, iron, aluminium, metallic oxides. They play the roles of binders, fillers reinforcements, frictional modifiers and friction stabilizers. For this study, the options available for the replacement of the functional materials in the brakepad and its effects on the performance of automobile consumable such as brakepads has been revived which are derived from various processing industries.
Keywords: *Industrial Process Wastes, Brakepad, Friction, Wear*

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Track-6: Others

OTH01: Interdisciplinary Fitness and Education: A Pathway to Holistic Student Development and Well-being

Avinash Machhindra Handal[1], Dr. Manik Maherban Rathod[2]
[1]B.Sc. Agril. Biotech; M.A; M.P.Ed; SET; Ph.D. Pursuing, M.S.M's - College of Physical Education, Khadkeswar, Chhatrapati Sambhajinagar
[2]M.S.M's - College of Physical Education, Khadkeswar, Chhatrapati Sambhajinagar, Dr. Babasaheb Ambedkar Marathwada University, Chhatrapati Sambhajinagar
*Corresponding Author Email ID: avinashhandal07@gmail.com

Abstract: This paper explores the symbiotic relationship between fitness and education, emphasizing their combined role in fostering holistic student development. It examines how integrating physical fitness into educational curriculums enhances cognitive abilities, emotional resilience, and social skills, contributing to comprehensive student well-being. The study analyses empirical data, psychological frameworks, and case studies to highlight the benefits of a fitness-inclusive education model. Additionally, it addresses the challenges of implementation and provides actionable recommendations to create an inclusive framework that benefits students across all demographics.

Keywords: *Fitness, Education, Holistic Development, Cognitive Abilities, Emotional Resilience, Social Skills, Well-being, Physical Activity, Student Engagement, Academic Success, Physical Education Policy, Inclusive Education.*

OTH02: Artificial Intelligence in Human Resource Management Transforming Talent Acquisition, Employee Engagement, and Organizational Efficiency

VAIBHAV MOHANLAL MORE[1]
[1]Shri Chhatrapati Shivaji Maharaj College of Engineering, Ahilyanagar
*Corresponding Author Email ID: vaibbhav.moore@scoea.org

Abstract: Artificial Intelligence (AI) is rapidly redefining the landscape of Human Resource Management (HRM) by enabling data-driven decision-making, enhancing operational efficiency, and transforming the employee lifecycle. This paper explores the multifaceted role of AI in revolutionizing key HR functions, particularly talent acquisition, employee engagement, and organizational efficiency. In talent acquisition, AI-powered tools such as predictive analytics, natural language processing, and automated screening systems streamline recruitment processes, reduce bias, and improve candidate-job fit. The study further examines how AI-driven platforms facilitate personalized employee experiences through real-time feedback systems, sentiment analysis, and intelligent chatbots, thereby strengthening employee engagement and retention. Additionally, the paper highlights the impact of AI on organizational efficiency by automating routine HR tasks, optimizing workforce planning, and enabling strategic decision-making through advanced analytics. While emphasizing the benefits, the research also addresses critical challenges, including ethical considerations, data privacy concerns, algorithmic bias, and the need for human-AI collaboration. Through a comprehensive review of existing literature and emerging industry practices, this study provides insights into how organizations can effectively integrate AI into HRM to gain a competitive advantage. The findings suggest that AI, when implemented responsibly, has the

potential to not only enhance HR processes but also create a more agile, inclusive, and performance-driven organizational environment.

Keywords: *Artificial Intelligence, Human Resource Management, Talent Acquisition, Employee Engagement, Organizational Efficiency, Predictive Analytics.*

OTH03: Balancing Rights and Responsibilities of College Students: A Legal and Educational Perspective

Mr. N. P. Kute[1]

[1]Shri Chhatrapati Shivaji Maharaj College of Engineering, Ahilyanagar

[*]Corresponding Author Email ID: nikhilkute@gmail.com

Abstract: This study examines the relationship between rights and responsibilities among college students within a legal and educational framework. Education is not only a tool for academic growth but also for developing responsible citizens. The research is based on a descriptive analysis of secondary data, including legal provisions, educational policies, and scholarly literature. It highlights key constitutional rights such as equality, freedom of expression, and access to education, along with corresponding duties like respect for laws, discipline, and social harmony. The paper emphasizes that rights and responsibilities are interconnected and must be balanced to maintain a healthy academic environment. It also discusses challenges faced by students today, including misuse of technology, lack of discipline, and weakening moral values. The study suggests that value-based education, awareness programs, and effective policy implementation are essential to address these issues. The findings conclude that a clear understanding of both rights and responsibilities plays a crucial role in the overall development of students and contributes to building a responsible and progressive society.

Keywords: *Student Rights, Fundamental Duties, Right to Education, Constitutional Law, Higher Education, Legal Awareness.*

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